

CHAPTER FOUR: In-School Veterinary Science Classes for High School (HS)

1 HS and University-level Veterinary Science Class

For high school students, the youth and sled dog program can be part of the curriculum as an elective course, as a science course, or as a course with dual high school and University-level credit for the student participant.

- For an elective credit, veterinary science classes with a kennel workshop component can be taught by any teacher.
- For credit in science toward high school graduation, only an Alaska Board of Education-approved class, taught by a certified animal husbandry teacher or science teacher with extensive animal experience will qualify.
- For dual high school/college credit, the University of Alaska Veterinary Medicine Department needs to certify the course.

2 HS Veterinary Science State of Alaska Board of Education-approved curriculum for dual high school and/or college credit

~ [King Career Center's Veterinary Science I & II curriculum](#) by Leesa Wingo

~YKSD 's Veterinary Science I, II, III, & IV curriculum by Peggy Bruno

3 HS Veterinary Science lesson plans

~YKSD's Veterinary Science lesson plans by Peggy Bruno

~HS Math Science ideas

~GPS use

4 University-level Veterinary Science & Dog Mushing curriculum & lesson plans

~University of Alaska Anchorage's Dog Mushing curriculum/lesson plans by Lisbet Norris

~Alaska Pacific University's Dog Mushing curriculum/lesson plans by Annie Grenier

5 University of Alaska-Fairbanks Department of Veterinary Medicine Involvement

The Department of Veterinary Medicine has offered assistance in developing curriculum and lesson plans including dog health, nutrition, and exercise physiology and performance. Classroom lectures may be provided by video, live video-conferencing, and/or on-campus class. Tours of the veterinary student facilities can be scheduled. Visits to the University give students the opportunity to see campus life and to talk to administrators and educators about degree programs before they graduate from high school. Student trips to UAF during the winter can be potentially combined with students' shadowing veterinarians at local sled dog kennels. Trips

made during the spring allow students to shadow veterinarians and dog mushers at the adult championship sprint sled dog races.

~[Example of Student Trip to UAF/ONAC in March 2016](#)

Dog Health and Kennel Manuals for teaching HS

~ [Mushing with Pride](#)

~[Ravenwood Veterinary Kennel Management Handout](#)

References for HS teachers

~[Curriculum and References about Culture/Dog Mushing](#)

Learning Outcomes At a Glance

QUARTER 1	QUARTER 2
1. Demonstrate safety in office, surgery, laboratory and kennel settings. 2. Complete OSHA General Industry 10 Hour Safety Training with 80%+ average. 3. Read and interpret a veterinary chart and progress notes correctly 4. Organize, develop and run a doggie daycare. 5. Create and maintain digital records of members of the doggie daycare. 6. Use a digital office system to manage the clients of the doggie daycare. 7. Wash/Groom 5 dogs 8. Manage inventory for doggie day care. 9. Research the <i>veterinary team</i> requirements for education, licensure, employment, continuing education, and retirement, including job-shadowing members of the veterinary community, phone interviews and Internet research. 10. Communicate effectively through writing, speech, and graphical displays 11. Demonstrate the capability to work both independently and in cooperation with others 12. Construct a functional resume and portfolio 13. Analyze diet requirements by species. 14. Use the correct restraint technique for the required veterinary process. 15. Teach basic obedience skills to an untrained dog.	1. Develop an owner/patient education program about current health issue (eg. Reproductive health, dental, vaccines) 2. Perform and interpret blood sugar analysis, urinalysis, and hematocrit. 3. Perform stool float analysis for Alaska Dog and Puppy Rescue group's puppies 4. Research and present the top 10 most common surgical procedures 5. Observe a surgery, journal about the experience 6. Demonstrate sterile techniques 7. Investigate the overlapping types of "clean" 8. Interpret and evaluate veterinary information and technical writing. 9. Critically evaluate ethical scenarios. 10. Disassemble, clean and assemble surgical equipment and sets. 11. Sort drugs according to schedule and 12 categories. 12. Demonstrate correct storage and dispensing of veterinary pharmaceuticals. 13. Calculate correct dosages of veterinary medications. 14. Calculate percent concentrations of solutions. 15. Explore HOSA's veterinary component

Essential Concepts	Essential Vocabulary (only a sample shown)	Standards	Assessments	Instructional Resources	Instructional Considerations
Quarter 1: Enduring Understanding – Communication is the Key Essential Questions - What makes an excellent veterinary clinic? - How are biology and business interrelated? Main Text Book: Text: Vanhorn, B., & Clark, R. (2011). <i>Veterinary assisting: fundamentals & applications</i>. (1 ed.). Clifton Park, NY: Cengage. Good references: Lawhead, J., & Baker, M. (2009). <i>Introduction to veterinary science</i>. (2nd ed.). Clifton Park, NY: Delmar Romich, Janet A., (2009). <i>An Illustrated Guide to Veterinary Medical Terminology, Third Ed.</i> Clifton Park, NY: Delmar					STANDARDS OSHA=Occupational Safety Health Administration HOSA=Health Occupations Student Organization CTC=Common Technical Core HL=Career Cluster Health HL-DIA=Health Pathway Diagnostic CCEL=Common Core English Language CCSM= Common Core Science Math SC=Concepts of Life Science SG=History and Nature of Science SA=Science as Inquiry and Process ACS=Alaska Cultural CRP=Career Ready Practices AAI=All Aspects Industry
1) Safety practices are essential, and have protocols to ensure health of all concerned.	OSHA PPE Sharps container Physical Hazards Chemical Hazards Chemotherapy drugs Insecticides Anesthetic gases Biological hazards Zoonotic hazards Brucellosis Canine distemper Ringworm Coccidia Rabies	Medic First AID/CPR CTC HL-2,3,6 CCEL SL.11-12.1,2 RST.11-12.1-4 CCSM [11]SA1.1, 1.2, 2.1, 2.3, 3.1 [11]SC1.1, 1.2, 2.1, 2.2, 2.3 A-CED-4 CRP 1,2,3,4,8 AAI 3,4,6,8 American Red Cross Canine First Aid	Observation Skill rubric, Online Exam	Text, HOSA Website www www.osha.gov OSHA 10 hr. General Industry online Safety certification course (\$25 per student) <i>Gloves, gowns, goggles, hats, shoe covers, drapes, cleaning solutions</i>	1. Create Room, Program, Building Safety Plans. 2. Pass OSHA Test with combined score of 80%+ 3. Devise animal emergency evacuation plans for school and for home use. 4. Assemble animal first aid kits for student with animals. 5. Role play for Vet1 students in their first aid training. 6. Monitor safety in the program/classroom/kennel weekly, and discuss any issues that appear.

	Lyme Disease Rocky Mountain Spotted Fever Leptospirosis Tetanus Parvovirus Hepatitis Toxoplasmosis Tick Paralysis MSDS, Right-to- Know Station				7. Design PPE paper dolls or puppets for each area of the program 8. Make a short video about safety in the workplace. Safety should take the first 4 days, then ongoing throughout the semester.
2) Veterinary Medical Language is made up of Latin and Greek Prefixes, Roots, and Suffixes, and knowing the basics is the key to understanding records, charts and notes	(Small sample) Lagomorph Murine DSH Dysuria V/D Dysuria NPO NAVTA Cystocentesis OU TID SOAP Sx	HOSA Clinical Practices HL-2 HL-DIA-1,2 CCEL SL.11-12.1A, RST.11-12.1,2,3,4,5 CRP 1, 3 AAI 7	Observation, written exam	<i>Textbook</i> <i>Classroom Wiki</i> <i>Business partners (vet offices)</i> <i>Romich</i> <i>Terminology Book</i>	1. Use wiki to reinforce vocabulary, and present case studies/records, charts and notes. 2. Learn Vocabulary by systems and topics. 3. Learn Common abbreviations and acronyms. 4. Use business partners to give real case notes. 5. Have students bring in copies of own animals' vet records and make copies for the class to interpret. 6. Make walk-around classroom activities to practice voc. ie. Tic-tac-toe Have students review last semester's vocabulary, then continue with topic specific

					vocabulary from Romich book
3) Running a doggie daycare combines animals, health, fitness, training, behavior and communication.	Conditioning, inappropriate elimination, instinctive behavior, learned behavior, modeling, modification techniques, pheromones, submissive, aggression, dominance/fear/aggression, Styptic powder Anal gland, keratin, buccal, labial	HOSA Clinical Practices CTC HL-1,2,6 HL-DIA-2,6 CCEL WHST.11-12.4 CCSM [11]SA3.1 [11]SC2.2, 2.3 [11]SE3.1 A-REI-10 ACS A6, B1 CRP 1,2,11 AAI 2,7,8	Customer evaluation, Rubric, Observation	Notes from the grooming Table by Melissa Verplank Need Dog friendly flooring, dog washing sink, rest area, grooming tables, equipment, exam tables, pens, kennels, toys, treats	1. Investigate the services in the community to see best practices. (involve admin.) 2. Design a brochure for your services, include what you 'll offer, how long it will take, what the industry charges, etc. 3. Create an area in which to do this. 4. Students determine what roles are needed, and write job descriptions. 5. Do it. Ongoing once set up. Keep in mind scheduling process, vacation days, fire alarms, and staffing.
4) Maintaining records are a necessity for running a business successfully.	VCPR Health certificate Confidentiality Cage card Numerical filing system Flow schedule Policy manual Procedures manual Wave schedule CPU Inventory	HOSA Clinical Practices WHST.11-12.2,4,6 ACS A-1, 4, B-1 CTC HL 1,2 HL-DIA 2 CRP 1,2 AAI 7	Customer evaluation, Rubric Observation	Digital pet record keeping programs. Microsoft Excel	1. Find industry standards 2. Evaluate if economically feasible. 3. Choose most economical yet efficient choice 4. Have students practice with own animals/staff animals 5. Go public Tied to Doggie Daycare, so

	Consumable Packing slip Automated inventory system				make sure students have assigned roles, and that there is a designated leader.
5) There are a multitude of ways to learn and share about the veterinary team, education, roles and responsibilities.	AVMA AAHA Continuing Education NAVTA CVT, LVT, RVT Veterinary Hospital Manager Kennel Attendant Receptionist Isolation ward Surgical Suite Invoice Travel sheet Veterinary Practice Act	HOSA Veterinary Careers SL.11-12.4, 5 RST.11-12. 3 ACS B-2, E-6 CTC HL 2,4,5,6 CRP 2,4 AAI 8	Pre-, Post- test, Rubric	Internet Community Partners Barriers to employment in Health care from Dept. of Labor. Free Resources - The College of Veterinary Medicine at Cornell	1. Compare/Contrast Education, roles, responsibilities, training through Internet research, and job shadowing. 2. Only students with perfect attendance and professionalisms scores can go to job shadowing experiences. Present this at the beginning of the semester, along with barriers to employment. 3. Identify Training programs and schools, locations and costs 4. Phone interviews of business partners in non-clinical settings Job shadows will last 2 hours each, 2 days each. One student at a time, at 3 different locations.
6) Teamwork is required in the clinic and in the classroom	Communication Verbal, nonverbal, sender, receiver, feedback, channel, body language,	[11]SA3.1 SL4, SL5 CTC HL-DIA 2 CRP 1,4,12 AAI 3	Observation, Self and Peer evaluations	http://www.recover-from-grief.com Midnight Sun Service Dogs	1. Team building activities to learn names, build rapport, make cleaning more fun, etc. 2. Role-play communication

	written communication sympathy vs. empathy kindness, patience, tactfulness speech communication, listening, conversation skills grief, euthanasia, humane, shock, denial, bargaining, acceptance, anger, sorrow, depression, guilt			Business partners	games. - Discuss grief process, but alert your counselor, as many students have unfinished business in dealing with their own grief. - Guest Speakers from Midnight Sun Service Dogs - Guest speakers – Office or Clinic Managers bring horror stories of communication with clients - Pair share on specific animal topics Spend 4 days on Communication, but return to it weekly for review and to use as a tool for other concepts.
7) Brand and sell yourself to get the career you want	Portfolio Functional Resume	WHST.11-12.4 SL.11-12.4, 5 ACS A-2 B-2, D-6, E-6, E-8 CTC HL-DIA 2 CRP 4 AAI 3	Audience evaluation, Parent, self, and teacher rubric	AKCIS.org Department of Labor personnel 1” Binders Clear Sleeves Dividers	- Sort personal characteristics and write concrete examples for each character trait. - Incorporate traits into personal essay - Construct functional resume - Create Scholarship Portfolio - Refine and rewrite it - Participate in mock interviews

					1 week in class, rest at home
8) What's for Dinner? Form Fits Function!	(sample)Amino acids Antioxidants, balanced ration, biological value, body conditioning score, calories, fatty acids, hydrolysis, lactation diet, nonessential fatty acids, trace minerals, trimester	HOSA Animal Nutrients CTC HL-DIA 4 RST.11-12.3 WHST.11-12.4 [11]SA3.1, [11]SC2.1, [11]SE2.1, 3.1 A-CED-4 N-Q.1 ACS C1,2 CRP 2 AAI 3	Lab report rubric	Text book Internet Alaska Mill and Feed Jaws, Teeth, Lab Equipment for nutrient analysis	1. Research types of teeth/jaws and correlate form and function. 2. Diagram the digestive system of a pig, chicken, horse, cat, dog, and cow. Describe the adaptations of poultry. Describe the differences between ruminants and non-ruminants. 3. Perform dissections if possible. 4. Research and analyze the nutrients in commercial dog and cat foods. Compare to Science Diet available from your veterinarian.
9) Restraint techniques match their purposes.	Restraint, sedative, tranquilizer, muzzle, squeeze cage, hobble, twitch, stanchion, square knot, Reefer's knot, Slip knot, Half Hitch, Recumbency, Sternal/lateral/dorsal recumbent, cephalic venipuncture restraint, jugular vein, saphenous vein, anesthesia chamber,	HOSA II, VIII CTC HL-DIA 2 [11]SC2.1 CRP 1,3,4 AAI 7	Observation, feedback from partners	Text Book Noose Leashes Soft Muzzles Wire Muzzles Gauze Towels Cat Bag Animal models	1. Practice on models 2. Practice on compliant animals at home, take videos and bring in. 3. Practice on compliant animals in day care. Initially 2-3 days, but practice intermittently as restraints are needed all semester.

	cat bag, squeeze chamber, scruff technique, stretch technique, Kitty Taco, Rabies/snare pole, fractious				
10) Dog training using positive reinforcement and intermittent reinforcement works fastest.	Sit, Stay, Wait, Up, Off, Down, Sit Pretty, Spin, Tunnel, Weave, Touch, Heel, Release, Operant Conditioning, Positive Reinforcement, Negative Reinforcement, Extinguish	HOSA Animal behavior CTC HL-DIA 4 CCEL SL.11-12.1 RST.11-12.1,3 CCSM [11]SC2.1, SC2.2, SC2.3 ACS C2 CRP 1, 3, 4 AAI 7	Observation, Skills test for dog	Training information/demo	1. Practice Obedience with doggies in daycare. 2. Practice at home with own dogs. Take video for proof. 3. Use Instructor's dog for demo. 4. Volunteer with AK Dog and Puppy Rescue Initial 3-4 days to review all commands, but ongoing practice reinforces success
11) Research HOSA and identify which schools run student groups and how to join.	Parliamentary procedure, Officers, HOSA bowl Knowledge test Skills-based test	HOSA All ACS B CRP 1,4,6,8,9,10,12 AAI 3,4,5,6,7	Rubric, Presentation	HOSA.org	1. Create a flow chart of the officer responsibilities in HOSA. 2. Visit a HOSA meeting and investigate how to join 3. Report findings to class 3 days and on-going

Essential Concepts	Essential Vocabulary (only a sample is shown)	Standards	Assessments	Instructional Resources	Instructional Considerations
Quarter 2 Enduring Understandings: Problem-solving requires protocols, persistence and principles. Essential Questions: - If the animal is in distress, can it be helped? - What is considered “clean”? - How do we minimize human error?					
1) Educate the owners to solve a multitude of problems.	Vocabulary will depend on the issue for public education.	HOSA Clinical Practices CTC HL 5,6 HL-DIA 1 CCEL RST.11-12.3 WHST.11-12.4,5,6 CCSM [11]SA1.1, SA3.1, [11]SC2.3 [11]SE2.1,SE3.1 SMP2,3 7. RP- 2 ACS B3, B4, C2,3,4 CRP 1, 3 AAI 4	Audience pre-post test	AVMA.org website Business partners Audience	1. Interview a business partner to gather potential issues for public education. 2. Judge the most pertinent and interesting topic. 3. In teams of two, design a pamphlet and corresponding education presentation. 4. Deliver to at least one audience with pre- and post test. 5. Evaluate the effectiveness of your presentation. 1 week
2) Accurately perform Laboratory tests to identify medical conditions, diseases and parasites.	Reference Lab, centrifuge microhematocrit, refractometer Serologic test kits, reagent Coplin jar Fecal	HOSA I, Disease Prevention CTC HL-DIA 2 CCEL SL.11-12.1 RST.11-12.1,3 CCSM [11]SA1.1 SA2.1, SA3.1, [11]SG3.1 2 MD-1, A-CED-4 A-REI-10	Skills rubric	Text Book PPE Samples from Business Partners IDEXX SNAP tests Lab Equipment: Microscope,	1. Use a microscope to examine pre-made bacterial slides, then identify unknown samples. 2. Examine Fecal Samples for parasites. 3. Make Blood Smears from samples obtained from

	Gross examination fecal flotation Vacutainer Tubes hemolysis Serum plasma, anticoagulant packed cell volume capillary action buffy coat, Total protein, Electrolytes, Hematology SNAP Test urinalysis, cystocentesis, midstream flocculent specific gravity chem strip, decant gram stain, culture and sensitivity Necropsy Ronguers	S-ID-6a,b,c N-Q.1 CRP 2, 3 AAI 7		slides, coverslip, lens paper, immersion oil, centrifuge, centrifuge tubes, microhematocrit tubes, refractometer, Reagents, Coplin jars, stains, Fecal sample solutions, trays, applicator sticks, wax pencil, Lugol's solution, methylene blue, saline, timer, Vacutainers, SNAP test examples, Chem Strips, Paper Towels, cadaver bag, suture material. needle, ID tag, tape, pathology forms	Business partners 4. Calculate total protein using a Hematocrit. 5. Collect a Voided Urine Sample from Guest Dog 6. Complete a urinalysis. 7. Prepare a sample as a gram stain. 8. Prepare a sample for culture and sensitivity test. 9. Prepare for a necropsy, and clean up afterwards. If taught as a unit, this may take 2+ week. You may want to intersperse these throughout the semester. *Cystocentesis is not appropriate in the high school setting. Build your own urine catcher by nailing a clean tuna can on the end of a broomstick. Cover the can with an inverted new Ziploc bag.
3) "Clean" can mean different things in different settings, and sterile technique helps control the spread of infection.	Sanitation, cleaning, disinfecting, sterilizing, cold sterilization, cold tray, dry heat, incinerator autoclave, radiation, ultrasound, filtration, antiseptics,	HOSA I, Disease Prevention CTC HL-DIA 2 CCEL SL.11-12.1 RST.11-12.1,3 CCSM [11]SA1.1 SA2.1, SA3.1, [11]SG3.1 2 MD-1 A-CED-4 CRP 2, 3	Observation, Skills rubric	www.osha.gov Becker bottle to show proportion	1. Practice dilution with colored water, then with disinfectant. 2. Test hand cleaning efficacy with glo-germ. 3. Order the areas to clean in a clinic. 4. Maintain clean animal

	hand hygiene, Asepsis, sterile technique, isolation ward, aseptic, microorganisms, sterile, contamination, pathogen, 'clean', disinfect, sterile field	AAI 7			enclosures through cleaning daily. - Clean all surfaces in facility. - Fold clean linens and clothing, and put away. - Maintain cleanliness and sterility in facility - Clean and sterilize surgical instruments, grooming equipment and toys - Demonstrate asepsis during mock surgery. - Create a wiki page to compare the different "levels" of clean. - Identify the most common veterinary nosocomial and zoonotic diseases. Share this information in a booklet, poster, poem or song. 2-3 weeks, then ongoing practice and use.
4) Surgery may be a vital part of the treatment plan.	Intubation, extubation Laryngoscope Laparotomy scrub pack open gloving closed gloving non-absorbable	HOSA I, HOSA Principles of Surgery CCEL WHST.11-12.4 SL.11-12.4, 5, CCSM [11]SC2.3 [11]SE2.1, 3.1 CTC HL 5,6	Presentation Rubric Journal	Dog and cat models, Intubation Tubes, Mock Surgery Suite and equipment, surgery packs and supplies	- Disassemble, clean and assemble surgical equipment and sets. - Observe a surgery, and share your experience with the class and with in a journal.

	suture swedged on needles suture reel “ought” surgical staples surgical glue palpebral reflex scavenger hose instrument milk	HL- DIA 2 CRP 2 AAI 8			- Investigate the top 10 surgeries performed on pets in the United States, and present to the class. - Gown for surgery using aseptic technique. - Prepare and drape a sterile surgical field. - Intubate and extubate the cannikin and catikin. - Identify and state functions of surgical tools.
					2-3 weeks
6) Technical writing and information has a specific style that still needs evaluated.	(Any terminology or procedures will show up here depending on which journal article is chosen)	[11]SG2.1 [11]SA3.1 HOSA Disease Prevention RST.11-12.1,2,3,4,5 CTC HL 1 HL-DIA 2 ACS C2,3,4 E8 CRP 1,5,7 AAI 7	Written Evaluation Discussion Rubric Panel Discussion	Access to veterinary journals through AVMA Access to veterinary articles.	Read Veterinary journals. Evaluate for threat to validity as to how the data was collected. Check 5 questions: Was there a good comparison group? Was there a plan? Who knew what when? Who was left out? How much did things change? Was there a control group? Concurrent or Historical? Was it observational or did they have experimental design? Did the research have a narrow focus? Was it a double blind trial? Statistically valid?

					Who knew what when? Who was left out? Sample size? Measurements have units? 2. Read veterinary articles for content and interest. 3. Write original entries and responses on blog about articles/journals on Edublog, or district's blog sites. Introduce over 2 class periods, ongoing as interesting articles arise.
7) Veterinary Ethics are moral principles that apply values and judgments to the practice of vet medicine.	Misrepresentation, slander, confidential, Veterinary Practice Act, State Board of Veterinary Medicine, Liability, Malpractice, Negligence, OSHA, ADA, Fair Labor Standards Act, Child Labor Law, Animal Welfare Act, FDA, DEA, Controlled Substances	CCEL RST.11-12.3 WHST.11-12.4,5,6 ACS D6, E8 CRP 7,10 AAI All	Written Evaluation Discussion Rubric Panel Discussion	Internet Business partners Ethical Issues to explore: Euthanasia Informed consent Complementary and Alternative Medicine Cosmetic interventions Non-therapeutic mutilations Negligence Professionalism Religious influences Research ethics Confidentiality Selective breeding Professional (self) Regulation	1. Research Ethical issues, and form a panel discussion of topic. 2. Critique public policy concerning ethical issues 3. Interview Business Partner for most pressing ethical issue for which they deal. 4. Respond to activities on class blog. 1-2 weeks initially, ongoing as hot topics arise.
8) Pharmacology in	aural, balling gun,	HOSA Clinical Practices	Written Test and Quizzes, Skills Rubric	Text	1. Read and interpret a

Vet Med is similar to that in human medicine, except that it is prescribed by the Veterinarian and sorted/dispensed by the Vet Techs and Vet Assistants.	controlled substance, dram, efficacy, hematoma, OTC, ophthalmic, pet piller, pinna, Rx, scheduled drug, topical	CTC HL 2, 3 HL-DIA 3 CCSM [11]SA1.1 ACS B1 CRP 1,5 AAI 3		Placebo Greenies Business Partners Fake Prescriptions	prescription by a veterinarian. 2. Prepare a prescription using a counting tray. 3. Create a display showing the Scheduled drug classes and examples. 4. Demonstrate how to administer oral medication using a placebo 5. Demonstrate how to administer aural medication 6. Demonstrate how to administer topical medication 7. Investigate common animal medications and for what they are used. 8. Calculate medication dosages for weights. 9. Calculate and mix solutions. 2 week introduction, longer to memorize drug names.
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Key Vocabulary

Word	Definition
Quarter 1 (a small sample)	
Amino acids	Building blocks of protein that form in chain-like structures
Antioxidant	Vitamin's that boost the body's immune system
Brucellosis	Reproductive disease that is highly contagious and has no treatment; also known as Bang's disease
Cystocentesis	Surgical puncture into the bladder using a needle to collect a urine sample
DSH	Domestic Short Hair
Dysuria	Difficult or painful urination
Euthanasia	The process of humanely putting an animal to sleep
Fractious	Ornery, disruptive
Hydrolysis	Chemical process of breaking cells down into smaller particles
Lactation	The process of milk production
Lagomorph	Species name for rabbits
Murine	Veterinary term for mice or rats
NAVTA	National Associations of Veterinary Technicians in America; sets the standard of veterinary technology and the care of animals
NPO	Nothing by mouth, as in food or water
OU	Both Eyes
Physical Hazards	Safety concerns that may cause physical injuries such as bites, kicks, scratches, or lifting heavy objects
PPE	Personal Protective Equipment
Stanchion	Head gate that holds the head of a cow in place during restraint
Styptic powder	Chemical used to stop bleeding
Trimester	The period of three months of pregnancy when additional nutritional needs must be provided and the development of the fetus occurs
V/D	Ventral-dorsal; position where the animal is restrained on its back with the x-ray beam traveling through the ventral aspect first and the dorsal aspect second
Veterinary Practice Act	Legal document that outlines rules and regulations of veterinary professionals
Zoonotic hazards	Safety concerns that allow contagious organisms to be spread to humans, causing infections, viruses, bacterial, fungal, and parasitic transmission

Quarter 2 (a small sample)	
Aural	Pertaining to the ears
Balling gun	A metal device with a long handle that has a plunger at the base, used to hold medicines to administer to large animals
Centrifuge	Piece of equipment that uses high speeds to separate liquid portions into solids
Controlled substance	Drugs that have the potential for abuse and addiction and must be kept in a locked cabinet by law
Dram	The measurement size of vials or bottles
Efficacy	The strength of the drug
Hematoma	The rupture of a blood vessel causing a fluid-filled pocket of blood
Intubation	The process of placing a tube into the trachea to establish an airway and allow the patient to continue to inhale gases that keep the patient under general anesthesia as long as necessary
Laparotomy	Surgery cutting into the abdomen
Laryngoscope	A tool made of heavy metal that has a light source on the end to help light the airway view
Microhematocrit	Measurement of the percentage of red blood cells in whole or unclotted blood; also called a PCV
Ophthalmic	Having to do with the eyes, usually medication
OTC	Over-the-counter drugs; medications that do not require a prescription
Pinna	External portion of the ear
Rx	Prescription
Scheduled Drug	Medicine with the potential for abuse and categorized from I to V according to strength; I is highest potential and V lowest potential for abuse
Topical	A drug that is administered to the skin

YKSD 's Veterinary Science I, II, curriculum by Peggy Bruno

A Basic Animal Husbandry for Veterinary Sciences

Animal Husbandry for Veterinary Sciences I

Animal Husbandry for Veterinary Sciences II

CTE Course Description & Cross-walk

Course Name: Basic Animal Husbandry for Veterinary Sciences Course Number: VTS 140 Date: August, 2010	District Name: Yukon-Koyukuk School District Developed by: Gina Hrinko with UAF-IAC Adapted from: UAF-IAC Course	EED USE ONLY	Date: Approval Status:
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■ Career Cluster/Pathway: Agriculture ■ Is this the final course offered by district? YES ☐ NO ☐ → Next Course In Sequence: ■ High School Credit = 0.5 ■ This course will be offered: X every year? <u>OR</u> ☐ every other year? ■ Technical Assessment Title & publisher: ■ Is this course part of a CTE Program of Study agreement? YES X NO ☐ If Yes, list postsecondary institution & attach the CTEPS Agreement: University of Alaska Fairbanks – Interior Aleutians Campus	■ Prerequisite Course(s): VTS 101 prior to or concurrent with ■ Is this course brokered through another institution or agency? X No <u>OR</u> ☐ Yes If Yes, list institution/agency: ■ Tech Prep: ☐ No <u>OR</u> X Yes If Yes, (1) List Postsecondary Institution: University of Alaska – Interior Aleutians Campus (2) Name of Course: VTS 140 Basic Animal Husbandry for Veterinary Sciences Number of postsecondary credits 3
■ Eligibility for Nationally Recognized Skill Certificate(s)/State License or credential at conclusion of this course? X No <u>OR</u> Yes If Yes, identify Certificate(s) or License:	

■ Course Description (a brief narrative description as published in the school's registration materials *OR* attach the district syllabus): This course provides an understanding of animal husbandry for canine, feline, goat, sheep, pig, horse, reindeer, musk oxen, bison, and cattle. Through lectures and a three day practicum students will explore animal behavior, proper restraint, handling, and management. Insight will be provided on how to adapt techniques to different situations, such as the Alaskan climate and geography.

■ Content Headings/Topics:

- **Breed and Species Identification** – This covers identification of large and small animals by species and breed.
- **Animal Behavior** – Includes tips for animal feelings, animal groups, and animal perceptions.
- **Restraint** – Various restraint techniques for dogs, cats, cattle, sheep, goats, swine, bison, musk ox, reindeer, and equine, including animal and personnel safety, handling, fencing, and chemical restraint. Hands on herding, capturing, and restraint of beef cattle, musk ox, reindeer and bison.
- **Basic Feeding Principles** – Addresses concepts of seasonal nutrition concern, feeding schedules, and feeding methods.
- **Animal Handling** – This includes hands on learning of handling, examining, and caring for domestic animals in a safe manner.
- **Housing** – Covers the basics of facilities, physical plant properties, and ventilation principles for dogs, cats, cattle, sheep, goats, swine, bison, reindeer, musk ox, and equine.
- **Management** – Addresses concepts of management for swine, beef and dairy cattle, sheep, goats, reindeer, bison, musk ox, equine, dogs, and cats. Empathesis is placed on nutrition, reproduction, disease and parasite control, and optimal environment. Participation in management of dog kennel with evaluation of housing and animals. Immunization, hoof trimming, fur and antler harvest, and nail trimming will be put into practice.
- **Marketing and Slaughter** - Covers slaughtering techniques and requirements, as well as current market products.

Source of Occupational Skills Standards

Source/Organization/Agency	Acronym	Website or Location of Information	Section, Chapters, etc. referenced in Performance Standards	Date of Edition or Version
States Career Cluster Initiative	SCCI	www.careerclusters.org	(use appropriate pathway)	2008
National Agriculture, Food, and Natural	AFNR	www.teamaged.org/councilindex.cfm	Animal Systems Career	2009

CTE Course Description & Cross-walk

Resources Career Cluster Content Standards				Pathway		
Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standards	Alaska Reading, Writing, Math, Science 4 th Ed. PSGLE's	Alaska Employability Standards	Alaska Cultural Standards	All Aspects of Industry	How Will Students Be Assessed? (Formative Level)
Students will learn about and be able to demonstrate knowledge on: Breed and Species Identification - Large animals - Geographic differences - Small animals - Domestic avian species - Exotic breeds	AS.02.01.02.a	R4.2.2;4.3.2; W4.2.2;4.2.4; SC2.1	A1-2	A4; B2; C4; D1; E1	Technical and Production Skills Health, Safety, and Environment	Written assignments; Class participation; Completion of practicum, Completion of Institutional Animal Care and Use Committee (IACUC) Module 1; Quizzes, Final Examination
Students will learn about and be able to demonstrate knowledge on: Animal Behavior - Animal groups - Animal perceptions - Animal feelings	AS.02.03.02.b	R4.2.2;4.3.2 4; W4.2.2;4.2.4; SC2.2;3.2	A1-7	A4; B1; E4	Technical and Production Skills Community Issues	Class participation; Practical Exercises; Completion of practicum, Completion of Institutional Animal Care and Use Committee (IACUC) Module 1; Quizzes, Final Examination
Students will learn about and be able to demonstrate knowledge on: Restraint - Animal safety - Human safety - General methods - Chemical restraint - Species specific restraint	AS.06.01.01.a-c	R4.3.1 W4.2.2;W4.2.4; SE2.2 &3.1	A1-2; A5-7	A4; B3; C4; E2	Health, Safety, and Environment Technical and Production Skills	Written assignments; Practical Exercises; Class participation; Completion of practicum, Completion of Institutional Animal Care and Use Committee (IACUC) Module 1; Quizzes, Final Examination
Students will learn about and be able to demonstrate knowledge on: Basic Feeding Principles - Feeding Schedules - Feeding Methods - Seasonal nutrition concerns	AS.04.01.01.a AS.04.01.02.a	R4.3.1 W4.2.2;4.2.4; M10.4.1 & M10.4.2; SA3.1; SC3.2; SF1.1-3.1	A1-7	A4; B3; E1	Health, Safety, and Environment Technical and Production Skills	Practical Exercises; Class participation; Completion of practicum, Completion of Institutional Animal Care and Use Committee (IACUC) Module 1; Quizzes, Final Examination
Students will learn about and be able to demonstrate knowledge on:	AS,06.01.01.b;	R4.3.1; W4.2.2;4.2.2;	A1-7	A1; B1; E2	Health, Safety, and	Written assignments; Practical Exercises;

CTE Course Description & Cross-walk

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standards	Alaska Reading, Writing, Math, Science 4th Ed. PSGLE's	Alaska Employability Standards	Alaska Cultural Standards	All Aspects of Industry	How Will Students Be Assessed? (Formative Level)
Animal Handling ○ Restraint and handling ○ Physical techniques ○ Mechanical techniques	AS.07.01.02.a	SC3.2			Environment Management Planning	Class participation; Quizzes; Completion of practicum, Completion of Institutional Animal Care and Use Committee (IACUC) Module 1; Final Examination
Students will learn about and be able to demonstrate knowledge on: Housing ○ Space requirements ○ Ventilation and sanitation ○ Shelter ○ Climate control ○ Fencing and enclosures	AS.07.01.01.a	R4.3.1; W4.2.2;4.2.4; M10.4.1;10.4.2; SA3.1; SC3.2	A1-7	A1; B3; E2	Planning Management Health, Safety, and Environment	Practical Exercises; Class participation; Completion of practicum, Completion of Institutional Animal Care and Use Committee (IACUC) Module 1; Quizzes, Final Examination
Students will learn about and be able to demonstrate knowledge on: Management (Concepts and Species Specific) ○ Feeding schedules ○ Handling and stress ○ Waste management ○ Disease management ○ Ventilation and climate considerations	AS.03.01.02.a; AS.04.01.02.b; AS.07.01.02.b;	R4.3.1; W4.2.2;4.2.4; M10.4.1;10.4.2; SA3.1; SC3.2; SE3.1	A1-7	A1; B1; D2; E1	Management Planning Finance	Class participation; practical exercises; Completion of practicum, Completion of Institutional Animal Care and Use Committee (IACUC) Module 1; Quizzes, Final Examination
Students will learn about and be able to demonstrate knowledge on: Marketing and Slaughter ○ Marketing and slaughter of large animals ○ Slaughtering techniques ○ Current market for domestic animal products	AS.06.02.01.b	R4.3.1; W4.2.2;4.2.4; M10.4.1;10.4.2; SA3.1; SE3.2 SF1.1-3.1	A6, B2-3	A1; B3; C2; E1	Finance Health, Safety, and Environment	Practical Exercises; Class participation; Completion of practicum, Completion of Institutional Animal Care and Use Committee (IACUC) Module 1; Quizzes, Final Examination

(Hint: Use the "tab" key at the end of this table to add more rows.)

List of Major Instructional Resources: (websites, textbooks, essential equipment, reference materials, supplies)

Alaska Reading, Writing, and Math Standards www.eed.state.ak.us
Alaska Content Standards www.eed.state.ak.us
Alaska Employability Standards www.eed.state.ak.us

CTE Course Description & Cross-walk

Cultural Standards for Students – Alaska Native Knowledge Network www.ankn.uaf.edu/stustan.html
National Agriculture, Food, and Natural Resources Career Cluster Content Standards – Animal Systems Career, www.teamaged.org/councilindex.cfm, 2009.
Dairy Housing and Equipment Handbook, MWPS-00007, UAF Cooperative Extension Service Publications
Beef Housing and Equipment Handbook, MWPS-00006, UAF Cooperative Extension Service Publications
Horse Housing and Equipment Handbook, MWPS-00015, UAF Cooperative Extension Service Publications
Natural Ventilating Systems for Livestock Housing, MWPS-00033, UAF Cooperative Extension Service Publications
Estimated Costs for Livestock Fencing, FM 1855, ISU Cooperative Extension Service Publications
Freestall Housing for Livestock, PM 1610, ISU Cooperative Extension Service Publications
Improving Winter Environments in Open-Ceiling Buildings for Pigs, PM 1760, ISU Cooperative Extension Service Publications
Poest, Grodon, *Raising Reindeer for Pleasure and Profit*, Morris Publication: Kearney, NE, 1996.
Haynes, B.N., *Keeping Livestock Healthily: A Veterinary Guide to Horses, Cattle, Pigs, Goats, & Sheep*, 4th Ed., Storey Publishing, LCC: North Adams, MA, 2001.
Groves, P., Musk ox Husbandry: A Guide for the Care, Feeding and Breeding of Captive Muskoxen, Biological Papers of the University of Alaska. Special Report No. 5, 1992.
Grandin, Temple, *Animals in Translation*, Simon & Schuster Publication: New York, NY, 2005.
Elluminate Live Software
Handset with earpieces and microphone
Restraint tools: snout snare, nose pliers, lariat, twitch, squeeze cage.
Breed charts: canine, feline, equine, bovine, ovine, caprine

Student Evaluation:

Quizzes	30%
Participation	5%
Written Assignments	25%
Final Examination	40%

CTE Course Description and Standards Crosswalk

Course Information	
Course Name	Animal Husbandry for Veterinary Science
Course Number	642
Number of High School Credits	0.5
Sequence or CTEPS (You must first have the Sequence or CTEPS entered into the EED-CTE system.)	Agriculture, Food & Natural Resources
Date of district Course Revision	May 2013
Career & Technical Student Organization (CTSO)	
CTSO embedded in this sequence	None
Occupational Standards	
Source of Occupational Standards	National Council for Agricultural Education
Names/Numbers of Occupational Standards	National Agriculture, Food and Natural Resources (AFNR) Career Cluster Content Standards
Registration Information	
Course Description (brief paragraph – as shown in your student handbook or course list)	This course provides an understanding of animal husbandry for canine, feline, goat, sheep, pig, horse, reindeer, musk oxen, bison, and cattle. Students will explore animal behavior, proper restraint, handling, and management, with a focus on canines. Insight will be provided on how to adapt techniques to different situations, such as the Alaskan climate and geography. Students will have the opportunity to become Pet First Aid certified.
Instructional Topic Headings (please separate each heading by a semi-colon)	Breed & Species Identification; Animal Behavior; Restraint; Basic Feeding Principles; Animal Handling; Housing; Management; Careers & Employability Skills
Summative Assessments and Standards	
Technical Skills Assessment (TSA)	Pet First Aid
Course addresses:	
New Alaska ELA and Math Standards	Yes
Alaska Cultural Standards	Yes
All Aspects of Industry (AAI)	Yes
Core Technical Standards	Yes
Employability Standards	Yes
Employability Standards	
Source of Employability Standards	Alaska Employability Standards
Tech Prep	
Current Tech Prep Articulation Agreement? (Y/N)	No
Date of Current Agreement	N/A
Postsecondary Institution Name	N/A

DISTRICT NAME: Yukon-Koyukuk School District

Postsecondary Course Name	N/A
Postsecondary Course Number	N/A
# of Postsecondary Credits	N/A

Additional CTE Course Information

Author	
Course developed by	Previous UAF-IAC Course
Course adapted from	Basic Animal Husbandry for Veterinary Science UAF-iAC course
Date of previous course revision	Spring 2008
Course Delivery Model	
Is the course brokered through another institution or agency? (Y/N)	No

Standards Alignment

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
Breed & Species Identification - Large Animals - Geographic Differences - Small Animals - Exotic Animals	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a; AS.02.01	AG-ANI6	RI.9-10.1-2,4,10; RI.11-12.1-2,4,10; W.9-10.2,4-10; W.11-12.2,4-10; SL.9-10.1-2,4-5; SL.11-12.1-2,4-5;L.9-	S-ID.5.; S-IC.6.	A4 B2; C4; D1; E1	A1-2	Tech Skills; Health/ Safety	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
			10.1-6; L.11-12.1-6; RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-12.1-10					
Animal Behavior - Groups - Perceptions - Feelings	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a.	AG-ANI2	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-12.1-10	A-REI.1.; F-LE.5.; S-ID.5.; S-ID.6.	A4; B1; E4	A1-7	Tech Skills; Community	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams
Restraint - Animal Safety - Human Safety - General Methods - Chemical Restraint - Species Specific Restraint	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a;	AG3; AG-ANI2; AG-ANI7; AG-	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10;	N-Q.1.-3.; A-CED.1.-2.,4.; A-REI.1.; G-	A4; B3; C4; E2	A1-2; A5-7	Health/ Safety; Tech Skills	Class Participation; Observation; Quizzes; Presentations ; Homework;

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
	CS.06.02-03; CS.08.01.	NR1-4	WHST.11-12.1-10	CO.1.; G-MG.3.				Exams
Basic Feeding Principles - Feeding Schedules - Feeding Methods - Seasonal Nutrition Concerns	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a; CS.03.02.01-02.a.; CS.06.02-03; AS.02.03.02.a; AS.04.01.01-02.a.	AG-ANI2; AG-ANI3; AG-ANI7; AG-NR1-4	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-12.1-10	N-Q.1.-3.; A-CED.1.-2.,4.; A-REI.1.; G-MG.3.	A4; B3; E1	A1-7	Health/ Safety; Tech Skills	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams
Animal Handling - Restraint and Handling - Physical Techniques - Mechanical Techniques	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a.; CS.06.02-	AG-ANI2; AG-ANI7 ; AG-NR1-4	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-	N-Q.1.-3.; A-CED.1.-2.,4.; A-REI.1.; G-CO.1.;	A1; B1; E2	A1-7	Health/ Safety; Management; Planning	Class Participation; Observation; Quizzes; Presentations ; Homework;

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
	03.; CS.08.01; AS.06.01. 01.a.		12.1-10	G-MG.3.				Exams
Housing • Space Requirements • Ventilation and Sanitation • Shelter • Fencing and Enclosures	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a.; CS.08.01; AS.07.01.01.a.	AG-ANI2; AG-NR1-4	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-12.1-10	N-Q.1.-3.; A-CED.1.-2.,4.; A-REI.1.; G-CO.1.; G-CO.12.; G-MG.2.-3.	A1; B3; E2	A1-7	Planning; Management; Health/Safety	Class Participation; Observation; Quizzes; Presentations; Homework; Exams
Management • Feeding Schedules • Handling and Stress • Waste Management • Disease Management • Ventilation and Climate Considerations	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a.; CS.06.02-03;	AG-ANI2; AG-ANI3; AG-ANI5; AG-ANI7;	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-	N-Q.1.-3.; A-CED.1.-2.,4.; A-REI.1.; G-CO.1.; G-	A1; B1; D2; E1	A1-7	Management; Planning; Health/Safety; Finance	Class Participation; Observation; Quizzes; Presentations; Homework; Exams

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
	AS.02.03.01.a; AS02.03.02.a.; AS.03.01.01-03.a.; AS.04.01.01-02.a.	AG-ENV4-5; AG-NR1-4	12.1-10	MG.1.-3.; S-ID.5.; S-IC.3., 6.			S	
Careers & Employability Skills - Education & Training - Job Duties - Employability Skills	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a; CS.01.05.01.a.; CS.02.02.03.a.; CS.02.03.01 & 03.a.	AG5; AG-ANI1	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-12.1-10	N-Q.1.; A-REI.1.; G-CO.1.G-MG.1.-3.	A1; B1; B2	A1; A3; A7; B1-5	Work Habits; Tech Skills; Labor	Class Participation; Observation; Presentations ; Homework; Exams

Instructional Resources

:youth program:(4) Web 4 •CHAPTER 4- IN School HS on 05.04.16:(4) Web 4. Example Curriculum (2. High School teachers):Example HS curriculum for VET SCIENCE Class- Peggy Bruno:Animal Husbandry for Veterinary Sciences I.docx Form #05-13-028

DISTRICT NAME: Yukon-Koyukuk School District

List the major instructional resources used for this course: (websites, textbooks, essential equipment, reference materials, supplies)

Videos

"The Toughest Race on Earth" by Discovery Channel

"See You in Nome"

"Veterinary Technology"

"Canine Cephalic and Saphenous Veins

"Canine and Feline Catheter Placement"

"Mushing in Fairbanks, Alaska"

"Lab Safety" by 4ACT

"Dissecting Cat Procedures" by Carolina Lab

"WebMed"

"Animal Health Care video: Receptionist Training"

"Animal Restraint"

"The Reindeer Queen"

Websites

yukonquest.com

<http://reindeer.salrm.uaf.edu>

Animal Care Technologies at site.4act.com

specollege.edu

Reading

"Reindeer in Alaska"

"Reindeer Research: Working to Preserve Alaska's Reindeer Industry"

"Concerns for Reindeer, Caribou Threaten Seward Peninsula Reindeer Herds

"Introduction to Animal Science" Leland S. Shapiro, Prentice-Hall, Copyright 2001; ISBN: 0-13-920992-1

Other Materials

Owl Pellets

CTE Course Description and Standards Crosswalk

Course Information	
Course Name	Animal Husbandry for Veterinary Science II
Course Number	644
Number of High School Credits	0.5
Sequence or CTEPS (You must first have the Sequence or CTEPS entered into the EED-CTE system.)	Agriculture, Food & Natural Resources
Date of district Course Revision	November 2013
Career & Technical Student Organization (CTSO)	
CTSO embedded in this sequence	None
Occupational Standards	
Source of Occupational Standards	National Council for Agricultural Education
Names/Numbers of Occupational Standards	National Agriculture, Food and Natural Resources (AFNR) Career Cluster Content Standards
Registration Information	
Course Description (brief paragraph – as shown in your student handbook or course list)	This course provides students a better understanding of animal anatomy, terminology, handling, feeding, and management. A focus is placed on canines. This course provides students a better understanding of animal anatomy, terminology, handling, feeding, management, and breeding. A focus will be placed on canines. Students will spend 10 hours during the semester doing a practicum with a local dog musher. Additionally, students will build their employability skills and learn more about vet science related careers.
Instructional Topic Headings (please separate each heading by a semi-colon)	Terminology; Anatomy; Handling; Feeding; Management; Breeding; Careers & Employability Skills; Practicum
Summative Assessments and Standards	
Technical Skills Assessment (TSA)	None
Course addresses:	
New Alaska ELA and Math Standards	Yes
Alaska Cultural Standards	Yes
All Aspects of Industry (AAI)	Yes
Core Technical Standards	Yes
Employability Standards	Yes
Employability Standards	
Source of Employability Standards	Alaska Employability Standards
Tech Prep	
Current Tech Prep Articulation Agreement? (Y/N)	No
Date of Current Agreement	
Postsecondary Institution Name	

DISTRICT NAME: Yukon-Koyukuk School District

Postsecondary Course Name	
Postsecondary Course Number	
# of Postsecondary Credits	

DISTRICT NAME: Yukon-Koyukuk School District

Additional CTE Course Information

Author	
Course developed by	Previous UAF-IAC Course
Course adapted from	Basic Animal Husbandry for Veterinary Science UAF-iAC course
Date of previous course revision	Spring 2008
Course Delivery Model	
Is the course brokered through another institution or agency? (Y/N)	No

Standards Alignment

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
Terminology - Anatomy - Handling - Management - Breeding - Genetics - Feeding - Nomenclature	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a; AS.02.01.01.a.	AG-ANI6	RI.9-10.1,4; RI.11-12.1,4; SL.9-10.1; SL.11-12.1; L.9-10.4-6; L.9-10.4-6; RST.9-10.1-10; RST.11-	N-Q.1.-2.; G-CO.1.	B1	A1-2	Technical Skills	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
			12.1-10; WHST.9-10.1-10; WHST.11-12.1-10					
Anatomy • Directional Terms • Systems ○ Muscles ○ Skeletal ○ Nervous ○ Circulatory ○ Respiratory ○ Digestive ○ Nervous ○ Excretory	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a.; AS.02.02.	AG-ANI2,6	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-12.1-10	N-Q.1.-3.; G-CO.1.1-5.	A4; B1; E4	A1-2	Technical Skills	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams
Handling • Restraint • Handling • Physical Techniques	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a.; CS.06.02-03.; CS.08.01;	AG3-4; AG-ANI2; AG-ANI6-7; AG-NR1-4	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-12.1-10	N-Q.1.-3.; A-CED.1.-2,4.; A-REI.1.; G-CO.1.; G-	A4; B3; C4; E2	A1-2, A6	Health/ Safety; Technical Skills; Work Habits	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
	AS.06.01.01.a.			MG.3.; S-IC.3.,6.				
Feeding - Anatomy of Digestive System - Essential Nutrients - Feed Nutrients	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a; CS.03.02.01-02.a.; CS.06.02-03; AS.02.03.02.a; AS.04.01.01-02.a.	AG1-4; AG-ANI2-3,5,7; AG-NR1-4	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-12.1-10	N-Q.1.-3.; A-CED.1.-2.,4.; A-REI.1.; G-MG.3.; S-IC.3.,6.	A4; B3; E1	A1-2, A6	Health/ Safety; Technical Skills; Work Habits	Class Participation; Observation; Quizzes; Presentations; Homework; Exams
Management - Feeding Schedules - Handling - Disease Prevention - Breeding - Sustainability	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a;	AG1-4; AG-ANI2-3,5,7; AG-NR1-4	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10;	N-Q.1.-3.; A-CED.1.-2.,4.; A-REI.1.;	A1; B1; E2	A1-2, A6	Health/ Safety; Management; Planning;	Class Participation; Observation; Quizzes; Presentations; Homework;

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
	CS.06.02-03; AS.02.03.01.a; AS02.03.02.a; AS.03.01.01-03.a; AS.04.01.01-02.a.		WHST.11-12.1-10	G-CO.1.; G-MG.1.-3.; S-ID.5.; S-IC.3.,6.			Work Habits	Exams
Breeding - Cell & Genetic Material - Hereditary Defects - Mating - Reproductive Anatomy - Estrous Cycle - Fertilization - Gestation	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a.; CS.08.01; AS.02.02.03-04.a.; AS.05.; AS.07.01.01.a.	AG1-4; AG-ANI1-7; AG-NR1-4	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-12.1-10	N-Q.1.-3.; A-REI.1.; F-IF.6.; G-MG.1.-3.; S-ID.1.; S-ID.5.; S-IC.3.; S-IC.6.	A1; B3; E2	A1-2, A6	Planning; Management; Health/Safety; Work Habits; Technical Skills	Class Participation; Observation; Quizzes; Presentations; Homework; Exams

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
Practicum - Handling - Feeding - Management	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a.; CS.08.01; AS.04.01; AS.06.; AS.07.01.01.; AS.08.	AG1-4; AG-ANI1-7; AG-NR1-4	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-12.1-10	N-Q.1.-3.; A-CED.1.-2.,4.; A-REI.1.; G-CO.1.; G-MG.1.-3.; S-ID.5.; S-IC.3.,6.	A1; A6; B1-2; C1; D1; D3; D4-6; E3-8	A1-2; A5-7	Planning; Management; Health/Safety; Work Habits; Technical Skills	Class Participation; Observation; Quizzes; Presentations; Homework; Exams
Careers & Employability Skills - Education & Training - Job Duties - Employability Skills	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a.; CS.01.05.01.a.; CS.02.02.03.a.;	AG1,5; AG-ANI1; AG-NR2	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-12.1-10	N-Q.1.; A-REI.1.; G-CO.1.G -MG.1.-3.; S-ic.6.	A1; B1; B2	A1-2; B1-5	Work Habits; Technical Skills; Labor	Class Participation; Observation; Presentations; Homework; Exams

DISTRICT NAME: Yukon-Koyukuk School District

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
	CS.02.03.01 & 03.a.							

Instructional Resources

List the major instructional resources used for this course: (websites, textbooks, essential equipment, reference materials, supplies)

Videos

"The Toughest Race on Earth" by Discovery Channel
 "See You in Nome"
 "Veterinary Technology"
 "Canine Cephalic and Saphenous Veins"
 "Canine and Feline Catheter Placement"
 "Mushing in Fairbanks, Alaska"
 "Lab Safety" by 4ACT
 "Dissecting Cat Procedures" by Carolina Lab
 "WebMed"
 "Animal Health Care video: Receptionist Training"
 "Animal Restraint"
 "The Reindeer Queen"

Websites

yukonquest.com
<http://reindeer.salrm.uaf.edu>
 Animal Care Technologies at site.4act.com
<http://vanat.cvm.umn.edu/specollge.edu>

:youth program:(4) Web 4 •CHAPTER 4- IN School HS on 05.04.16:(4) Web 4. Example Curriculum (2. High School teachers):Example HS curriculum for VET SCIENCE Class- Peggy Bruno:Animal Husbandry for Veterinary Sciences II.docx

Form #05-13-028

Alaska Department of Education and Early Development

DISTRICT NAME: Yukon-Koyukuk School District

Reading

"Reindeer in Alaska"

"Reindeer Research: Working to Preserve Alaska's Reindeer Industry"

"Concerns for Reindeer, Caribou Threaten Seward Peninsula Reindeer Herds

"Introduction to Animal Science" Leland S. Shapiro, Prentice-Hall, Copyright 2001; ISBN: 0-13-920992-1

"Saunders Veterinary Anatomy Coloring Book" Saunders, Copyright 2011, ISBN: 978-1-4377-1439-5

Other Materials

Owl Pellets

YKSD 's Veterinary Science III, & IV curriculum by Peggy Bruno

Animal Husbandry for Veterinary Sciences III

Animal Husbandry for Veterinary Sciences IV

CTE Course Description and Standards Crosswalk

Course Information	
Course Name	Animal Husbandry for Veterinary Science
Course Number	642
Number of High School Credits	0.5
Sequence or CTEPS (You must first have the Sequence or CTEPS entered into the EED-CTE system.)	Agriculture, Food & Natural Resources
Date of district Course Revision	May 2013
Career & Technical Student Organization (CTSO)	
CTSO embedded in this sequence	None
Occupational Standards	
Source of Occupational Standards	National Council for Agricultural Education
Names/Numbers of Occupational Standards	National Agriculture, Food and Natural Resources (AFNR) Career Cluster Content Standards
Registration Information	
Course Description (brief paragraph – as shown in your student handbook or course list)	This course provides an understanding of animal husbandry for canine, feline, goat, sheep, pig, horse, reindeer, musk oxen, bison, and cattle. Students will explore animal behavior, proper restraint, handling, and management, with a focus on canines. Insight will be provided on how to adapt techniques to different situations, such as the Alaskan climate and geography. Students will have the opportunity to become Pet First Aid certified.
Instructional Topic Headings (please separate each heading by a semi-colon)	Breed & Species Identification; Animal Behavior; Restraint; Basic Feeding Principles; Animal Handling; Housing; Management; Careers & Employability Skills
Summative Assessments and Standards	
Technical Skills Assessment (TSA)	Pet First Aid
Course addresses:	
New Alaska ELA and Math Standards	Yes
Alaska Cultural Standards	Yes
All Aspects of Industry (AAI)	Yes
Core Technical Standards	Yes
Employability Standards	Yes
Employability Standards	
Source of Employability Standards	Alaska Employability Standards
Tech Prep	
Current Tech Prep Articulation Agreement? (Y/N)	No
Date of Current Agreement	N/A
Postsecondary Institution Name	N/A

DISTRICT NAME: Yukon-Koyukuk School District

Postsecondary Course Name	N/A
Postsecondary Course Number	N/A
# of Postsecondary Credits	N/A

Additional CTE Course Information

Author	
Course developed by	Previous UAF-IAC Course
Course adapted from	Basic Animal Husbandry for Veterinary Science UAF-iAC course
Date of previous course revision	Spring 2008
Course Delivery Model	
Is the course brokered through another institution or agency? (Y/N)	No

Standards Alignment

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
Breed & Species Identification - Large Animals - Geographic Differences - Small Animals - Exotic Animals	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a; AS.02.01	AG-ANI6	RI.9-10.1-2,4,10; RI.11-12.1-2,4,10; W.9-10.2,4-10; W.11-12.2,4-10; SL.9-10.1-2,4-5; SL.11-12.1-2,4-5;L.9-	S-ID.5.; S-IC.6.	A4 B2; C4; D1; E1	A1-2	Tech Skills; Health/ Safety	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
			10.1-6; L.11-12.1-6; RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-12.1-10					
Animal Behavior - Groups - Perceptions - Feelings	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a.	AG-ANI2	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-12.1-10	A-REI.1.; F-LE.5.; S-ID.5.; S-ID.6.	A4; B1; E4	A1-7	Tech Skills; Community	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams
Restraint - Animal Safety - Human Safety - General Methods - Chemical Restraint - Species Specific Restraint	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a;	AG3; AG-ANI2; AG-ANI7; AG-	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10;	N-Q.1.-3.; A-CED.1.-2.,4.; A-REI.1.; G-	A4; B3; C4; E2	A1-2; A5-7	Health/ Safety; Tech Skills	Class Participation; Observation; Quizzes; Presentations ; Homework;

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
	CS.06.02-03; CS.08.01.	NR1-4	WHST.11-12.1-10	CO.1.; G-MG.3.				Exams
Basic Feeding Principles - Feeding Schedules - Feeding Methods - Seasonal Nutrition Concerns	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a; CS.03.02.01-02.a.; CS.06.02-03; AS.02.03.02.a; AS.04.01.01-02.a.	AG-ANI2; AG-ANI3; AG-ANI7; AG-NR1-4	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-12.1-10	N-Q.1.-3.; A-CED.1.-2.,4.; A-REI.1.; G-MG.3.	A4; B3; E1	A1-7	Health/ Safety; Tech Skills	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams
Animal Handling - Restraint and Handling - Physical Techniques - Mechanical Techniques	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a.; CS.06.02-	AG-ANI2; AG-ANI7 ; AG-NR1-4	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-	N-Q.1.-3.; A-CED.1.-2.,4.; A-REI.1.; G-CO.1.;	A1; B1; E2	A1-7	Health/ Safety; Management; Planning	Class Participation; Observation; Quizzes; Presentations ; Homework;

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
	03.; CS.08.01; AS.06.01. 01.a.		12.1-10	G-MG.3.				Exams
Housing • Space Requirements • Ventilation and Sanitation • Shelter • Fencing and Enclosures	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a.; CS.08.01; AS.07.01.01.a.	AG-ANI2; AG-NR1-4	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-12.1-10	N-Q.1.-3.; A-CED.1.-2.,4.; A-REI.1.; G-CO.1.; G-CO.12.; G-MG.2.-3.	A1; B3; E2	A1-7	Planning; Management; Health/Safety	Class Participation; Observation; Quizzes; Presentations; Homework; Exams
Management • Feeding Schedules • Handling and Stress • Waste Management • Disease Management • Ventilation and Climate Considerations	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a.; CS.06.02-03;	AG-ANI2; AG-ANI3; AG-ANI5; AG-ANI7;	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-	N-Q.1.-3.; A-CED.1.-2.,4.; A-REI.1.; G-CO.1.; G-	A1; B1; D2; E1	A1-7	Management; Planning; Health/Safety; Finance	Class Participation; Observation; Quizzes; Presentations; Homework; Exams

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
	AS.02.03.01.a; AS02.03.02.a.; AS.03.01.01-03.a.; AS.04.01.01-02.a.	AG-ENV4-5; AG-NR1-4	12.1-10	MG.1.-3.; S-ID.5.; S-IC.3., 6.			S	
Careers & Employability Skills - Education & Training - Job Duties - Employability Skills	CS.01.01.01.b.; CS.01.01.03-6.a.; CS.03.01.01.a; CS.01.05.01.a.; CS.02.02.03.a.; CS.02.03.01 & 03.a.	AG5; AG-ANI1	RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.1-10; WHST.11-12.1-10	N-Q.1.; A-REI.1.; G-CO.1.G-MG.1.-3.	A1; B1; B2	A1; A3; A7; B1-5	Work Habits; Tech Skills; Labor	Class Participation; Observation; Presentations ; Homework; Exams

Instructional Resources

youth program:(4) Web 4 •CHAPTER 4- IN School HS on 05.04.16:(4) Web 4. Example Curriculum (2. High School teachers):Example HS curriculum for VET SCIENCE Class- Peggy Bruno:Animal Husbandry for Veterinary Sciences I.docx
Form #05-13-028

DISTRICT NAME: Yukon-Koyukuk School District

List the major instructional resources used for this course: (websites, textbooks, essential equipment, reference materials, supplies)

Videos

"The Toughest Race on Earth" by Discovery Channel

"See You in Nome"

"Veterinary Technology"

"Canine Cephalic and Saphenous Veins

"Canine and Feline Catheter Placement"

"Mushing in Fairbanks, Alaska"

"Lab Safety" by 4ACT

"Dissecting Cat Procedures" by Carolina Lab

"WebMed"

"Animal Health Care video: Receptionist Training"

"Animal Restraint"

"The Reindeer Queen"

Websites

yukonquest.com

<http://reindeer.salrm.uaf.edu>

Animal Care Technologies at site.4act.com

specollege.edu

Reading

"Reindeer in Alaska"

"Reindeer Research: Working to Preserve Alaska's Reindeer Industry"

"Concerns for Reindeer, Caribou Threaten Seward Peninsula Reindeer Herds

"Introduction to Animal Science" Leland S. Shapiro, Prentice-Hall, Copyright 2001; ISBN: 0-13-920992-1

Other Materials

Owl Pellets

CTE Course Description and Standards Crosswalk

Course Information	
Course Name	Animal Husbandry for Veterinary Science IV
Course Number	647
Number of High School Credits	0.5
Sequence or CTEPS (You must first have the Sequence or CTEPS entered into the EED-CTE system.)	Agriculture, Food & Natural Resources
Date of district Course Revision	November 2013
Career & Technical Student Organization (CTSO)	
CTSO embedded in this sequence	None
Occupational Standards	
Source of Occupational Standards	National Council for Agricultural Education
Names/Numbers of Occupational Standards	National Agriculture, Food and Natural Resources (AFNR) Career Cluster Content Standards
Registration Information	
Course Description (brief paragraph – as shown in your student handbook or course list)	This course builds upon the skills and knowledge students have learned in Animal Husbandry for Vet Science I, II, and III. During this course students will learn specific skills and information needed to be a veterinary technician, while focusing on canines. Areas to be covered are physiology, veterinary office procedures, physical examination, medical nursing, animal emergency medicine, reproduction, physical therapy, and the stages of life. Throughout the course students will learn and use appropriate terminology. Students will spend 20 hours during the semester doing a practicum with a local dog musher and be responsible for at least one dog. Additionally, students will continue to build veterinary careers specific employability skills and continue to work on their career plan.
Instructional Topic Headings (please separate each heading by a semi-colon)	Physiology; Veterinary Office Procedures; Physical Examination; Medical Nursing; Animal Emergency Medicine; Reproduction; Physical Therapy; Stages of Life; Practicum; Employability Skills & Careers
Summative Assessments and Standards	
Technical Skills Assessment (TSA)	Pet First Aid
Course addresses:	
New Alaska ELA and Math Standards	Yes
Alaska Cultural Standards	Yes
All Aspects of Industry (AAI)	Yes
Core Technical Standards	Yes
Employability Standards	Yes
Employability Standards	
Source of Employability Standards	Alaska Employability Standards
Tech Prep	
Current Tech Prep Articulation Agreement?	No

DISTRICT NAME: Yukon-Koyukuk School District

(Y/N)	
Date of Current Agreement	
Postsecondary Institution Name	
Postsecondary Course Name	
Postsecondary Course Number	
# of Postsecondary Credits	

Additional CTE Course Information

Author	
Course developed by	Previous UAF-IAC Course
Course adapted from	Basic Animal Husbandry for Veterinary Science UAF-iAC course
Date of previous course revision	Spring 2008
Course Delivery Model	
Is the course brokered through another institution or agency? (Y/N)	No

Standards Alignment

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
Physiology • Respiratory • Circulatory • Digestive • Excretory • Regulative • Reproductive • Movement • Senses	CS.01.01.01.,03.a.; CS.03.01.a-b.; CS.09.02.01.a-b.; AS.02.02.01,02,04.a.,05,06.a.	CRP1-9, 11-12; HL1	RI.9-10.4,7,10; RI.11-12.4,7,10; SL.9-10.1-6.; SL.11-12.1-6.; L.9-10.1-4,6.; L.11-12.1-4,6.; RST.9-10.1-10; RST.11-	N-Q.1.-3.; A-SSE.1.; A-CED.4.; G-MG.1-1.; S-ID.3.	A6; B1-4; C4; E3-4	A1-2	Technol ogy; Technic al Skills	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams

DISTRICT NAME: Yukon-Koyukuk School District

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
			12.1-10; WHST.9-10.2,4-10; WHST.11-12.2,4-10					
Veterinary Office Procedures - Employee Positions - Facilities & Workflow - Clients & Client Services - Management of Veterinary Practice	CS.01.01.01.,03.a., CS.02.02.02.a.,03.a-b.; CS02.03.03.a.; CS.03.01.01.a-b.; CS.06.02.01.a.,04.01.a.; CS.08.01.01-02.,02.01.a.; CS.09.02.0a.a-b.; AS.06.01.01.a-c.-	CRP1-12; AG3,5; AG-ANI7; HL2,4; ST3,5	RI.9-10.1-4,10; RI.11-12.1-4,10; SL.9-10.1,4-6; SL.11-12.1,4-10; L.9-10.1-3,6; L.11-12.1-3,6; RST.9-10.1-4,7,9-10; RST.11-12.1-4,7,9-10; WHST.9-10.2,4,7-10; WHST.11-12.2,4,7-10	N-Q.1.-2.; A-SSE.1.; G-MG.1.	A1, A6; B1-4; C4; E7	A1-3, A5-7; B2	Plannin g; Manage ment; Health/ Safety; Technol ogy; Work Habits; Technic al Skills; Labor	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams

:youth program:(4) Web 4 •CHAPTER 4- IN School HS on 05.04.16:(4) Web 4. Example Curriculum (2. High School teachers):Example HS curriculum for VET SCIENCE Class- Peggy Bruno:Animal Husbandry for Veterinary Sciences IV.docx Form #05-13-028

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
	02.a.; AS.07.01. 01.a.,02. a-b.							
Physical Examination - Veterinary Technician Role - Documenting Information - Surroundings - Temperature, Pulse & Respiration - Systems Review	CS.01.01. 01.,03.a., CS.02.02. 02.a.,03. a-b.; CS02.03. 01.a.,03. a.; CS.03.01. 01.a-b.; CS.06.02. 01.a.,03. 01.a.,04. 01.a.; CS.07.01. 01.01.a- b.,04.02. a., CS.08.01. 01- 02.,02.01	CRP1- 12; AG- ANI2,7 HL- DIA2-4; ST2-3	RI.9- 10.4,7,10; RI.11- 12.4,7,10; SL.9-10.1- 6.; SL.11- 12.1-6.; L.9- 10.1-4,6.; L.11-12.1- 4,6.; RST.9- 10.1-10; RST.11- 12.1-10; WHST.9- 10.2,4-10; WHST.11- 12.2,4-10	N-Q.1- 3.; A- SSE.1.; A- CED.1,4 .; A- REI.1,3. ; G- MG.1- 3.; S- ID.1-4.	A1, A6; B1-4; C4; D1, D3, D5; E3- 4	A1-2, A5- 7	Health/ Safety; Technol ogy; Work Habits; Technic al Skills	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams

youth program:(4) Web 4 •CHAPTER 4- IN School HS on 05.04.16:(4) Web 4. Example Curriculum (2. High School teachers):Example HS curriculum for VET SCIENCE Class- Peggy Bruno:Animal Husbandry for Veterinary Sciences IV.docx
Form #05-13-028

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
	.a.; CS.09.02. 0a.a-b.; CS11.01.- 02.; AS.02.02. .01,02,04 .a.,05,06. a.; AS.06.01. 01.,02.01 .a.							
Medical Nursing - Veterinary Technician Practice Model - Patient Data - Identify & Prioritize Evaluations - Develop Nursing Care Plan - Re-evaluate Plan - Medical Records - Diseases	CS.01.01. 01.,03.a., CS.02.02. 02.a.,03. a-b.; CS02.03. 01.a.,03. a.; CS.03.01. 01.a-b.; CS.06.02. 01.a.,03. 01.a.,04.	CRP1- 12; AG- ANI2,7; HL- DIA2-4; ST2-3	RI.9- 10.4,7,10; RI.11- 12.4,7,10; SL.9-10.1- 6.; SL.11- 12.1-6.; L.9- 10.1-4,6.; L.11-12.1- 4,6.; RST.9- 10.1-10; RST.11- 12.1-10;	N-Q.1- 3.; A- SSE.1.; A- CED.1,4 .; A- REI.1,3. ; G- MG.1- 3.; S- ID.1-4.	A1, A6; B1-4; C4; D1, D3, D5; E3- 4	A1-2, A5- 7	Health/ Safety; Technol ogy; Work Habits; Technic al Skills	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams

DISTRICT NAME: Yukon-Koyukuk School District

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
	01.a.; CS.07.01. 01.01.a-b.,04.02. a., CS.08.01. 01- 02.,02.01 .a.; CS.09.02. 0a.a-b.; CS11.01.- 02.; AS.02.02. .01,02,04 .a.,05,06. a.; AS.03.01. 01a- b.,05.a-b.; AS.06.01. 01.,02.01 .a.		WHST.9-10.2,4-10; WHST.11-12.2,4-10					

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
Animal Emergency Medicine • Triage • Assessment • Initial Diagnosis • First Aid • Emergency Care • Shock & Systemic Inflammatory Response Syndrome • Respiratory System Support & Oxygen Therapy • Cardiopulmonary Arrest • Cardiopulmonary Cerebral Resuscitation • Patient Monitoring • Care of Patient	CS.01.01.01.,03.a., CS.02.02.02.a.,03.a-b.; CS02.03.01.a.,03.a.; CS.03.01.01.a-b.; CS.06.02.01.a.,03.01.a.,04.01.a.; CS.07.01.01.01.a-b.,04.02.a., CS.08.01.01-02.,02.01.a.; CS.09.02.0a.a-b.; CS11.01.-	CRP1-12; AG-ANI2,7; HL-DIA2-4; ST2-3	RI.9-10.4,7,10; RI.11-12.4,7,10; SL.9-10.1-6.; SL.11-12.1-6.; L.9-10.1-4,6.; L.11-12.1-4,6.; RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.2,4-10; WHST.11-12.2,4-10	N-Q.1-3.; A-SSE.1.; A-CED.1,4.; A-REI.1,3.; G-MG.1-3.; S-ID.1-4.	A1, A6; B1-4; C4; D1, D3, D5; E3-4	A1-2, A5-7	Health/Safety; Technology; Work Habits; Technical Skills	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
	02.; AS.02.02. .01,02,04 .a.,05,06. a.; AS.03.01. 04.a-b.; AS.06.01. 01.,02.01 .a.							
Reproduction - Female Reproduction - Male Reproduction - Estrous Cycle - Breeding - Gestation - Parturition - Breeding Soundness Examinations	CS.01.01. 01.,03.a., CS.02.02. 02.a.,03. a-b.; CS02.03. 01.a.,03. a.; CS.03.01. 01.a-b.; CS.06.02. 01.a.,03. 01.a.,04. 01.a.; CS.07.01.	CRP1- 12; AG- ANI2- 4,6-7; ST2-3	RI.9- 10.4,7,10; RI.11- 12.4,7,10; SL.9-10.1- 6.; SL.11- 12.1-6.; L.9- 10.1-4,6.; L.11-12.1- 4,6.; RST.9- 10.1-10; RST.11- 12.1-10; WHST.9- 10.2,4-10;	N-Q.1.- 3.; A- SSE.1.; A- CED.4.; G-MG.1- 1.; S- ID.3.	A1, A6; B1-4; C4; D1, D3, D5; E3- 4	A1-2, A5- 7	Health/ Safety; Technol ogy; Work Habits; Technic al Skills	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams

DISTRICT NAME: Yukon-Koyukuk School District

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
	01.01.a-b.,04.02.a., CS.08.01.01-02.,02.01.a.; CS.09.02.0a.a-b.; CS11.01.-02.; AS.02.02.04.a.,05,06.a.; AS.06.01.01.,02.01.a.; AS.05.01.01.,02.01.,02.02.; AS.05.03.01.a-b.; AS.06.01.01.		WHST.11-12.2,4-10					

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
Physical Therapy • Exercise Based Therapy • Therapeutic Exercises • Hydrotherapy • Land Treadmill • Manual Therapies • Passive Range of Motion • Electrical & Magnet Based Therapies • Light & Sound Based Therapies • Assistive Devices	CS.01.01.01.,03.a., CS.02.02.02.a.,03.a-b.; CS02.03.01.a.,03.a.; CS.03.01.01.a-b.; CS.06.02.01.a.,03.01.a.,04.01.a.; CS.07.01.01.01.a-b.,04.02.a., CS.08.01.01-02.,02.01.a.; CS.09.02.0a.a-b.; CS11.01.-	CRP1-12; AG-ANI2,7 HL-DIA2-4; ST2-3	RI.9-10.4,7,10; RI.11-12.4,7,10; SL.9-10.1-6.; SL.11-12.1-6.; L.9-10.1-4,6.; L.11-12.1-4,6.; RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.2,4-10; WHST.11-12.2,4-10	N-Q.1.-3.; A-SSE.1.; A-CED.4.; G-MG.1-1.; S-ID.3.	A1, A6; B1-4; C4; D1, D3, D5; E3-4	A1-2, A5-7	Health/Safety; Technology; Work Habits; Technical Skills	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
	02.; AS.06.01. 01.,02.01 .a.							
Life Stages • Neonatal • Puppy • Adolescence • Adulthood • Senior	CS.01.01. 01.,03.a., CS.02.02. 02.a.,03. a-b.; CS02.03. 01.a.,03. a.; CS.03.01. 01.a-b.; CS.06.02. 01.a.,03. 01.a.,04. 01.a.; CS.07.01. 01.01.a- b.,04.02. a., CS.08.01. 01- 02.,02.01	CRP1- 12; AG- ANI2,7; ST2-3	RI.9- 10.4,7,10; RI.11- 12.4,7,10; SL.9-10.1- 6.; SL.11- 12.1-6.; L.9- 10.1-4,6.; L.11-12.1- 4,6.; RST.9- 10.1-10; RST.11- 12.1-10; WHST.9- 10.2,4-10; WHST.11- 12.2,4-10	N-Q.1.- 3.; A- SSE.1.; A- CED.4.; G-MG.1- 1.; S- ID.3.	A1, A6; B1-4; C4; D1, D3, D5; E3- 4	A1-2, A5- 7	Health/ Safety; Technol ogy; Work Habits; Technic al Skills	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
	.a.; CS.09.02. 0a.a-b.; CS11.01.- 02.; AS.05.02. 01.a-b.; AS.08.02. 01.a-b.							
Practicum • Handling • Nutrition • Management • Breeding	CS.01.01.01.; CS.01.01.03-6.a.; CS.03.01.01.a.; CS.08.01; AS.02.03.02.01,02; AS.04.01.01.c; AS.06.01.01.; AS.08.02.01,c.	CRP1-12; AG4; AG-ANI2-3,5,7; ST2-3	RI.9-10.4,7,10; RI.11-12.4,7,10; SL.9-10.1-6.; SL.11-12.1-6.; L.9-10.1-4,6.; L.11-12.1-4,6.; RST.9-10.1-10; RST.11-12.1-10; WHST.9-10.2,4-10; WHST.11-	N-Q.1.-3.; A-SSE.1.; A-CED.4.; G-MG.1-1.; S-ID.3.	A1; A6; B1-4; C1; D1; D3-6; E3-8	A1-2; A5-7	Plannin g; Manage ment; Health/ Safety; Work Habits; Technic al Skills	Class Participation; Observation; Quizzes; Presentations ; Homework; Exams

youth program:(4) Web 4 •CHAPTER 4- IN School HS on 05.04.16:(4) Web 4. Example Curriculum (2. High School teachers):Example HS curriculum for VET SCIENCE Class- Peggy Bruno:Animal Husbandry for Veterinary Sciences IV.docx
Form #05-13-028

DISTRICT NAME: Yukon-Koyukuk School District

Student Performance Standards (Learner Outcomes or Knowledge & Skill Statements)	Specific Occupational Skills Standard	Common Technical Core Standards	New Alaska ENG/LA Standards	New Alaska Math Standards	Alaska Cultural Standards	Employability / Career Readiness Standards	All Aspects of Industry/ Systems	Assessment
			12.2,4-10					
Employability Skills & Careers - Education & Training - Job Duties - Employability Skills	CS.01.01.01.; CS.01.01.03-6.a.; CS.03.01.01.a.; CS.01.05.01.a.; CS.01.06.01.a-b,05.; CS.02.02.02.,03.a-b.; CS.02.03.01.,03.	CRP1-12; AG5; HL1-2; ST5	RSLSTS1-10; WSLHSSSTS1-10	N-Q.1.-3.; A-SSE.1.; A-CED.4.; G-MG.1-1.; S-ID.3.	A1-5; B1-5 C3; D1; D3; D5-6; E8	A1-5; B1-5	Technoloy; Work Habits; Technical Skills; Labor	Class Participation; Observation; Presentations ; Homework; Exams

Instructional Resources

List the major instructional resources used for this course: (websites, textbooks, essential equipment, reference materials, supplies)

Videos

"Veterinary Technology"

"Canine Cephalic and Saphenous Veins

:youth program:(4) Web 4 •CHAPTER 4- IN School HS on 05.04.16:(4) Web 4. Example Curriculum (2. High School teachers):Example HS curriculum for VET SCIENCE Class- Peggy Bruno:Animal Husbandry for Veterinary Sciences IV.docx Form #05-13-028

Alaska Department of Education and Early Development

DISTRICT NAME: Yukon-Koyukuk School District

"Canine and Feline Catheter Placement"

"Lab Safety" by 4ACT

"WebMed"

"Animal Health Care video: Receptionist Training"

"Animal Restraint"

Websites

yukonquest.com

Animal Care Technologies at site.4act.com

<http://vanat.cvm.umn.edu/>

specollge.edu

Alaska Career Information Systems (AKCIS) at https://acpe.alaska.gov/STUDENT-PARENT/College_Career/AKCIS

Colorado State Virtual Canine Anatomy at <http://www.cvmb.colostate.edu/vetneuro/>

Books

McCurnin's Clinical Textbook for Veterinary Technicians, Eighth Edition;, Joanna M. Bassert & John A. Thomas, ISBN: 978-1-4377-2680-0

McCurnin's Clinical Textbook for Veterinary Technicians Workbook, Eighth Edition;, Joanna M. Bassert & John A. Thomas, ISBN: 978-1-4557-2671-4

Ideas for High School Math Lesson Plans

Anonymous teacher

Math Science possible lessons

- Measure angles on various sled stations and height and length of the sled basket, and length of runners used for hauling weight, sturdiness, and lightweight flexible types. Determine center of gravity.
- Measure line lengths and compare for different trail types (twisty narrow trails with trees vs. open river miles)
- Study parasite life cycles and determine the most efficient method of control
- Sustainable numbers of sled dogs based on food availability (all subsistence, or mixed dry and subsistence fish) using ecology trophic dynamics.
- Figure speed by using a known length of trail and time taken to cover the trail.
- Determine body heat in doghouse with insulation, with just straw, no insulation and with both straw and insulation. Look at the calories saved in feeding the dog by insulating a house.
- Weigh dogs before and after running to determine hydration loss during training. Does this vary with temperature and with individual dogs?
- Compare conditioning programs for various forms of dog team use. Work, recreational, sprint, mid distance, and long distance racing. Use retired dog to show how to count respiration and heart rates.
- Use studies on dog DNA to determine breeds in village dogs to introduce genetics and DNA.

COURSE CONTENT GUIDE
UNIVERSITY OF ALASKA ANCHORAGE
COMMUNITY AND TECHNICAL COLLEGE

Department: HPER

Date: April 4, 2005

Course Number: PER A167

Course Title: Dog Mushing

Credits: 1 (one)

I. Course Description:

Introduces the practice of dog mushing, including the sport's history, dog breeds and characteristics, their training and feeding needs, kennel-management routines and dog handling skills.

Special note: Requires ability to function comfortably in extremely cold and/or inclement weather

II. Course Design

- A. Designed for individuals interested in obtaining an introduction to all aspects of current and historical dog mushing. Combines lecture material with hands-on application of material during day outings(s).
- B. One credit.
- C. Total time of student involvement: 52.5 hours
 - 1) Lecture: 7.5 hours
 - 2) Lab: 15 hours
 - 3) Outside 30 hours
- D. Status of course relative to a degree or certificate program: N/A
- E. A fee will be assessed.
- F. May be scheduled in any time frame, but not less than one week.
- G. This is a revised course.
- H. Military Programs, Mat-Su, Kenai Peninsula, Kodiak, Health Sciences and UAA List Serv.
- I. Course level justification: Course outcomes meet the criteria listed in the Curriculum Guide for a 100 level course.

III. Course Activities:

This course will include lecture, skill development and field application.

IV. Course Prerequisites:

There are no prerequisites for this course.

V. Course Evaluation:

Grades will be A-F based on written/oral examinations, written assignments, skill proficiency, class attendance and participation. Specific grading criteria will be discussed during the first class.

VI. Course Curriculum

- 1.0 Course Introduction
 - 1.1 Class and campus safety
 - 1.2 Appropriate apparel & footwear
 - 1.3 Travel and transportation

- 2.0 Introduction to Risk Assessment and Hazard Evaluation
 - 2.1 Environmental Risks
 - 2.1.1 Weather-cold challenges, white-outs, etc.
 - 2.1.2 Terrain-rivers, trees, and obstacles
 - 2.1.3 Animal encounters-other dogs, moose, etc.
 - 2.2 Hazards specific to the sport
 - 2.2.1 Injuries from lifting and falling
 - 2.2.2 Injuries from dogs and handling dogs
 - 2.2.3 Cold-weather injuries-frostbite and hypothermia
 - 2.3 Human factors
 - 2.3.1 Developing critical thinking, learning to think ahead
 - 2.3.2 Knowing your limits
 - 2.3.3 Learning to take care of yourself
 - 2.3.3.1 Personal protection/clothing
 - 2.3.3.2 Nutrition/hydration

- 3.0 History of Sled Dog Mushing
 - 3.1 Travel
 - 3.2 Racing
 - 3.3 Exploration
 - 3.4 War
 - 3.5 Service

- 4.0 Sled Dogs
 - 4.1 Characteristics of a good sled dog
 - 4.2 Common types/breeds

- 5.0 Mushing equipment
 - 5.1 Vehicles
 - 5.1.1 Sled types
 - 5.1.2 Wheeled rigs
 - 5.1.3 Rigging your vehicle
 - 5.2 Dog Accessories
 - 5.2.1 Harnesses
 - 5.2.2 Protective devices
 - 5.3 Securing devices
 - 5.3.1 Snowhook
 - 5.3.2 Snubline
 - 5.3.3 Quick-release

- 6.0 Personal Equipment
 - 6.1 Musher clothing
 - 6.2 Personal survival equipment-the 10 essentials
 - 6.3 First aid kits for humans and dogs
 - 6.4 Repair kits
 - 6.5 Equipment for animal encounters
- 7.0 Kennel Management
 - 7.1 Location/size
 - 7.2 Securing dogs
 - 7.3 Sanitation
 - 7.4 Shelter
 - 7.5 Health care
 - 7.6 Breeding
 - 7.7 Puppy care
 - 7.8 Regulations
 - 7.8.1 City, Federal, and Borough
- 8.0 Sled Dog Psychology
 - 8.1 Communicating with the animals
 - 8.2 Maintaining control
 - 8.3 Using positive and negative reinforcement
 - 8.4 Importance of consistency
 - 8.5 Training vs. exercise
 - 8.6 Physical vs. mental needs
- 9.0 Dog Nutrition and Feeding
 - 9.1 Food sources- what's available and where to obtain it
 - 9.2 Food and diet composition-carbohydrates, proteins, etc.
 - 9.3 Caloric needs
 - 9.3.1 Effect of weather
 - 9.3.2 Effect of exercise
- 10.0 Dog Sled Handling
 - 10.1 Starting
 - 10.2 Steering
 - 10.3 Stopping
 - 10.4 Securing
- 11.0 Frontcountry vs. Backcountry Mushing
 - 11.1 Frontcountry etiquette
 - 11.1.1 At the truck
 - 11.1.2 On the trail
 - 11.2 Backcountry travel
 - 11.2.1 Overnight needs and equipment
 - 11.2.2 Navigation

11.2.3 Dealing with emergencies

- 12.0 Racing Sled Dogs
 - 12.1 Sprint racing
 - 12.2 Middle-distance racing
 - 12.3 Long-distance racing
 - 12.4 Stage racing
 - 12.5 Skijoring/pulka
- 13.0 The Next Step
 - 13.1 Clubs and other local resources
 - 13.2 Venues
 - 13.3 Literature and biography

VII. Suggested Textbook:

Barve.L (2001) Chechako's Guide to the Art of Dog Mushing, Todd Communications.

VIII. Bibliography

Bern, K. (1997) Complete Winter Sports Safety Manual: Staying Safe & Warm Snowshoeing, Skiing, Snowboarding, Snowmobiling, & Camping. Bear Klaw.

Flowers, P. and Dixon, A. (2001). Alone Across the Arctic: One Woman's Epic Journey by Dog Team. Graphic Arts Center Publishing Company.

Salisbury, G. and Salisbury, L. (2005) The Cruellest Miles: The Heroic Story of Dogs and Men in a Race Against Epidemic. Norton, W.W. & Company, Inc.

Weiss, H. (1992). Secrets of Warmth: Warmth for Comfort or Survival. Brooklyn, NY: Vibe Publications.

Wendt, R. (1995) Alaska Dog Mushing Guide: Facts & Legends. Goldstream Publications.

IX. Instructional Goals, Student Outcomes and Assessment Procedures

Instructional Goals:

Presents concepts, skills and safety elements associated with introductory dog mushing.

Student Outcomes	Assessment Procedures
After successful completion of the course, the student will be able to:	
Identify potential hazards they may face during classroom activities, while on campus, and associated with travel to and from class or an outing.	Discussion Demonstration

Identify the most common hazards of dog mushing including environmental factors (such as weather, sport-specific hazards, moose encounters, encounters with other dogs on the trail) and subjective hazards (such as decision making).	Discussion Demonstration
Identify how the practice and sport of mushing dogs has changed through the years	Written exam
Identify the characteristics of a good sled dog, as well as the types and breeds most commonly used.	Discussion
Identify the types of sleds and vehicles, dog accessories, and securing devices that are common to the sport.	Written exam
Identify the equipment most commonly used by mushers, including personal clothing, survival gear, first-aid kits for humans and dogs, and repair kits.	Demonstration Written exam
Identify the importance of and steps to good kennel management including keeping a clean kennel, dog breeding considerations and practices and caring for puppies.	Written assignment
Identify the importance of musher/dog communication styles, disciplining techniques and how to get the most out of the dogs.	Discussion
Identify the correct food and diet composition for proper animal nutrition and maintenance and how weather and exercise affect a dog's caloric needs.	Written exam
Understand techniques used in handling dog sleds, including starting, stopping, steering, and securing the sled/dogs.	Demonstration Written exam
Identify the common rules and etiquette for frontcountry mushing.	Demonstration Written exam
Identify equipment, navigation skills and procedures for traveling in the frontcountry vs. backcountry.	Demonstration Written exam
Identify the different types of sled dog races, including sprint, middle-and long distance races, stage races, and skijoring.	Written exam
Identify the types of resources available to help them continue to develop their interest in, and knowledge of, dog mushing.	Discussion

DOG MUSHING OS11100

Curriculum Outline

Sled Dog Confirmation and Breeds

- Allow 15 minutes for students to share field day experiences and observations
- Dog “combinations”
 - Siberian
 - Bred for arctic climate, eats little food, strong for small size, slow but steady, independent
 - Pointers
 - Strong Barrels, fast over short distances, willing to please, short coats, eats lots of food
 - Sight hounds
 - Fast, deep chest cavity, short coat, not as strong as Siberians or pointers

*The type of dog you desire can change based on the specific mushing sport. Sprint racing dogs and distance dogs are very different, and many dogs that may not be race worthy may still make a great rec sled dog.

- Sled dog “shape”
 - Sloped croup, low tail set, tall withers, strong chest
 - Scapula and pelvis should be the same size
 - Humerus and femur should be the same size
 - Elbows set under the dog
 - No “cow hocks” or flat feet

*If possible have a couple dogs of different builds to show students the difference.

Kennel care

- Allow 15 minutes for students to share field day experiences and observations

Kennel

- Follow all municipal ordinances
- Keep it clean
 - Helps prevent parasites, bacteria build up, and run off pollution
 - Allows you to analyze stool (what comes out of the dog is a good indication of what is happening inside the dog)
- Noise control
 - Maintaining schedules and keeping dogs adequately exercised helps prevent them from getting bored or barking at random times
- Population control
 - Responsible breeding

- Adequate tie out or kennel to prevent accidental breeding
- Spay and neuter if it is known that a dog is not going to be bred
- Clean houses, bowls and buckets

Confining

- Kenneling
 - Cleaner if there is a deck underneath that can be hosed
 - Dogs tend to get less bored
 - Easier to play with other dogs
 - Freer movement
 - More expensive
 - If a dog gets sick it is more easily spread and it is harder to tell which dog may have loose stool
- Tie outs
 - Cheap
 - Dogs get used to lines and understanding how to untangle themselves
 - Dogs get dirtier
 - Should have enclosure around perimeter to protect the dogs and keep loose dogs from running off
- Housing
 - Insulated dog houses for short haired dogs
 - Should not be too large so that dogs can heat the space
 - Adequate straw
- Exercise outside of mushing
 - Sled dogs need mental breaks like any athlete but still need to burn energy
 - Free play groups, or bikejoring are fun for dogs and outside of normal “work”

*Provide pictures of several different dog yards. Ask students to point out what they see as pros and cons of each.

Health care and injuries

- Regular vet care
 - Many mushers do their own shots for their dogs, but rabies must be done by a licensed veterinarian
 - Each dog should receive a brief physical post-run looking for stiffness, lameness, dehydration, feet issues, etc. from the musher
- Common injuries/ailments
 - Feet
 - Broken toes, torn pads, torn nails
 - Apply ointment and rest dogs as needed for injuries to heal
 - Pulled muscles

- Often seen as lameness shown more in movement of limb rather than applied pressure
 - Stretch, apply heat and animal safe massage oils
 - Bite marks
 - Depending on severity, trim hair and clean minor bites regularly checking for infection, for severe bites seek veterinarian attention
 - Frostbite
 - Reheat and keep from refreezing, for severe frostbite seek veterinarian attention
 - Heat exhaustion
 - Catch early!, bag dog if on a run, cool with water/snow, keep out of sun
 - Diarrhea and vomiting
 - Depending on severity, for minor cases remove food for 24 hours to allow stomach to settle and monitor for worsening conditions, for severe cases seek veterinarian attention especially when blood is present or dog is incapable of keeping fluids
- Other injuries or trauma
 - Broken bones
 - Stabilize and seek veterinarian attention
 - Choking
 - Abdominal thrusts, CPR
- Stabilizing
 - Muzzle even the best dogs, pain may cause them to bite as a reflex
 - Support spine and injured limbs

*The hardest job for a musher is having to decide when it is too much for the dog and when it is time to let go.

*Have dogs available for students to practice physicals on

Dog Mushing OS11100

Risk Management Plan

Spring 2013

Description

An in-depth introduction to dog powered sports. Topics include dog sledding history, veterinary care, racing technique, and equipment. This course builds student experience through hands-on field classes and kennel visits, assisting with a dogsled race, and a dogsled/ski trip. There will also be a multi-day trip to the Kellogg farm over spring break. Students will follow a slow progression when learning to mush. Before any dogs are harnessed and lined out, students will find their balance while being towed on a sled by a snow machine. When it is finally time to hook up dogs, student will not use more than 4-6 dogs depending on the size and ability of the student. They will be running a 2 mile, relatively flat trail in Willow. Students will also have a classroom session each week where safety habits will be reinforced and studied.

Objectives

For Students to understand:

- History of dog sledding and mushing styles
- Dog and equipment care
- Racing kennel operations
- Mushing techniques
- The literature of dog mushing

Participants

- **Instructors:** Debra and James Castro:
 - Owners/Operators of Susitna Sled/Kayak
 - James has helped teach OS Swiftwater
 - Kennel is clean, organized with friendly dogs
- **TA:** Annie Grenier (First Aid and CPR certified)
- **Students: (tentative, as of 2.1)**

Risk Management Plan

This class will have many inherent risks including but not limited to, transportation, potential animal inflicted injuries, cold weather injuries, and athletic injuries caused by handling or mushing dogs. Every effort will be made to mitigate these risks, so the students can learn in a fun, safe, and controlled environment.

Transportation

All transportation of students will be in APU vans driven by an APU certified driver. All seatbelts will be worn at all times and all speed limits will be followed. There is a good chance that there will be snow on the roads on travel days. Drivers will not drive faster than is safe to drive on such conditions. If conditions are severe, field days will be canceled. If the weather gets worse while we are at the kennel in Willow, the class has been given permission to spend the night in home of the kennel owners. The instructors will ensure that no unnecessary risk will be taken when transporting students.

Animal Inflicted Injuries

Because of the unpredictable nature of animals, it is impossible to eliminate the chance of a bite wound or other animal inflicted injuries. However, every opportunity to mitigate that risk will be taken. Students will be given direct instructions on how to handle sled dogs in a way that the dogs are used to being handled. Students will also be under constant supervision while handling dogs. Any behavior from a student that will knowingly provoke a dog to bite will not be tolerated. No dog with a known bite history will be mushed or handled by a student of in the class. Susitna Sled and Kayak Kennels was chosen specifically for their well mannered dogs.

Athletic Injuries

Athletic injuries are hard to prevent, but students will be required to wear proper gear and will be reminded to remain focused by both instructors and peers. All students will wear helmets when mushing. One instructor will be on a snow machine in the immediate area of mushing students to provide quick response in case of injury or loose team.

Cold Weather Injuries

All students will be required to dress appropriately for outdoor activities. Any student not dressed properly will not be allowed to participate. Instructors are providing hand warmers and a cold weather safety class, so that all students will be aware of the signs, symptoms, and effects of cold weather and cold weather injuries. All overnights will be spent in a heated house, and most activities will be done within short distances of houses or other heated structures. Any student showing signs of cold weather related injuries will be able to enter any of these and stop the injuries before they become serious.

Communications: Instructors will have radios to communicate at events and field days. Instructors will also have cell phones at all times. Paul Twardock will be the in town contact for the course in the event of an emergency.

Emergency Response Plan

The course will follow APU's Emergency Response Plan in the event of an injury.

Syllabus:

OS 11100 Dog Mushing 2 credits Spring 2013

Instructors: Debra Castro, James Castro, Annie Grenier

Course Description:

An in-depth introduction to dog powered sports. Topics include dog sledding history, veterinary care, racing technique, and equipment. This course builds student experience through hands-on field classes and kennel visits, assisting with a dogsled race, and a multiday dogsled/ski trip. Lab fee required.

Learning objectives: For students to understand:

History of dog sledding and mushing styles

Dog and equipment care

Racing kennel operations

Mushing techniques

The literature of dog mushing

Course Schedule:

2/5: Intro Equipment, Cold Injuries;

2/7: Willow

2/14: Exon Mobile Open Race

2/21: Willow

2/27: Fur Rondy Races: optional volunteer/spectator day

2/28-2/29 Extra Credit Volunteer Opportunities for Fur Rondy

3/7-3/8: Iditarod Saturday or Sunday

Spring Break: 3/8 – 3/12: Kellogg Campus and Willow: Susitna Sled and Kayak

3/21: Willow

3/28: Final

Brief Course Outline:

Introduction to Equipment

Introduction to Mushing history and literature

Kennel Component:

- Dog anatomy, breeding, and care
- Equipment use and care
- Sled designs and history
- Kennel operations
- Dog handling
- Training techniques and scheduling

Trip Component:

- Trip planning
- Personal Care including basic cooking and meal prep
- Canine and human equipment and food preparation
- Sled and dog handling
- Route finding
- Field care of dogs and humans
- Risk Management

Primary Text: Book: Miki and Judy Collins: Dog Driver, 2nd Ed. Alpine Publications Crawford CO ISBN: 978-1-57779-094-5

Secondary Readings:

1000 Miles By Dogsled: Hudson Stuck

Dog Drivers Manual: Mikki and Julie Collins

Yukon Alone: John Balzar

Mushing Magazine (available at grocery stores)

Method: The class will meet one weekday evening in a classroom, and Saturdays in the field. The course includes a 4 day trip over spring break. The class will stay at the Kellogg Campus, but students will be sleeping in a partially heated yurt and cooking their own meals.

Grading: Grades will be based on attendance in class, degree of improvement in skills, and quality of written work.

Attendance and Participation: 80%

Journal: 10%

Homework: 10%

A=100-90%

B=89-80

C=79-70

D=69-60

F=50 or less

Requirements:

1. Be on time for class and not miss more than 2 classes.
2. Keep a journal on the class. Include each day's activities, class notes, what you learned, what went well and what you need to work on.
3. Be an active participant in the class, which includes positive 'expedition behavior.'
4. Complete any readings or other assigned homework.
5. Complete the 4 day spring break trip.

Risk Management: This is an active class requiring students to be in the outdoors during all types of weather. Risks include injuries from dogs, sledding cold weather, animals on trails, and driving to and from events. Injuries and illnesses may include physical/mental trauma, frostbite and immersion foot, hypothermia, loss or damage to clothing and equipment, and death. Students are responsible for their own well being and informing the instructor of injuries and illness.

Equipment: Students are responsible for acquiring their own gear. Some gear may be available to rent from Outdoor Programs, or borrowed from Outdoor Studies. Note: Outer garments like jackets, pants, and gloves will get trashed by the dogs jumping on you, pawing you, etc. They will also have the distinct "kennel" smell: not great. So DO NOT bring your brand new Patagonia puffy jacket. If you do bring a tough old rain or ski jacket to go over it! See equipment list below.

Food: Students are responsible for being well fed. Being in the outdoors in a cold environment requires a healthy diet with plenty of calories. On the spring break trip students will be responsible for bringing their own meals to be cooked at the Kellogg Campus. Students will organize themselves to coordinate meals.

Equipment List:

For Mushing: outer garments must have a tough outer layer like Cordura or thick nylon.

Name: _____ Expedition _____

Have Need

Footwear

- ☐ ☐ Socks- 3 pair
 - Wool or synthetic
- ☐ ☐ Winter (insulated) boots: Sorrels, mukluks, Bunny Boots, Plastic Mountaineering boots
- ☐ ☐ Gaiters
- ☐ ☐ Camp shoes
 - Running shoes and neoprene
 - socks or Crocs and neoprene
 - or Galoshes

Upper Body (3-4 layers)

- ☐ ☐ Lt. Weight top
- ☐ ☐ Mid. Weight top
- ☐ ☐ Fleece Jacket
- ☐ ☐ Non cotton sweater or shirt
- ☐ ☐ Nylon ski like jacket
- ☐ ☐ Sports Bra
- ☐ ☐ Synthetic insulated parka

Lower Body (1-2 layers)

- ☐ ☐ Lt. Weight bottom
- ☐ ☐ Mid. Weight bottom
- ☐ ☐ Nylon ski pants
- ☐ ☐ Synthetic insulated Pants
- ☐ ☐ Nylon Underwear

Head

- ☐ ☐ Synthetic ski hat
- ☐ ☐ Neck Gaiter, Buff, or Face Mask
- ☐ ☐ Ski Helmet

Hands

- ☐ ☐ Liner gloves
- ☐ ☐ Thick Ski Gloves and/or Mittens

Sleeping (for Spring Break Trip)

- ☐ ☐ 20°F synthetic sleeping bag
- ☐ ☐ Sleeping bag compression sack
- ☐ ☐ Full length closed cell foam or
inflatable sleeping pad

Toiletries (for Spring Break trip)

- ☐ ☐ Toilet paper
- ☐ ☐ Handy wipes
- ☐ ☐ Hand disinfectant
- ☐ ☐ Toothbrush and paste

Eating and Drinking (For Spring Break Trip)

- ☐ ☐ Bowl
- ☐ ☐ Mug
- ☐ ☐ Spoon

Miscellaneous

- ☐ ☐ Sunglasses or Ski goggles
- ☐ ☐ Headlamp
- ☐ ☐ Journal/pencil
- ☐ ☐ Bandanna
- ☐ ☐ Lip balm

- ☐ ☐ Sunscreen
- ☐ ☐ Personal first aid kit (band aids, anti-inflammatory, athletic tape)
- ☐ ☐ Chemical Foot and hand warmers (buy in bulk @ Costco)

Optional

- ☐ ☐ Camera
- ☐ ☐ Crazy creek chair
- ☐ ☐ Binoculars
- ☐ ☐ Paperback novel
- ☐ ☐ Personal listening device (for van rides)

Food

Lunch and snacks for each Saturday outing

Insulated water bottle

For Spring break

4 dinners, 5 lunches, and 4 breakfasts. Dinners and Breakfasts will be cooked in either small groups or as a

March 15 – 21, 2016: Activity report for three FAYSDP Students attending Veterinary Science Class at UAF and sled dog care/help during the ONAC

Chaperones: Kathy Turco (907-687-3655) and Joe Bifelt (907-371-6099)

•Tuesday, March 15th (DAY 1)-

~~~~Kathy T. goes food shopping in early morning and then to UAF Veterinary Science Dept to get sleeping bags

~~~~Kathy T. meets UAF Dorm manager to pick up keys and drop off sleeping bags and food for the week~~~~

- 2:30pm: Three FAYSDP/JHS 9th grade students leave Huslia to fly to Fairbanks
- 4:30pm: KT meets students at WRIGHTS and takes to UAF dorm and see Joe Bielt
- 5:30pm: Students walk from Dorm to see Wood Center and buildings on lower campus
- 6:00pm: Eat pizza with Grandma Madeline Williams in the UAF Dorm with kitchen
- 7:30pm: Students go to UAF Recreation Center with Joe to practice basketball
- By 9:00pm: Students back to Dorm to go over schedule and talk about possible ways share they can their experience with students back in Huslia (i.e. journal notes, photos, etc). Students stay in Dorm public area to enjoy playing pool & games

•Wednesday, March 16th (DAY 2)-

- 8:00am: Eat breakfast at UAF Dorm with kitchen and make sandwiches for lunch
- 9:00am: Students go to Veterinary Science Building on upper campus for a Arleigh Reynolds lecture on dog breeding, health, & racing, tour the Vet Science facility including study areas and labs, and meet faculty and vet students
- 12:00pm: Students go to Musher's Hall on Farmer's Loop to meet ONAC mushers, help with sponsors 3-dog race, and eat lunch
- 2:00 – 4:00pm: All head to YKSD office to attend annual board meeting. Elders Madeline Williams and Pollock Simon, and other community members attend to meeting to voice support of the FAYSDP
- 4:30pm: Arrive back to UAF dorm to snack and get dog truck to participate in parade at Carlson Center. Attend ONAC start order drawing at AlaskaLand.
- 6:00pm: Eat dinner then go to UAF Recreation Center with Joe to practice basketball
- By 9:00pm: Students back to UAF Dorm to talk about day & write in journals

•Thursday, March 17th DAY 3)-

- 8:00: Eat breakfast at UAF Dorm with kitchen and make sandwiches for lunch
- 10:00am: Students are welcomed by UAF Dorm Head Resident. All walk to lower campus to Duckering Bulding to visit Rural Student Services and talk to advisors. Students see more of lower campus. All go back to UAF Dorm kitchen for lunch.
- 11:00am: Students help Grandma Madeline get to Wedgewood, then eat lunch downtown
- 12:30pm: Students visit Dog Mushing/Community Museum and the Yukon Quest Office and Grandma Lorna at Week House.
- 4:00pm: All drive to Patty Imus's house to help Tom Bird care for dogs – just arrived from Canada so dog box needed cleaning. Helped Tom repack gear to run the ONAC.
- 6:00pm: Stop and meet old timer carpenter who build dog boxes
- 7:00pm: Students swim at Hamme Pool- Kathy there
- 8:30pm: Eat at Fast Food place on way back to Dorms
- By 9:00pm: Students back to UAF Dorm to talk about day and stay in Dorm public area to enjoy playing pool & games

•Friday, March 18th (DAY 4)-

- 8:00: Eat breakfast at UAF Dorm kitchen and make sandwiches for lunch
- 9:00am: Students write in Journals and talk about upcoming race day
- 11:00am: Drive to ONAC held at Musher's Hall to meet Tom preparing to race. Visit with mushers and shadow Arleigh Reynolds caring for race dogs. Eat Lunch. Help Tom and other mushers get dog teams to starting line. Race Day #1 is 1- 3pm.
- 4:00pm: Drive to Patty Imus's house to help Tom with dog care including feeding, rub-downs, work on feet, clean up, etc.
- 6:00pm: Eat dinner at Patty's house with Tom and family (Patty cooks). Also sew FAYSDP patches on jackets
- 7:00pm: Students back to UAF to attend North American basketball games with Joe
- By 9:00pm: Students then to UAF Dorm to talk about day and stay in Dorm public area to enjoy playing pool & games

Saturday, March 19th (DAY 5)-

- 8:00: Eat breakfast at UAF Dorm kitchen and make sandwiches for lunch.
- 9:00am: Students write in Journals and talk about upcoming race day
- 11:00am: Drive to ONAC held at Musher's Hall to meet Tom preparing to race. Visit with mushers and shadow Arleigh Reynolds caring for race dogs. Eat Lunch. Help Tom and other mushers get dog teams to starting line. Race Day # 2 is 1- 3pm.
- 4:00pm: Students go to Bentley Mall so can buy things with spending money from family
- 5:00pm: Drive to Patty Imus's house and students wait for Tom (he was visiting dog mushers). Eat dinner (Kathy cooks). While waiting, watch a Yukon Quest and also an Iditarod documentary.
- 9:00pm: Students help Tom drop and water dogs last time for night, rub-downs for last day of racing, work on feet, clean up, etc.
- By 10:00pm: Students back to UAF Dorm to talk about day & stay in Dorm public area to enjoy playing pool & games.

Sunday, March 20th (DAY 6)-

- 8:00: Eat breakfast at UAF Dorm kitchen and make sandwiches for lunch.
- 9:00am: Students write in Journals and talk about upcoming LAST race day
- 11:00am: Drive to ONAC held at Musher's Hall to meet Tom preparing to race. Visit with mushers and shadow Arleigh Reynolds caring for dogs. Eat Lunch. Help Tom and other mushers get dog teams to starting line. Race Day #3 is 1- 3pm. Congratulate all mushers and thank ONAC volunteers before leaving
- 4:00pm: Still at racetrack, help Tom with photo shoot for Mushing Magazine (*Super Dogs*).
- 6:00pm: Students attend ONAC banquet at Westmark downtown with Tom and family, etc. Students immediately attend a meeting with Grandpa Alfred Attla
- 8:30pm: Students go to UAF Recreation Center with Joe to practice basketball
- By 9:00pm: Back to UAF Dorm to talk about day, write in journals, and pack to leave Monday morning.

Monday, March 21st (DAY 7)-

- 8:00: Eat breakfast at UAF Dorm kitchen and clean up
- 8:30am: Check in one student at WRIGHTS- flight leaves at 9:30am
- 9:30am: Take one student downtown to stay with parents. Make stop at Coldspot Feed for care supplies needed for sled dogs in Huslia
- 11:30am: Check in the last remaining student at WRIGHTS- flight leaves at 12:30pm.
- ~~~~Kathy goes back to UAF Dorm to finish cleaning, return keys, and all sleeping bags~~~~



MUSH WITH P.R.I.D.E.

Sled Dog Care Guidelines

3rd Edition, 2009

Introduction to the 3rd Edition

It has been just over a decade since the 2nd Edition of the Mush with P.R.I.D.E. Sled Dog Care Guidelines were published. During that time, scientists have made great strides in their understanding of dog physiology, psychology and behavior. Researchers have studied working sled dogs, with the support of their mushers, some of whom are also Mush with P.R.I.D.E. members. Many of these research projects have validated sled dog care methods that have been practiced for more than a century. Others have challenged traditional ideas that have persisted among mushers for many generations. This research, in addition to the experience of dog mushers from around the world, has contributed to a better understanding of sled dogs' needs, and will lead to an increased level of care not only for our sled dogs, but also for companion dogs everywhere.

In this edition of the Guidelines we are following the lead established by Mush with P.R.I.D.E.'s founders and describing sled dog care practices that are humane, practical, relevant and that reflect the best available current information.

Introduction to the 2nd Edition

This book is based on a consensus of dozens of the world's most experienced dog mushers. Here they describe what they consider to be responsible, humane, and practical sled dog care and training methods. Since it was first published in 1993, this book has become one of the world's most respected and widely distributed basic resources on good sled dog care.

What is Mush with P.R.I.D.E?

P.R.I.D.E. stands for Providing Responsible Information on a Dog's Environment. The relationship between sled dogs and humans is one of the oldest bonds of its kind. Modern sled dog owners are proud of their dogs as canine athletes that are bred and trained to do what they love – run as part of a team. Mush with P.R.I.D.E. supports the responsible care and humane treatment of all dogs and is dedicated to enhancing the care and treatment of sled dogs in their traditional and modern uses.

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The Dog Yard and Housing

BASIC CONSIDERATIONS:

Space and Location

Beginner mushers are sometimes surprised by the amount of space needed for a kennel. A dog yard for 10 dogs will require at least 1,000 square feet of pens, or 1,450 square feet for tethers. These figures do not include space needed for walkways, out buildings or other facilities. (Note 1)

A dry, well-drained area makes life pleasant for both dogs and musher. These conditions are also best for the dogs' feet and for disease control. A location both visible and audible from a house window allows you to enjoy the company of your animals and alerts you to problems or emergencies.

Locating the kennel on a slight slope or on a high spot will greatly improve springtime drainage. A low-lying flat area may seem perfectly dry in the summer or winter but a few weeks of standing water during spring thaw will make life miserable for both you and your dogs.

In summer, shade helps keep the dogs cool, and a breezy location helps keep bugs away. During winter, a sunny area that is protected from wind helps conserve the dogs' energy. It is best to lay out the dog yard so the ground is exposed to full sunlight for at least part of the day. Direct sun (ultraviolet light) is one of the best natural means of controlling disease organisms. However, you should try to provide at least one shady spot for each dog to retreat from the sun's heat. Ideally, your kennel should be located on a southern aspect adjacent to hardwood (deciduous) trees. The trees will provide summer shade, and after leaf fall the winter sun will improve the microclimate of the kennel.

Planning your dog yard in a way that allows you to do your chores efficiently also allows for more time to care for and interact with your dogs. If you are able to run dogs directly from the yard, it's well-worth planning a safe takeoff area for runs. Some kennels are set up to allow the musher to leave from the middle of the dog yard to facilitate harnessing. Other considerations include access by vehicles for loading up dogs and for maintenance.

Surface

The ideal dog yard surface depends upon its location and the method of confinement. Soil is fine in areas with good drainage. However, keep in mind that soil can harbor disease organisms and therefore requires more diligent feces pick-up. Soil is relatively easy to work with and is easily manipulated to meet your needs. Excessive silts and clays in the surface will produce a rock-hard surface when dry but will slow drainage and become slick and sticky when wet. Adding sand to soil improves its ability to absorb water and also reduces dust.

For wetter locations, sand, wood chips, coarse wood shavings, wooden platforms, or fine, smooth gravel less than $\frac{3}{4}$ inch in diameter are good surfacing alternatives. Excessive amounts of decaying bedding material increases water retention and can increase the amount of fungi, mites and other organisms that may be harmful to your dogs. Beware of large gravel and stones in the dog yard. If your dogs are rock-eaters, remove rocks larger than 1 inch from the soil. Although many dogs swallow rocks without incident, there have been cases of dogs that have died from rock ingestion.

Typical sled dogs love to dig in dirt. Because digging is an instinctive “species typical” behavior many mushers accept the extra work of filling in holes rather than trying to thwart the action. Other mushers prefer to prevent digging; there are several methods of doing so. One popular method to prevent digging is to cover the ground with sturdy fencing or concrete reinforcing mesh before adding the surface material. Another method is to house dogs on a concrete or plywood surface. This not only precludes digging, it also (and most importantly) prevents your dogs from eating rocks. Plywood floors work well in dry climates and are softer to stand on than concrete. They are also easy to clean and repair, but eventually breakdown and need replacement.

Pros and Cons of Plywood Surfaces

Pros:

Prevents digging and rock eating.

Easily cleaned with high-pressure water hose and disinfectants.

Less likely to cause chronic joint injuries than paved surfaces.

Cons:

Can harbor infectious bacteria and fungi within its pores.

Can be difficult to keep dry.

Deteriorates over time, and must be periodically replaced.

Pros and Cons of Concrete Surfaces

Precludes digging and prevents rock eating.

Easily cleaned with high pressure water hosing and disinfectants.

Cons

Can be difficult to keep dry

Can harbor infectious bacteria and fungi within its pores.

Hard surface can cause chronic injuries to dog's joints

Is abrasive and can cause excessive wear to dogs' feet and coats.

Is caustic and can cause excessive drying.

The Mush with P.R.I.D.E. Guidelines Committee firmly stresses that dogs should NOT be continuously housed on concrete surfaces. Dogs housed on concrete should be allowed to exercise on other surfaces several hours each day.

Kennel Upkeep

Cleaning up dog waste at least once every day makes the dogs' environment pleasant and helps control diseases such as parvovirus and intestinal parasites. Waste management or "scooping poop" is one of the daily chores all mushers must undertake.

Locate permanent waste disposal or temporary waste storage sites away from water drainages and from any location that may cause ground water contamination, such as wellheads and areas uphill from natural springs. Methods of disposing of dog feces include composting, burial, or removal to a landfill.

Landscaping around your kennel can be both attractive and practical. Begin by removing brush that is an attractant to moose, which have little fear and often have animosity toward dogs. Remove foxtails and other grasses with barbed or brush-shaped heads that are prone to getting lodged in the soft flesh of dogs' ears, eyes, throat and respiratory system. Identify other noxious plants found in your area and remove them from your dog yard.

CONFINEMENT SYSTEMS:

General Considerations:

Even mushers who primarily house their dogs in their own homes need some sort of outdoor confinement system. Mushers with larger teams usually confine some or all of their sled dogs in an outdoor "dog yard". Whether confining members of a two-dog skijoring team or a 100+ dog racing kennel, the general considerations and methods of confining the dogs remain the same.

The confinement system you choose must provide a reliable and safe means of preventing the dog from escaping. It must allow enough room for the dog to move around freely and engage in "species typical" behaviors such as running or jumping. Materials and hardware used in your confinement system should be durable, reliable and maintained in good condition. Chains or cables used in tethering systems should contain at least one swivel to prevent tangles that can potentially choke your dog.

It is recommended that kennels include a sturdy fence around the perimeter to contain any dogs that may get loose from their primary confinement and to keep unwanted people, wildlife and stray domestic animals away from your dogs. All dog yards should also include fenced pens or runs to confine females in heat, dogs that display dog-directed aggressiveness, sick dogs, or puppies too small to collar and tether. Many mushers incorporate a fenced "play yard" into their kennels where compatible dogs can run and play together.

When planning a dog yard, consider including one or two "extra" spaces that can be used to house dogs while making repairs or modifications to the dog's normal housing area.

Post and Swivel Tethering Systems:

Tethering is a common, traditional and economical means of confining multiple sled dogs. The only controlled scientific study comparing sled dogs confined by tethers to those confined in pens found no evidence that tethering is either unsafe or inhumane (Haupt K). The most common tethering systems used by mushers allow dogs to interact more directly with their surroundings, musher and handlers, and with teammates.

The tethering method preferred by most mushers involves attaching a chain to a rotation device at the top of a post or pipe, thus allowing the chain to travel in a full circle around the post. One simple rotation device uses a piece of rebar with a 90-degree angle bend and an eye for the chain welded on the end. A hollow iron or steel pipe is driven into the ground to serve as the post. In use, the long arm of the rebar slips inside the pipe allowing the rebar to swing in a complete circle. With this system the post can be easily lengthened in deep snow by slipping a taller pipe of larger diameter over the shorter summer post. Another method to allow for rotation is to bolt the end ring of a chain to the top of a beveled solid wooden post.

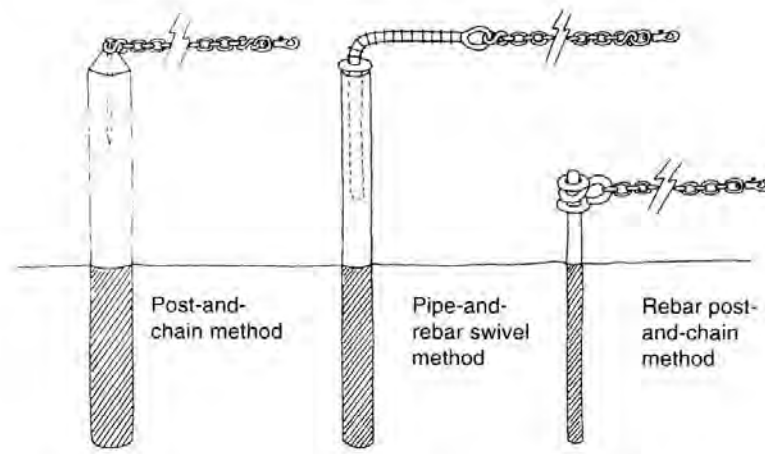
Using a top-mounted post and swivel chain system, each dog needs a strong chain of 5 to 7 ft (1.5 to 2 m) in length rotating on a post of about 3 to 4 ft (0.9 to 1.2 m) in height, with at least another 3 ft (1 m), preferably more, buried in the ground. A pole of this height will hold the chain above most snow accumulations. If snow is deeper, provide taller poles and longer chains. Never use cable to tether dogs to their posts. Cable is much too likely to tangle around legs (in an armpit or hock) and can cinch up like a snare. Cables also have a tendency to fray and break.

The optimal length of the chain is somewhat longer than the height of the pole or post. If the chain is too short the dog will not have enough space to lie down or move around comfortably. If the chain is too long it will drag on the ground too much, increasing the chances for a tangle and spreading and breaking up feces before they can be cleaned up. For soil-based kennels that use tethers, it is best to use elevated tethers to minimize the amount of time that the chain drags on the ground.

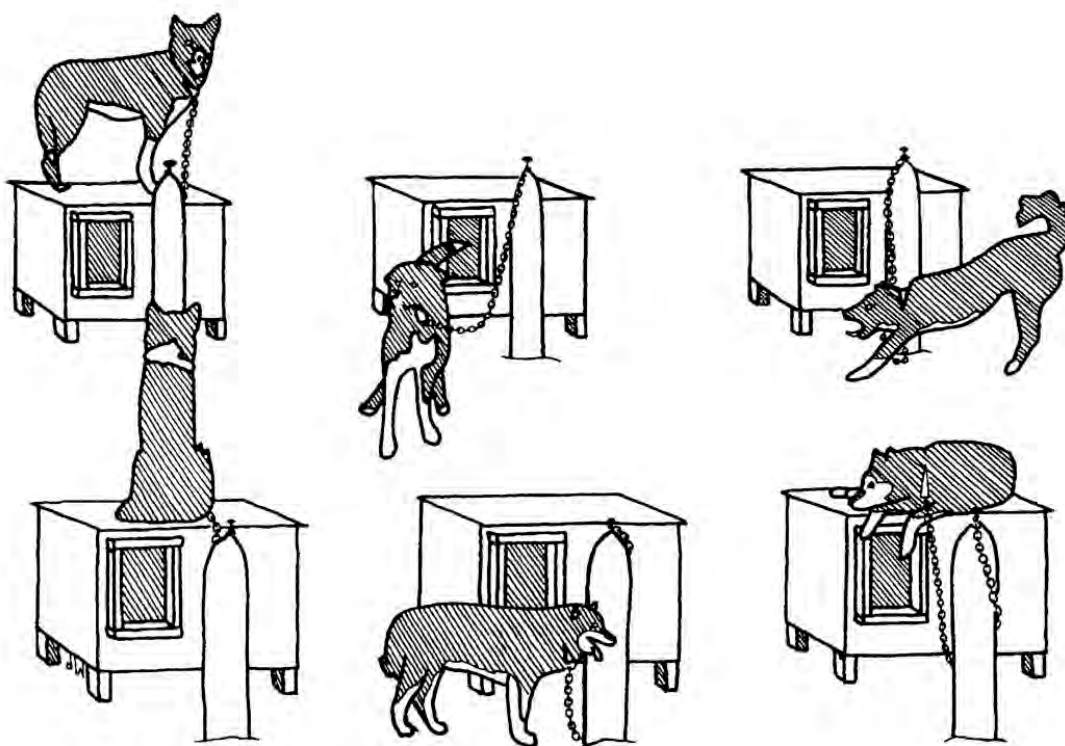
The simplest method of tethering sled dogs is the post and loop, or post and chain method. This involves looping a chain around a solidly buried post or pole. The chain should have a large loop, or preferably a large welded steel ring securely built into one end with an S-hook or quick link. The loop or ring should be at least twice the diameter of the post to minimize binding. The post may be either wood or steel, but it must be smooth to allow the chain to rotate freely. The post must also be tall enough so that the chain loop or ring cannot fly up and over the top, especially when the dog jumps up on top of its house. A 5 ft (1.5 m) post is generally adequate. Where posts cannot be reliably buried, a 100-lb (45.5 kg) concrete block with an eyebolt cast in the center and a swivel attached will adequately secure a 5 ft chain.

Although the post and chain method is easy to set up, it has a few major drawbacks. The chain drags entirely on the ground, stirring up a dust cloud, spreading feces around, and making cleanup much more difficult. Also, the chain often freezes to the ice and snow when the dog urinates on its post. The chain is also more prone to binding around the post than in other methods, so it must be checked several times each day.

Various dog yard tethering methods.



Whichever tethering method you use, space the posts so that adjacent chains can't overlap and so that dogs can't tangle or strangle each other. Strong hardware is also essential. Look for snaps that are durable, easy to open with a gloved hand yet difficult for dogs to activate. A bull snap is reliable because it requires the gate to be rotated outward. Snaps and chains do wear out, so replace them before they cause problems. We recommend using a snap with a swivel on each chain to avoid tangles and possible choking, as well as to save wear and tear on the chain and snap. It is preferable to use two swivels on each chain to provide a backup in case one fails (ices up, for example). A snap at both ends of the tether also gives you an instant "leash" when moving dogs from place to place. This is important in the unlikely event that you need to evacuate the dogs from your kennel. Having each dog with its own tie-out makes emergency kenneling in a safe location easier. The drawback is that snaps have a shorter life-span than solid links but the added convenience is well worth the extra cost.



Using either of the tethering systems described here with 5 ft chains gives each dog an area of slightly more than 78 square feet in which to exercise. With 6 ft chains, the dog's play area is increased to about 113 square feet, and 7 ft chains allow each dog a personal playground of nearly 155 square feet.

Fenced Runs or Pens:

Runs or pens must be large enough to allow dogs to perform most behaviors that are typical of their species. It is recommended that pens provide at least 100 square feet of space for each dog housed within them. Many certified behaviorists have observed that dogs spend more time exercising in rectangular pens rather than in square, so a pen measuring 10' X 20' would be very effective for two dogs housed together. (Rollet J)

Chain link or sturdy woven wire fencing with walls buried 6 to 12 inches (15 to 30 cm) into the ground provide a sturdy barrier that discourages digging. A fence height of 5 ft (1.5 m) or higher is recommended to discourage climbing. Place doghouses so the roof cannot be used as a platform from which a dog can climb or jump over a fence.

Runs and pens should be equipped with gates that are wide enough to permit easy entry and exit, but which can be closed quickly if necessary to prevent a dog from "rushing" the gate and escaping. Gates should be installed with thresholds that allow for snow to accumulate without interfering with the gate opening and closing. These thresholds should also be removable to allow for access with equipment like wheelbarrows and carts. Gates should be equipped with latching devices that are easy for the musher to manipulate while wearing gloves, but difficult for dogs to manipulate. During winter it is important to shovel snow well away from gates in order for them to swing fully open when necessary, especially if thresholds are not incorporated in the design. Thresholds of 1 ft high that are removable for maintenance will almost completely eliminate the need for snow shoveling.

Some mushers have used “hot wire” electrical fencing successfully to prevent burrowing under and climbing over fences. The goal of electric fencing is to modify the behavior of the dog and provide a visual cue to restrict the animal. “Invisible” underground fencing has seen only limited success in sled dog kennels where no other fence or physical barrier is used.

Heat Pens

The most reliable way of preventing unplanned litters is to spay or neuter all dogs you do not intend to breed. Regardless of the primary confinement system used, if your kennel includes any intact (unspayed) female you will need a heat pen to prevent unplanned litters. Plan enough space in the pen or pens to contain all of your intact females at the same time. Females in season (heat) tend to stimulate other intact females’ heat cycles. A secure gate and walls at least 5 ft high are minimum requirements. A fenced roof keeps climbers out and burying about 2 ft of fencing or lining the pen with boulders helps discourage digging. Chain and house the female dogs in the pen so they can’t jump over the fence and to prevent “through fence” breeding. If only one intact female is in season, she may be housed with another compatible female or neutered male to provide company and mental stimulation. If you decide to breed your dogs, it’s also nice to have a pen large enough to accommodate a pair of dogs. Note that heat pens can also serve as puppy pens later on, so be sure that the fencing material used is fine enough to prevent puppies’ heads and adult dogs’ feet and legs from getting trapped.

Social and Psychological Stimulation in the Dog Yard:

Recent research indicates that both social and psychological stimulation in the housing area may be even more important for maintaining physical and mental health in dogs than providing adequate space. (Hubrecht 1995, Hughes & Campbell 1998). This doesn’t mean that space is not important, but rather stresses the importance of providing a stimulating environment for your dogs. The quality of life of a sled dog is not based only on its environment and confinement method, but also on what the dog does outside of the tether and the dog yard. The following ideas may help to improve the dog’s quality of life while it is in the yard.

Dogs are very social creatures. They thrive in an environment in which they can interact with their teammates. Whenever feasible, dogs should be able to see, smell and safely play with each other. Isolating dogs from the company of their teammates has been associated with an increased incidence of behavioral abnormalities. (Hetts et al. 1992). Research has shown that dogs housed in a way that allows them to interact with at least one companion spend a similar amount of time interacting with each other as dogs kept in groups of 5-11 animals. (Hubrecht 1993). If you must isolate a dog from his or her teammates because of health issues, aggression or to isolate a bitch in season, try to keep the duration to a minimum. Isolated dogs should be given extra human interaction and housed within sight of other dogs. (Hubrecht 1993).

Generally, you should provide dogs with a stimulating, non-barren environment. Toys, chew bones and other safe objects with unique smells and placement can provide psychological stimulation. Offer a variety of appropriate items and rotate them frequently between dogs. Many mushers incorporate a “play area” in their kennels in which compatible dogs can interact under supervision.

Spend time interacting with each of your dogs while doing chores and include additional time to play with your dogs and train them to perform simple behaviors. Try to make all such interactions as positive and rewarding for the dogs as possible. Most importantly, take your dogs on frequent training and conditioning runs. Working with other team members, physical exercise and the unique sights and scents of the trail are the best possible form of stimulation for working dogs.

Dogs as Good Neighbors

If you live near other people, it is important to teach your dogs to be quiet. Respect for the rights of other people for peace and quiet makes for happy neighbors and promotes a positive image of the sport. Dogs bark for a reason; it could be to get your attention, to communicate to other dogs, or to announce a visitor (moose, fox, human, etc). A daily howl can express happy communication in the dog yard, but dogs can be trained to be quiet most of the time.

Once you discover the reason for the barking, it may be possible to train them to bark only at acceptable times. Some methods that work include (1) providing a positive experience when they are quiet. This takes a lot of work, but the success is most satisfying, both to dogs and to mushers, and/or (2) providing consistent training when they do bark—a dousing from a squirt gun in summer, hitting the top of a dog house, or a vocal reprimand. You must be consistent and correct them, even at 3 a.m., in order to be successful. If all efforts fail and you can't seem to get a dog to be quiet, discuss the issue with a professional trainer or your veterinarian.

On-the-Road Housing

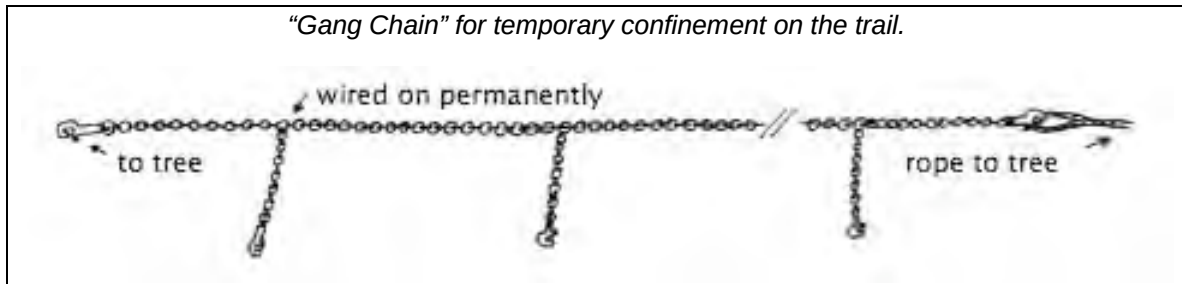
Dogs should travel in safety and comfort in dog boxes or airline crates. Dogs need to be restrained during travel to prevent injury to themselves and to other occupants of the vehicle. Dogs left in the back of pickup trucks and in the back seat in the cab are risks to themselves, the occupants of the vehicle and to other vehicles on the road.

Dog boxes or crates should be large enough for the animal to stretch and turn around in. They should have dry bedding and adequate ventilation. Dog boxes should be well-constructed, have user-friendly latches and locking devices and should be securely attached to the vehicle.

Boxes should not be open or vented in the back of the truck because the vacuum created behind the truck can suck in exhaust fumes. Many mushers modify the exhaust systems of their vehicles so they discharge above the dog box to reduce the exposure to toxic fumes. Mushers in regions noted for extremely hot or humid conditions may equip their dog trucks with ventilation systems for cooling. These ventilation systems should be designed so they draw clean air from the front of the box and exhaust air towards the back to prevent back-drafting of vehicle exhaust.

Some mushers believe that dogs are more comfortable when doubled-up in larger boxes. If you wish to do this, be sure to match compatible companions. When traveling, dogs should be taken out (“dropped”) several times a day. The dogs can be safely left for eight hours at night as long as the vehicle is not moving. Some dogs require their bedding to be changed every day; others, not until the straw has broken down. Drop chains should be kept short to avoid entanglement. Leaving the drop chains or plastic-coated cables in the boxes or on the dogs while traveling keeps the snaps thawed.

While on the road, as at home, keep water buckets clean. Avoid parking where other dogs have been in order to avoid exposure to diseases and parasites. It is best not to store smelly food and equipment in motel rooms, for the sake of your hosts. It is also important that mushers rake up straw and other waste from wherever they have dropped their dogs and dispose of it properly.



Notes:

- (1) The figure cited for pen housing is extrapolated from Table 1 appearing in Hubrecht R., "Comfortable Quarters for Laboratory Dogs"; UNIVERSITIES FEDERATION FOR ANIMAL WELFARE, 8 HAMILTON CLOSE, SOUTH MIMMS, POTTERS BAR, HERTFORDSHIRE, EN6 3QD, UNITED KINGDOM. This table lists 8.0m as the minimum floor space per dog for dogs weighing 35 or more lb., kept in research facilities. The Mush with P.R.I.D.E. Guidelines Committee chose to recommend the larger, 100 square feet per dog figure because sled dogs tend to be more active than dogs in research facilities (Haupt K).

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Feeding and Watering

Choosing Feeds:

Sled dog sports today include a wide variety of dog breeds and sizes doing different activities in almost every kind of climate. There is no single perfect diet that will meet the requirements of every sled dog under every condition. The ideal diet for a dog depends on the dog's genetic makeup, age, physical state, training regimen, environment and the food sources that are available.

Sled dog diets usually consist of commercial dry food, meat-based food, or a combination of the two. Dry foods are convenient to feed and store, requiring only a cool, clean, dry location. Fresh meat products require refrigeration or freezing. Feeds marked with an expiration date should be consumed prior to that date to provide maximal nutritional value.

Meat feeds are extremely palatable to dogs. They may help maintain hydration because they contain up to 75% water by weight. High-quality meat-based feeds are readily available in all but the most remote locations.

Commercial dry food provides vitamins, minerals and carbohydrates. Some mushers prefer to mix their own meat ration and add some commercial dry food to it as a source of vitamins, minerals, and carbohydrates. If you choose this route, be sure to enlist the help of an experienced musher or nutritionist, as it is not always easy to balance a ration this way. Recently a few commercial dry products designed to be fed as supplements with meat have become available. These products are enriched with vitamins and minerals and help take some--but not all--of the guesswork out of feeding a non-commercial meat-based diet.

When choosing a product or combination of products to feed your dogs, remember that a sled dog's nutrient requirements change significantly depending on age, environment, and physiological state. You may wish to choose different products that will meet your dogs' requirements for each of these situations, or you may choose a feed that will act as a base that can be supplemented as necessary. In either case, choose a product that is relatively high in fat (15% minimum), relatively high in protein (25% minimum), fresh, and of the highest quality available. Poor quality commercial pet foods do not provide adequate nutrition to meet the needs of working sled dogs.

Determining a Working Dog's Dietary Needs

The best way to monitor your dog's body condition status is to run your hands over him or her. Perform this examination at least every two or three days. (In extreme conditions, it is important to keep an even closer eye on a sled dog's weight.) The ribs, spine, and hip bones should not be buried under an inch of fat, nor should they protrude. Rather, they should be easy to feel. A well-conditioned sled dog should be lean and muscular—neither skinny nor obese. (See Appendix). If you are unsure of what the ideal appearance and feel of your individual dogs should be, solicit advice from an experienced musher or veterinarian. Take every opportunity to feel and look at dogs from other kennels that are doing well in your particular mushing discipline.

Formulas and tables on dog food labels will give you a place to start, but they should not be relied upon for long-term feeding guidelines. There is too much variation in metabolism among dogs, their working environments and their various levels of performance to rely on "average" requirement guidelines. Most mushers agree that it is crucial to monitor your dogs' weight and body condition with your hands.

Meeting the Changing Demands of Training

Maintaining a dog's optimal weight requires frequent adjustments to the amount of food he or she is given. When you begin training and each time you increase the workload, your dogs will require more food. During cold or wet weather they will need more food just to maintain their normal body temperature. One of the most difficult times of year to maintain a dog's body weight is during the fall when the weather is often cold and wet and training miles are increasing. During such periods, anticipate your dogs' increasing nutritional needs and begin feeding them more before they start to lose weight. During the most demanding times, a sprint dog may require two to three times more food than during the offseason; a long-distance racing dog may require three to six times its offseason requirement.

Feeding During the Off Season

Recent research indicates that dogs that continue to receive high-quality rations through the off-season are better prepared to resume training because their bodies are more able to mobilize and burn fat during exercise. It also takes several weeks for a dog's metabolism to adapt to a high fat diet. The drawback of feeding premium dog food year-round is that it can be easy for dogs to become overweight in the offseason; watch your dogs closely and adjust their portions as necessary.

Life Stages

Dogs also have different nutrient requirements during pregnancy, lactation, growth, and old age and their diet and food intake must be adjusted accordingly during these times.

Pregnancy and Lactation: A female should be maintained on a performance type ration throughout pregnancy and lactation. She can be fed at maintenance levels for the first four weeks; however, from the fifth to the ninth week, her intake should be increased by 10 percent each week so that when she whelps, she is getting about 1 1/2 times what she was eating in the maintenance state. As a rule of thumb, her food intake should be increased by 30 percent of maintenance for each puppy she is nursing. Thus, if she only has one puppy, she should be fed 130 percent of maintenance. These suggestions are just guidelines-; remember to run your hands over her regularly and adjust her food intake as needed. A lactating dog should be neither skinny nor obese.

Puppies: Puppies usually weigh between 10 and 14 ounces at birth and should gain weight every day after their third day of life. Weight gain is an excellent way to monitor the nutritional and overall health status of a litter of pups. Slow or negative puppy weight gain can be the first noticeable sign of a health problem with the mother or pups, and supplemental feedings may be required. Enlist the help of a veterinarian or an experienced musher the first time you attempt to raise orphan pups or even supplement nursing ones.

Puppies can begin to eat solid food at three weeks of age. Puppy food or a high-quality performance food with a small kibble size is recommended for at least the first four months of their lives. A flat pie pan with soaked dry food or a meat ration is a good way to entice them to start eating. As they walk through the food, they will get bits of food on their paws, lick them, and realize it is something good to eat. Over the next three to four weeks, they will consume more food, so they can usually be weaned between six and seven weeks of age. Before, during, and after weaning, be sure that less assertive pups are maintaining a normal rate of growth. Since there is no standard rate, compare the growth rate of the less assertive pups and their littermates.

After four months of age, pups should be fed a premium food at a rate that keeps them in optimal body condition but not so much that they become fat or grow too fast. (Maximum growth rate of 2 1/2 pounds per week for huskies, 3 to 3 1/2 pounds per week for larger Northern breeds).

Older dogs: The aged dog may have a slightly decreased ability to digest and absorb nutrients. It may also take an older dog longer to move a meal through its gastrointestinal tract. Most older dogs will do well on the same ration as younger dogs in the offseason. Occasionally, a dog will have trouble digesting all the fat in this ration or may become constipated. If so, try feeding a diet lower in fat or higher in fiber. When in doubt, ask your veterinarian about specific diet regimes and supplements for your geriatric dogs.

Water

Water is the most essential part of a feeding regime. While deficiencies in protein, fat, vitamins, or minerals will affect a dog's health, it may take days or months before such problems are noticeable. In contrast, dehydration affects an animal's health immediately and in extreme cases can even lead to death within hours if left untreated.

A dog gains water by drinking it directly, by eating foods that contain water, and by generating water through metabolism. Water is lost each day through urine, feces, and water vapor in the breath. Anything that increases a dog's daily water loss will increase its daily requirement. Dogs also lose a significant amount of water through panting when the weather is warm. Exercise leads to increased water loss not only through the breath, but also through the stool and urine. A dog's water requirement may double if it participates in open-class sprint racing and increase three to five-fold if it participates in long-distance racing. Medical problems such as diarrhea and vomiting also increase water loss.

It is difficult to know exactly how much water each dog requires. Understanding how environment, training, and illness may affect the dog's hydration needs allows you to anticipate these changes and offer your dogs more water when they need it. During warm weather it is best to have clean, fresh water available at all times. When the temperature drops below freezing, water consumption can be encouraged by offering warm, baited water. The bait can come from any source that will increase palatability such as dry food, meat or cooked fish. The bait should mix well in water and must not be spoiled or soured.

About 1 1/2 quarts of water should be offered two to three hours before training. Some dogs will not drink this amount all at once but will readily consume several smaller portions offered within a short period of time. Small amounts (about a pint) can be offered immediately after exercise to help cool the dogs down followed by more water (about a quart) after they have completely cooled down. Offering 1 to 1 1/2 quarts of baited water before feeding or mixed in with a dog's food can further encourage water intake.

These recommendations are a starting point and should be adjusted according to the needs of your dogs. Monitor your dogs' hydration status by observing their hunger for snow and by examining their skin and gums. In a well-hydrated dog, the tent made by lifting up the skin on the shoulder blades should disappear within one to two seconds and the white spot made by pressing on a pink area of the gums should disappear in one second or less. If either of these processes takes longer, the dog is probably dehydrated and in need of fluids.

Monitoring Your Dogs

These guidelines are intended to help you begin your feeding and watering programs. The best feedback on how well you're doing will come from the dogs themselves. Watch them carefully and learn as much as you can from experienced mushers who you respect. Proper dog nutrition is a blend of science and art. It's easy to get a brain-full of science by reading books and articles on the subject, but you can only develop the actual skills with hands-on practice. So keep your eyes and ears open, and go have fun with your dogs!

Training and Conditioning

Planning Your Dogs' Training and Conditioning Regimen

Many training principles are specific to the type of activity in which your dogs will be involved. Other variables include climate, terrain, age of the dog, breed of the dog, etc. All forms of mushing, with all the different variables, are wonderful ways of forming a close bond with your dogs.

In general, training can be separated into two categories: education and physical conditioning. When you are planning your training schedule, consider your goals and your dogs' abilities. Simply counting miles, for instance, can be deceptive. The type of conditions that the dogs encounter are important too, i.e. steep hills, trail breaking in heavy snow, extreme temperatures or wind. New mushers should consult books and experienced mushers for help, but also use common sense. Think about what your dogs have been trained to do and do not allow them to get carried away in their enthusiasm to run. Never ask your dogs to do more than you are reasonably certain they can accomplish.

Educating Your Sled Dogs

Anything you do repeatedly with a dog is educational. Be sure you want your dogs to learn what you are teaching. Think about the signals you are giving your dogs, and don't send mixed messages. For example, if you want your dogs to pass well, don't stop and chat with the neighbor every time you pass. Doing so trains your dogs to stop at every pass.

It is important that you never lose your temper with your dogs. Try to train them in a calm, consistent manner. If one method is not working, try another. For example, if a dog is not pulling well in a large team, reduce the size of the team and put that dog in wheel position for a week. If a dog continually plays with the dog next to it while running, run that dog alone for a few weeks. Remember that repetition is a great teacher. If your leader is not taking gees/haws well, go out with a very small team and work on commands. Always praise the dogs enthusiastically when they are doing what you want.

Recent research has proven that dogs learn much more readily with positive, reward-based methods than with methods that rely primarily on punishment. Positive methods also result in a closer bond between musher and team, and are much less likely to cause unwanted fearfulness or human-directed aggression in sled dogs.

A reward is anything that increases the likelihood that a behavior will be repeated. The value of a reward is determined by the dog, rather than the handler. Just because you think something should be rewarding doesn't necessarily mean your dog will agree. A food treat is only a reward if the dog is willing to repeat the behavior in order to earn another.

Most sled dogs place high value on tasty food treats and on running, but there are exceptions. If your dog doesn't respond to one type of reward, switch to something the dog is more willing to work for such as chance to play with a favorite toy. If your dog does not place a high value on running you may want to reconsider its suitability as a sled dog. You may both be happier if the dog becomes someone's pet.

In addition to the cues used while mushing, training classic 'obedience' cues is a great way to help socialize your dogs and provide psychological stimulation. Formal training classes expose your dog to new situations and introduce a positive image of sled dogs and mushers to the general public. The learning process will help your mushing training continue smoothly during the working season and the day-by-day, step-by-step progress will keep your dogs active during their

offseason. Different dogs will enjoy different activities, but each dog needs individual attention.

Physical conditioning.

Do not expect your dogs to do more than they are ready for. Watch each individual. Dogs work as a team but they have individual needs and abilities. Don't be afraid to be conservative and don't worry about how far other mushers are taking their dogs. Never push a dog to go any farther or faster than it is capable of going.

Have fun and build relationships with your dogs. Small teams are better for training. Depending on your preference and the size of your kennel, training teams might include only three dogs or as many as six to ten. Dogs can only run at their own pace and must never be pulled, whether by mechanical or other means.

Summer/Warm Season Training

Dogs need some form of physical exercise all year long. As long as your dogs are in good health, light training in the summer is fun and beneficial. Equipment options include a bicycle, cart, dog walker, ATV, or a leash. For some dogs and mushers, running a few dogs loose may be an option. It is important to always use proper harness sizes and gang line lengths. Always check each dog for foot problems or injuries after each run.

If you mush dogs in the summer, ensure they are well-hydrated before and after each run. Wetting them down beforehand with a hose or with creek or lake water can be effective. It is best to exercise sled dogs during the coolest time of the day, but even then you must watch carefully for signs of overheating. Signs of heat stress include heavy panting with an open trachea, gait change, wobbly legs and vomiting.

If you are concerned that a dog might have heat stress, remove the dog from the team and carry it in your sled or vehicle. If you need to cool a dog down during summer, wet it with cool water. During winter, pack its body in snow. During and after cooling, continue to monitor its temperature with a rectal thermometer. Dogs routinely have temperatures of 103 to 106 degrees F while running (normal is 101 to 102 degrees F). Recheck the temperature every fifteen to thirty minutes as the dog cools. If the dog's temperature is still not normal after you have attempted to cool the dog, call your veterinarian. This could indicate a serious problem.

Fall Training

Most mushers like to start on some kind of wheeled rig before the snow comes. Make sure the rig has good brakes to slow the dogs down and that there is some type of parking brake. ATV's are widely used because they give the musher complete control over the dogs' speed, and they steer more easily than a cart. They also have lights, which provide safety in darkness, especially along roads, and they make noise that warns wildlife of the team's approach. A speedometer is convenient on any type of training vehicle.

Start your fall training season with small, easily controlled teams and short runs, perhaps only ¼ to 3 miles in length. Early season runs may require frequent rest stops. Decrease the number of rest stops and gradually increase mileage in subsequent runs as the dogs get stronger and fitter. If you have run the same distance over repeated training sessions and the dogs are finishing strong and happy, it is time to move up to the next level. In early fall training, don't push the speed too much. The dogs' muscles are not well-toned yet and it is easy to injure them. The goal of early training is to build up the dog's muscle structure to prevent injury later in the season.

Any training schedule must include rest days to allow time to build muscle. You might run a dog every other day, or run two days followed by a day off. Water (or broth) your dogs when you return from a run, and check for worn pads, especially if they are running on gravel or pavement.

Winter Training

Once you are working on snow, continue to build slowly to the distance and speed of your choice. Always check each dog's feet and provide plenty of water or broth. It is not good enough to let them bite snow for their fluids.

Not all injuries are easy to detect. A dog does not always show a substantial limp, so watch carefully for subtle signs. If a dog seems weaker or slower one day than the previous week, it might be due to injury or illness. A back can be injured without causing a limp, or a dog can be so excited to run that it will not show any signs of injury while running. You may be able to detect problems by observing the dog at home.

Detecting injuries or illnesses early can keep your team healthy and working all season, and can save money on veterinarian fees. Check each dog over carefully at least once a week. Knowing each dog's "healthy" condition will make it easy to detect changes. Consult with your veterinarian or another expert if you suspect problems.

Booties should be used to prevent injuries on rough trails, including when snow crystals are abrasive in severely cold weather. If your dogs' feet develop any signs of worn pads or soreness, use booties on those feet until the problem is completely healed. You might consider not running the dog at all for a short while, depending on the severity of the problem. Be sure the booties fit well. A bootie that is too large flops around, picks up snowballs, and makes it difficult for the dog to run normally. A bootie that is too small can constrict the foot and be uncomfortable.

Be sure to check booties regularly. A bootie with a hole in it can cause more problems than no bootie at all. Also, pick off all snow and ice balls around the tops of the booties frequently, as these frozen clumps can cause severe chafing. If your dog has dewclaws, watch for signs of wear around them. Remember that booties are not a cure-all for every foot problem. Consult your veterinarian or an experienced musher for more advice.

In extremely cold or windy conditions, dogs can get frostbitten on some body parts. On a male, watch the sheath of the penis and the scrotum. On a female, watch the nipples, flanks, and vulva. Be extra careful with any female that whelped over the summer. Her nipples are usually somewhat enlarged throughout the winter, making them more susceptible to problems. Special dog jackets, belly pads, and fur sheath protectors are available and can help prevent cold-related injuries. Contact a mushing equipment company or other local mushers for ideas. Remember that males and females have very different problems, and the same equipment does not always suit all dogs.

Dog Fights

Minor spats and squabbles are relatively common among dogs, but serious dogfights are dangerous for both dogs and mushers. Dogs should be taught at a young age that fighting is unacceptable. It is essential to stop a dogfight before a dog is injured or killed. Fighting dogs must be separated and restrained, but be extremely cautious when handling highly aroused or aggressive dogs. In the heat of the battle, the dog may redirect its attack to you, inflicting serious wounds. Mushers have been severely bitten while breaking up fights and care should be taken when intervening.

Training and Conditioning Tips

Long-distance racing:

Before you consider running a long-distance race for the first time, evaluate your skills carefully. You must be good at winter camping with dogs, starting campfires at -50 degrees F with a strong wind blowing, applying first aid to dogs and yourself or another musher should you get caught between checkpoints, etc. You must have advanced skills in handling however many dogs you choose to start the race with. (In your first race, it is better to start with fewer dogs. A smaller team is easier to control and means fewer dogs to feed and care for.) You must also be an expert in feeding and foot care during high mileage situations. The time to learn these skills is during training, not out on the race trail.

In general, to run a thousand-mile sled dog race, you should have at least 1,500 miles of training on each dog. These miles should be accrued in no less than a six -month period. To run in a 200 to 500 mile race, you should have at least 750 miles of training on each dog. These miles should be put on in no less than a four month period. Much of the training should duplicate your proposed racing situation, with weight in the sled, some four to six hour runs, camping trips etc. It is inadvisable to run any dogs under 18 months old in a thousand mile race. The ability of each dog in the team should be fairly equal so that no one dog is being pushed too hard. Teach your dogs to eat, drink and sleep in harness before you race them. Feed them the same diet that they will race with, at least during the latter stages of training.

Sprint racing:

To create a quality team, sprint racers use the same training and conditioning techniques as those used for other types of mushing. Distance and speed should be built up slowly on a schedule determined by your dogs' progressive conditioning and willingness. It is better to err on the conservative side than to risk hurting a dog physically or mentally by demanding more than it is ready for.

While speed may be the primary objective in sprint racing, not every training run should be at "race pace." To prevent injuries in the fall, dogs should be physically conditioned with slower working runs before you allow them to run fast. Throughout the race season, vary your training speeds and your dogs will be more willing to go fast when asked.

Proper manners and well-behaved dogs are a must for a top-performing sprint team. Even the quickest stop for a tangle or problem dog is a major disadvantage in a race. Take the time required to teach your dogs the necessary behavioral skills. Some sprint mushers simply concentrate on maintaining enthusiasm in their team, but a well-behaved *and* enthusiastic team is possible and should be the ultimate goal.

Recreational mushing:

Training a recreational team can be extremely rewarding and satisfying. It can also be extremely expensive, both in time and money. Keep your priorities straight, share the work among family members, and have fun!

Before you begin, decide whether you want a dog team for your family to enjoy and consider everyone's goals for the team. If you have small children, you may want to select dogs that are small and gentle so the kids are comfortable with them. Some older, well-trained retired dogs from another team may be perfect for you, and they can help train younger dogs.

Make your dog time quality time for your family. Chart the accomplishments of each dog. In the summer, you might have a weekly dog show to demonstrate each dog's new tricks. Having a small number of dogs allows you to give each individual lots of attention. The dogs will learn that they have fun with you, and they will be eager to please. Seeing your dogs thrive on this special attention, watching your family share the responsibilities, experiencing the magic of bonding with animals and the satisfaction of a job well done are ample rewards.

In winter, plan methods of training and goals for your team with family members. Listen, talk, encourage, and reward. Have fun and don't be afraid to ask other mushers for help. Practice "whoa" and "come haw" repeatedly, until the dogs respond easily; this will give the less experienced members of your family a better sense of security. You may want to work with the dogs on a leash, rewarding them for correct behavior.

Take a family member with you on the sled. A less experienced passenger can help out and learn what you ask of the dogs. Make sure your passenger is comfortable. After the dogs have settled down, let your passenger drive the team on a safe stretch of trail. Always train with small, controllable teams. Gradually increase your distance over the winter. Explore new trails. This gives your dogs experience in different conditions: breaking trail, running into open water and on ice, and turning around. Take a picnic along. Stop along the trail and build a campfire.

Your family might enjoy working toward taking the team on an overnight trip, either camping out or staying in a remote cabin. This could be a spring celebration after a winter's training. Remember that you don't have to go a thousand miles. Plan according to the abilities and desires of your family and the endurance built up by your dogs.

Skijoring:

Skijoring is one of the simplest forms of dog driving, but common sense, patience, and general training principles still apply. Stay within your dog's capabilities for weight load, speed, and distance. Be aware that some dogs (including experienced sled dogs) can be quite frightened by the strangeness of the skis, and a dog may need extra time and lots of positive reinforcement before it will accept being followed by them. Avoid running into your dog with your skis or ski poles at all costs.

Proper equipment is important both for your own safety and for the comfort of your dog. Use a wide skijoring belt (at least 3 inches wide across the back) and a releasable skijoring line at least 7 ft long. Longer lines (up to about 15 or 20 ft) work well for recreational skijoring and hilly terrain. Shorter lines give better control and are favored for racing. Be sure the line is long enough to prevent the tips of your skis from hitting the dog. A line with a shock (bungee) cord incorporated into it will absorb the stress of sudden starts and stops, a benefit to both you and your dog. Use a properly fitted, standard X-back or H-back mushing harness. Weight pulling harnesses are not recommended.

A wide variety of dog breeds have been used successfully for skijoring. If you skijor with a non-Northern breed, watch carefully for foot problems. Some breeds of dogs, especially those with silky coats, are particularly prone to ice balls. Booties may be necessary in some cases. Also, a shorthaired dog may need a dog coat and/or a sleeping pad in very cold weather or when camping out.

Weight pulling:

A weight pull dog should be strong, sound, in good health and have a desire to please. Before a dog is entered in a weight pull competition, it should have at least basic training and be in good physical shape. A dog that is in poor condition might pull beyond its physical abilities simply because it wants to please or because of the excitement of the activity. Avoid heavy pulling until your dog is in top shape.

Conditioning can be accomplished in various ways: running in a team, running alongside a bicycle, skijoring, or pulling a tire. Perhaps the best method is for your dog to pull a tire with increasingly heavy loads. Be very careful to increase the loads gradually. This is important for mental as well as physical conditioning. A dog must know that when it is commanded to pull, the load will move. Many factors influence the ease with which a dog can pull: weight of the load, snow depth and conditions, and temperature. As you train, adjust the load downward if your dog has difficulty starting the load. It is important to condition the cardiovascular system as well as the building muscle. This is done by alternating days of pulling heavy and light loads.

You can't begin too early to train your dog to pull. Even a young puppy can have fun wearing a harness and pulling an empty box around. Use this time to teach some basic commands, such as those to pull, whoa, and perhaps gee and haw, as well as to sit, lie down, and stay. Be careful not to let the box bump into the dog or let the dragging noise frighten it. Gradually increase the weight the dog pulls and progress from a piece of firewood to a 12 inch tire, for instance. Give the command to pull, let the dog pull a short distance and lavish it with praise. Make it fun. Your dog will pull for sheer enjoyment and because it pleases you. Be sure to let your dog know that you appreciate its effort.

Dogs should not be entered into competition until they are at least a year old; large breeds should wait until a year and a half. This gives them time to reach skeletal maturity. When the time comes to enter a weight pull competition, your dog will know what is expected and will be ready to do it well.

Winter transportation:

Some mushers use their teams for traveling cross-country, doing fieldwork for their jobs, freighting supplies, running trap lines, and general winter transportation. Training these teams may focus on building endurance and strength and on mushing in severe weather conditions. It is critical that the dogs learn to whoa, wait in harness, and find old trails in drifts. They need to learn to follow along behind their musher when he or she is breaking trail on snowshoes, and they should learn to ignore animals caught in traps. These abilities come by working with small teams, day after day.

Mushers who depend on their dogs for winter transportation often have a very close relationship with their teams. The trust and appreciation that develops after many hours, many days, and many seasons together create a team that seems able to go anywhere and do anything. This is not magic. It is simply the result of clear communication, mutual respect and consistent, repetitive reinforcement.

Basic Health Care

Veterinary Care

Develop a good working relationship with your veterinarian. He or she will be a valuable source of information on current dog care practices as well as providing other services as needed. As the dog's owner, you can often perform routine procedures yourself but will also have to obtain professional care on occasion. The science of veterinary medicine is evolving; frequent contact with your veterinarian will help you stay informed about the latest changes in veterinary sciences. Many veterinarians also benefit from contact with mushers. Mushers bring a new dimension to animal health care professionals. Having regular contact with a vet also serves to present a positive image of dogsledding to the public.

Record Keeping

Record keeping is an essential part of any kennel operation and is a requirement for P.R.I.D.E. kennel certification. Often, your veterinary clinic keeps records of office visits and professional vaccinations, but most care and medical treatments are given by the musher. Keeping track of medications, vaccinations, illnesses and general health will allow you to identify health trends. Records can include breeding cycles, on and off season weight fluctuations and training and performance distances and times.

Adult Dogs

Daily care: Monitor each dog's appetite and water intake as well as its fecal and urine output. These changes, as well as changes in behavior, are frequently the first signs of a health problem. Running your hands over the dogs daily will allow you to recognize abnormal conditions.

Monthly care: Trim nails and groom each dog. If a dog is shedding, grooming keeps its skin and coat healthy. Administer heartworm preventives and external parasite control medications (for fleas, lice, ticks, etc) if needed.

Three to six month care: Work out a deworming program for your dogs with your veterinarian. The type of dewormer and frequency of administration will depend on the type and species of intestinal parasites in your area.

Yearly: Consult with your team's veterinarian to develop an appropriate vaccination schedule for your dogs based on the health problems common to your region and the demands of your race schedule or mushing goals. If you live in an area where professional veterinary care is not available, contact your state, provincial or national veterinary medical association for information to help you develop your own vaccination schedule. Also, consult the rules of races you may wish to run with your team. Many race-giving organizations (RGOs) specify vaccinations that dogs in their events must have received.

During the late summer of each year you should perform a thorough physical exam on each dog in the kennel. If you find any problems, you will have time to treat them before fall. (See Yearly and Pre-purchase Exam section below for guidelines.)

Consider having an annual veterinary house call to your kennel. This is one way of becoming a P.R.I.D.E. certified kennel. An annual kennel visit is a great way to have vaccinations administered and have all your dogs examined without transporting every dog in the kennel to the vet office. This is a great time to discuss each dog's specific needs with your vet, allowing every dog to reach his or her greatest potential.

Some veterinarians have backgrounds and training in large animal care but rarely get out in the field anymore. Your request for a kennel visit may become something they look forward to.

Basic Health Examination

The following guidelines are basic elements of a physical health examination for dogs. They can also be used when considering the acquisition of a new dog for your kennel or for a pre-season health assessment. Some of these elements should be performed daily or even both before and after running. (Elements marked with an asterisk (*) should be performed daily.)

General attitude*: A dog should be alert and interested in its surroundings.

Weight and coat*: A dog should be lean but not thin. It should have a healthy, shiny coat and skin that is a light pink with no raw areas or excessive flaking. Run your hand over the dog's whole body, checking for lumps, bumps and sores. A dog that is underweight/overweight or with an unhealthy coat may be showing signs of conditions such as hypothyroidism, parasitism (internal parasites such as worms or coccidia, or external parasites such as lice, fleas, or mange mites), malabsorption syndrome (an inability to absorb nutrients), or another ailment. However, keep in mind that all dogs do not always look their best. Even a beautiful coat looks rough during shedding, and a female will shed after she has pups.

Eyes*: The eyes should be clear, without excessive tearing, redness, or a gray or blue haziness on the cornea. The pupils should be symmetrical.

Ears*: The ears should be clean inside without a waxy or pussy discharge and without a foul odor.

Nose*: There should be no nasal discharge, raw areas, or dry, crusty buildup around the nostrils.

Mouth: The mouth and teeth should be clean without any strong odors or excessive tartar buildup. The gums should be pink without infection along the teeth gum border. Check for broken teeth or an uneven bite. Dental disorders may contribute to poor appetite, poor attitude, or chronic infection.

Respiration: A dog's normal heart rate is 100 to 130 beats per minute, and its respiratory rate should be about 22 breaths per minute. These may both be elevated in an excited dog, and both will be hard to evaluate in a panting dog. Listen for coughing, wheezing or other abnormal respiratory sounds.

Chest: The chest should expand and contract symmetrically. There should be no pain or tenderness when the dog's ribs are gently pressed. Use a stethoscope to listen over the lungs. Lung sounds should be equal from side to side with no abnormal noises. Because chest injuries can result in massive internal bleeding and respiratory system compromise, a veterinarian should evaluate any significant chest injury.

Abdomen: The abdomen should be symmetrical and not distended. The abdominal wall should be pliable when gently pressed toward the spine. A painful, tender, distended abdomen may be a sign of a potentially fatal problem that requires the intervention of a veterinarian.

Muscles and bones*: Check the dog for symmetry. Compare the muscles and joints of the two hind legs and of the two front legs. Swelling on the foot may be an indication of an old metacarpal fracture. Swollen wrists may be a sign of arthritis. Asymmetrical muscle masses may indicate lameness or an unequal use of limbs.

Feet*: Check the feet for signs of injury or excessive licking between the pads (mahogany, discolored hair). Examine the nails and dewclaws and trim when necessary.

Rectum: Check the rectum for open sores, growths or excessive swelling.

Veterinary considerations when acquiring a new dog.

Females to be used for breeding: Dogs intended for breeding deserve special consideration. Before purchasing a female, be sure to ask the following: Has she had regular heat cycles? Has she ever been on medications to delay or postpone heat cycles? Has she ever had a pregnancy terminated? Has she ever had pups? If so, what kind of mother is she? Examine the mammary glands for swelling; mammary tumors are not uncommon in older intact females. Examine her nipples for signs of frostbite. Severely frozen nipples are not functional. Examine the vulva for growths, swelling, or discharge.

Males to be used for breeding: Make sure that a male that might be used for breeding has two normal sized testicles. Check for excessive prepuce discharge. Ask if the dog has ever sired a litter. Has he ever had any medications? Anabolic steroids, for example, will reduce fertility. Low thyroid levels will decrease fertility in males as well as females. Brucellosis is a sexually transmitted disease that should be tested for if it is a problem in your area. (Some countries make a brucellosis test an entrance requirement.)

Other considerations: Before introducing a new dog into your yard, make sure that it is current on vaccinations and has recently been dewormed. Check the dog closely for lice, mange, and fleas. If you are buying a dog with parasites, isolate him/her from the others until the issue has been resolved.

KEEPING YOUR KENNEL THE RIGHT SIZE

Determining Your Needs

Any trainable dog can be a sled dog, depending on what you want to do with it. A musher must use appropriate care when asking any dog to work. A team of beagles can pull a sled, but they couldn't break trail in deep snow. A team of 30 lb border collies might pull well, but they should be outfitted with booties to protect their long-haired feet. A team of poodles can make good sled dogs but it isn't wise to ask them to camp out in severe weather.

Some dogs have a head start for some types of mushing. Northern breeds evolved specifically as sled dogs and they have physical adaptations that keep them comfortable in very cold weather. Thousands of years of selective breeding have given them a strong instinct to run and pull.

When deciding how many dogs you should own, consider how much money and time you can dedicate to your team, what your zoning laws and living situation will allow, and what it will take to do the type of mushing you want. There are different types of sled dogs and you will need fewer dogs if all of the dogs you own are suitable for what you want to do. If you keep fewer dogs, your costs will be lower, and you can give more attention and better care to the ones you have. Keep your kennel the size you can care for properly. Don't let numbers increase to the point that neither you nor the dogs are happy.



Preventing Breeding

The most effective method for preventing dogs from breeding is to spay or neuter all dogs you do not intend to breed. Spaying (ovariohysterectomy) or neutering (castration) are good options for dealing with heat cycles and for preventing unwanted litters. Sterilization can also make it easier to run females and males together, and can save money by reducing dogfights, health problems, and food requirements. Spaying and neutering can save a tremendous amount of frustration, energy and money in the long run. One unwanted litter or one serious dogfight is much more expensive than the cost of the surgery.

Some mushers are under the false impression that spaying or neutering will reduce the drive of the racing sled dog, but this is not the case. (Zink 2005) Many top long-distance and sprint mushers have successfully run neutered and spayed dogs in their racing teams with no decrease in performance. And, many races have been lost by having a bitch come into heat at an inopportune time. The only reason to not neuter or spay a dog is the desire to breed the dog.

Some veterinarians who specialize in canine athletes recommend spaying or neutering athletic dogs including working sled dogs any time after 14 months of age. Dogs sterilized prior to puberty seem to run a higher risk of injuries to bones and joints, to some types of cancers and to some behavioral problems, including fearfulness and aggression. (Zink)

If you own any female dogs that aren't spayed, you must have at least one heat pen. It should be capable of containing all dogs in season comfortably and securely at the same time. To be effective your heat pen should be either tall enough or roofed over so that dogs can't get in or out. Even if all of your male dogs are secured and under control, the heat pen is necessary to prevent breeding with stray dogs. (See *The Dog Yard and Housing* section for details on heat pen design.)

If you suspect a bitch has been accidentally bred, consult with your veterinarian as soon as possible. Your vet may be able to perform tests to determine whether or not she is actually pregnant. If she is pregnant you may abort the pregnancy and prevent future mishaps by having the vet spay the dog. If you strongly feel you want to breed her in the future, treatments are available to terminate pregnancy if given within a few days of breeding. Abortions can cause complications and aren't a substitute for prevention. Consult your veterinarian for details.

What to Do with Dogs You Don't Want to Keep

It is unfair to the dogs to own more than you can handle. Any musher only has so much time, space, and money, and those are divided by the number of dogs in the yard. Sled dogs are born to run and should not be kept on their chains all their lives. Don't keep them if you don't have the time to exercise them. Review your needs, honestly evaluate the dogs you already have and then decide the best course of action.

The most difficult part of owning dogs is figuring out what to do with the ones you can't keep. You might be able to sell your extra dogs, but don't assume so. The market is very limited except for sellers with top-notch kennel records or dogs from rare and highly desirable bloodlines. If you do sell dogs, be honest and try to make the right matches. Be sure that the new owner will care for the dog properly. Consider giving trial and return periods as a means to encourage adoption. Occasionally contacting and being available to assist new owners in the care of your old dog is a great way to maintain a positive relationship with the new owner and let you maintain a lifelong connection to the dog.

Another option is to give surplus dogs away to interested, reliable people. Consider recreational or junior mushers, skijorers, mushers competing in a different mushing discipline or less demanding classes, or pet owners looking for a companion. Be sure to fit the dog to the right person. Many sled dogs are challenging pets; some have an instinct to roam or kill livestock and are often more independent than some pet owners expect. Keep in mind that dogs that have been properly cared for and socialized have the best chances to be placed. Since a dog that is not good enough to keep is probably not good enough to breed, consider having the dog spayed or neutered before giving it away, or requiring that the person taking the dog have the operation performed.

If you are unable to find new homes for unwanted dogs you may be able to relinquish your surplus dogs to a sled dog rescue organization. Some sled dog rescue groups specialize in pure-bred dogs such as Siberian huskies or Alaskan malamutes, and others are willing to accept Alaskan huskies and other mixed breeds. A group that specializes in sled dogs will generally have a better chance of placing your working dogs in an appropriate home than a government operated animal shelter. Such shelters should only be considered for dire and/or emergency situations.

If you must relinquish your dogs to an animal control agency be aware that any dog not adopted within a limited time period will probably be euthanized no matter how friendly or well-socialized the dog is. Sled dogs are often misidentified as non-adoptable or overly aggressive by some shelters and can be euthanized immediately based on local ordinances or requirements. Your dog's chances of survival are much greater if you take responsibility for finding it a new home yourself.

Acquiring Dogs

It is important that you determine your mushing goals before acquiring even a single sled dog. Once your mushing goal is firmly established acquire only those dogs with physical and behavioral attributes that will help you achieve that goal. This will prevent you from acquiring unsuitable dogs that will need to be re-homed later.

Leasing or borrowing dogs may be an option if you aren't sure how committed you are or if you need extra dogs for only one race or one season. Shop carefully, as there are many options. Ensure your lease or loan agreement is clear about who is financially responsible for illnesses or injuries, and remember that borrowed dogs need the same responsible care as the ones you own. Another option is to volunteer to "foster" rescued sled dogs for a sled dog rescue organizations.

When you are ready to establish your own kennel, keep your mushing goal in mind. If your goal is to win sled dog races, it isn't enough to buy the best dogs you can afford. Instead, you must afford the best dogs you can buy. Today's sled dog races are extremely competitive. Only teams made up of exceptional dogs can win consistently.

Mushers with more modest goals have a much wider range of options. Experienced sled dogs suitable for a variety of mushing disciplines are frequently available through sled dog rescue organizations or from other mushers in your area who have surplus dogs that need to be re-homed. There are very good dogs available but you have to make sure the dogs you get are the right dogs for you. Don't make the assumption that a dog from a well-known kennel or bloodline will meet your needs. Evaluate the individual dog in relation to your goals. For help in evaluating the health of the dog you are considering acquiring, see the *Basic Health Care: Basic Health Examination* section.

Another option for building a team is raising puppies yourself if you have the time and energy for this process. Good dogs are easier and probably less expensive to buy than to raise. However, raising puppies is a fulfilling experience if you can afford to do it and have homes for each of the puppies. Breeding sled dogs should be viewed as a way to produce better dogs, not just more dogs.

If you do decide to breed dogs, remember that in less than six months you will have essentially full grown dogs, each needing a house and chain or a pen of its own. For Alaskan huskies, plan on more than six pups per litter. A litter can easily be as many as ten or as few as one. Two litters can therefore produce as many as twenty new dogs!

Before you breed dogs you must do two things: (1) Make a realistic plan for what you will do with every pup that is born. (2) Ensure that the dogs you breed have all the essential characteristics you want. If you don't have the right dogs, buy a good female, buy the service of a good stud, or offer to raise pups for a musher who has high quality dogs. Never breed dogs with any physical or behavioral defects. Undesirable attributes are as likely to be inherited by their offspring as the traits you wish to perpetuate.

Remember that good genes are responsible for only a portion of the final result. Raising excellent sled dogs requires excellent physical care, mental and physical conditioning, socialization and training. The more time you spend with your puppies the better sled dogs they will be.

Both male and female dogs become fertile at six to 12 months of age. The average interval between estrus cycles is about six months, but it varies widely. Some females come into heat every three to four months, others only once a year. Although a female may be bred in her first heat, many breeders prefer not to because it interrupts her growth and because young dogs can be poor mothers. It is also a good idea not to breed very young dogs so that you can be sure they have the traits you want. Older dogs can be bred, but fertility generally declines after about 10 years of age. Be cautious about breeding females over six years old that have not been bred for two or three years, as they more frequently have problems with whelping.

Be sure the female is adequately vaccinated and wormed before breeding. Be sure to disinfect the puppy pen, doghouse and whelping box before the pups arrive. Pups are usually born 60 to 65 days after the breeding. An experienced veterinarian can often tell if a dog is pregnant by palpating the abdomen 21 to 28 days after breeding. For more information, consult your veterinarian.

A pregnant female will need progressively more food starting the last three weeks of her pregnancy. The female should be in good condition and weight. Do not allow her to become obese, as this can cause trouble during whelping. See the *Feeding and Watering* section for more information.

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Whelping and Raising Puppies

Whelping and Puppy Pen

Give the female all the advantages possible so she can produce a healthy litter. She needs an extra large house in which she can easily stand up and turn around. She should have extra room all around her when she lies down so she won't lie on the pups or be restricted during whelping. The whelping house should be equipped with a hinged or removable roof to make it easier to access the mother and pups.

Most litters should be planned to arrive during the spring, summer, or fall so the outside temperature is not too cold at birth. If you decide to have a winter litter you may have to plan on having the female inside a building where it is at least above freezing. Whelping can take place outside at lower temperatures but extreme caution should be used, especially with a female whelping for the first time or under exceptionally harsh weather conditions.

Summer temperatures above 70 degrees F can also be dangerous for the puppies. During their first few weeks of life puppies cannot regulate body temperature. Mosquitoes can also be a very serious problem for pups born during summer. You may need to whelp and raise the puppies inside your home if you have a bad mosquito problem or high temperatures.

Keep the whelping pen clean and dry. Many pathogens that are lethal to puppies are soil borne. In areas where such pathogens are known to exist it is important to maintain a level of pathogen protection. Basic hand-washing before and after handling both mother and offspring will go a long way. Other practices such as shoe and boot pans placed outside of the whelping pen will also prevent various pathogens infecting the puppies. A boot brush and a solution made up of one part household bleach and 10 parts water will reduce the risk of introducing soil borne organisms into the puppy pen.

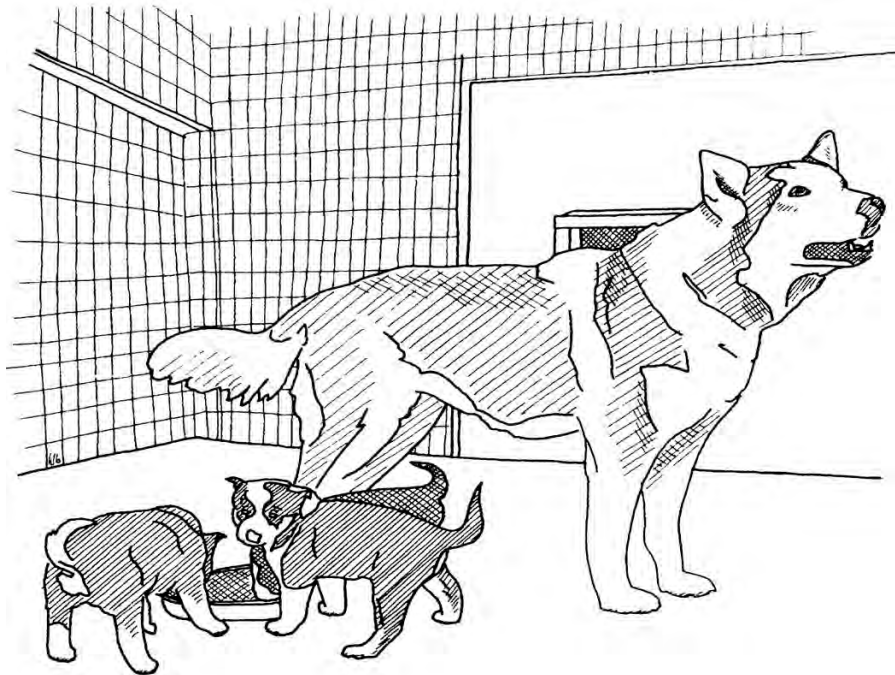
A whelping box provides a nest or den in which the female can whelp and begin raising her litter. It should be large enough to allow the female to stretch out without lying on or disturbing her puppies. It should be tall enough to contain the puppies but allow the mom to leave them when she desires.

The whelping box should be placed in a larger enclosure or pen either indoors or out as a primary containment for both the mom and puppies, but also isolate them from other dogs. Puppy pens should be a minimum of 100 square feet. If birds of prey might be a threat the pen should have a roof. Rawhide or hard rubber chews and balls are nice extras for the puppies' enjoyment. Ramps, tunnels and bridges provide mental stimulation. Be sure all additions are of sizes and made of materials that are safe for the puppies and mother.

The puppy pen needs to be cleaned at least once a day, or more often as needed. Whelping boxes need little bedding since the mother does most of the cleaning. If the puppies are reared in warmer months a smooth wooden floor will suffice.

Weaning and Feeding Puppies

Puppies should be offered gruel of ground and soaked kibble beginning at three weeks of age to supplement what they receive via nursing. Puppy food or a performance diet is recommended. Puppies under four months should be fed two to three times per day or free fed. Puppies should be fed enough to keep them fleshed out and to ensure they have enough energy to grow, but they should not be allowed to become obese. (See [Feeding and Watering section](#).)



Weaning

Mothers will usually wean puppies themselves when the pups are between four and a half and eight weeks old. If you wish to remove the mother from her pups at this time, you can. Ideally puppies should remain with the mother for the entire eight weeks. Emergency health situations may require early separation, but this should only be done under the direction of your veterinarian. If the pups are removed while the mother is still lactating the mother will need to be dried off. If the mammarys become firm, swollen, or red, consult your veterinarian.

Many mushers choose to leave the mother with the pups until the pups are individually tethered or penned. If the mother is still enjoying the pups and playing with them, this can be a good source of education for the puppies.

Reintroduce the mother to mushing slowly. She needs time to recover from nursing the pups. Short runs of 2 to 3 miles with the team are fine. Protect her enlarged nipples from cold weather for the whole season after whelping.

Puppies need to stay in the litter for at least eight weeks to ensure normal psychological development. During the fourth through sixth week, a puppy learns basic social behavior for dogs. If a puppy is removed from its family before six weeks it may have behavioral problems as an adult. When you rehome a pup, make sure you provide copies of all vaccination and deworming records to the new owner and caution him or her to change the pup's food slowly.

Puppy Health Care

Day 1: Examine each puppy for abnormalities. Check the mouth for cleft palate. Make sure that all puppies are nursing, as it is important for the puppies to receive the mother's colostrum, which flows for only a few days. If you have any questions or problems, call your veterinarian right away.

Day 2: Remove dewclaws, if there are any, from both the front and rear paws. This prevents trouble with booties later on and prevents the dewclaws from getting caught on something and damaged. Have your veterinarian perform this procedure, or have a veterinarian or an experienced musher show you how to do it yourself.

Three to four weeks: Deworm with the product recommended by your veterinarian. Continue deworming the puppies and mother on a schedule recommended by your veterinarian.

Eight weeks: Vaccinate with a combination vaccine that is recommended in your area. These may include distemper, parvovirus, and adenovirus. Work with your veterinarian to develop a vaccination program to meet the specific needs of your team.

Rearing

Just as children have formative years, puppies have formative months. Puppies need lots of human attention early. The more you put into your pups, the more you will get out of them as adults. Play with them at least a little every day so they don't become shy of people. Try to familiarize your puppies with as many different situations as possible by taking them on walks, bringing them inside, having children play with them, exposing them to crowds etc.

The most important time to develop a trusting, positive relationship with a pup is between its third to 16th weeks of life. Many people mistakenly believe that good genetics are all that are needed to produce a good sled dog. Without the proper care and training, a puppy with great potential can become a complete failure as a sled dog or pet. The following are some benchmarks in a puppy's development:

One to three weeks: During their first weeks of life, handle each puppy two or three times a day. Weigh them to ensure that no negative changes are occurring. Pet them and talk to them. Their relationships to humans can start from the day they are born.

Three to sixteen weeks: Introduce the puppies to as many unique experiences as possible. Between six and eight weeks is a particularly critical time for socialization with people. They can learn their names, learn to come when called, and develop a strong bond with humans during this stage of their development.

Four to six months: If tethering is the method of confinement this is the time to introduce them to this experience. Put a collar on each puppy and in the months following frequently check the collar's tightness and adjust it as the puppy grows. Place them on individual tethers.

Five to eight months: Harness training is most easily done during this stage of the puppies' development. Many methods are used: putting one or two pups in a small team with adults, or putting one adult leader with all the rest of the pups. Either way, the teams should be small (three to seven dogs), and the runs should be short (perhaps $\frac{1}{2}$ to 3 miles). It is best not to have a steep down slope, icy trail or open water on the puppies' first few runs in harness. It is easy to scare a puppy. The most important thing is to let the pups have fun. Mushing will be an important part of their lives and it should always be a positive experience. Puppies should never be dragged along or pulled by a machine; they should always be going forward on their own accord and have the option to stop if they become frightened or tired.

Some puppies will have a natural instinct to pull the first time they are harnessed. Other puppies will be overwhelmed by being tugged by the neck while at the same time running next to another dog. To avoid this, you might want to connect a pup and a reliable lead dog with a neckline and let them run around for a few minutes. Be sure to do this away from the dog yard to avoid tangles. Repeat the experience a few days before running the pup in the team. This helps a puppy to learn to jump over the ropes and accustoms it to the neckline. Be careful to match compatible dogs, and be ready to jump high when they come toward you at full speed.

Eight to twelve months: It is important to get the pups out often in harness so that they learn all the basics of mushing while they are young: not getting tangled in the traces, pulling hard, urinating and defecating on the run, not chewing harnesses and gang lines, how to cross ice and water, how to pull on hills, forward and whoa commands, how to pass other teams, and most importantly, to have fun with their owner out on the trail. All of these are easiest and best learned when they are young.

Twelve months: At this point, a dog has attained its basic size, although depending on the breed and genetic background, many dogs continue to fill out until about two and a half years old. Also remember that although a dog is one year old and looks mature, it is not mentally mature yet and still needs much more time to develop before it can be expected to behave and perform like an adult.

Everything you would like to teach your dog (in addition to mushing) is also best done at an early age. At 4 to 12 months, their minds are open and responsive. For example, if they will spend a lot of time inside as adults or if they need to be obedience trained, put in the effort training them while they are young and make each experience positive and educational.

Geriatric Dogs and End of Life Issues

When planning your mushing kennel it is important to recognize that, like humans, sled dogs grow old and eventually die. It is important that the musher have a plan for dealing with the special issues presented by geriatric dogs. As a dog matures beyond his or her physical prime you must decide whether to keep the dog for the remainder of his or her life, or find an appropriate new 'retirement' home for the dog.

Most sled dogs start showing physical signs of aging at around seven years of age, though there are plenty of exceptions to this rule of thumb. The first sign that many racing mushers see is that the dog is no longer able to run as fast or as far as his or her younger teammates. Competitive sled dog racers who do not want to support older, slower dogs should consider finding the dog a new home while it is still in good physical condition.

Gifts of older dogs in good physical condition are often greatly appreciated by junior mushers, beginners and mushers competing in less demanding disciplines or classes. You may also consider placing a retired dog into a home as a pet. Be sure the dog and the new owner are a good fit. Many sled dogs can be challenging pets; some have an instinct to roam or kill livestock and are often more independent than expected. Keep in mind that dogs that have been properly cared for and socialized have the best chances to be placed. Since a dog that is not good enough to keep is probably not good enough to breed, consider having the dog spayed or neutered before giving it away, or requiring that the new owner have the operation performed.

Many mushers prefer to keep their geriatric dogs and care for them until the end of their natural lives. Older dogs are especially valuable for helping train puppies and young dogs.

Housing Considerations for Geriatric Sled Dogs:

Older dogs often do not cope well with sudden changes in their environment. If you plan to keep your older dogs as house pets or change your confinement method, make the transition gradually, bringing the dog into the new environment for short visits and gradually increasing the amount of time until the dog becomes comfortable in his or her new setting.

Older dogs are often less tolerant of weather extremes than younger dogs. They may require additional bedding or even an insulated doghouse to be comfortable during cold weather. During warm weather, ensure that older dogs have easy access to shade and fresh, clean water.

Like younger dogs, geriatric dogs require adequate space and mental stimulation. (See the Dog Yard and Housing section.)

Feeding Considerations for Geriatric Sled Dogs:

As your aging dog's metabolic rate and general activity levels decrease, he or she will require less food to maintain a healthy body. Most older dogs will do well on the same ration you feed your younger dogs during the offseason. Occasionally a dog will have trouble digesting all the fat in this ration or may become constipated on it. If this occurs, try feeding a diet lower in fat or higher in fiber, respectively. It is important that you not allow your geriatric dog to get too fat. Obesity is the most common cause of major health problems in dogs, including kidney and liver diseases, diabetes and arthritis.

Monitor older dogs' weight just as you do younger dogs, and adjust the volume of feed accordingly. Consult a veterinarian if you have concerns or questions.

Health and Husbandry Issues of Geriatric Sled Dogs:

Geriatric dogs lose muscle mass and tone, long bones such as those in their legs become brittle, and arthritis frequently sets in. Providing your geriatric sled dogs opportunities for frequent short, slower runs with other older dogs or with puppy teams can help the geriatric dogs maintain a higher degree of flexibility, mobility and fun as they age.

Geriatric dogs are more prone to infectious and chronic diseases than young dogs. Work with your veterinarian to determine an appropriate vaccination schedule for your older dogs and consult with him or her if you notice any changes in the dog's behavior, activity level or appearance. Be especially alert for any of the following signs of disease in geriatric sled dogs:

- Sustained significant increase in water consumption or urination
- Weight loss.
- Significant decrease in appetite or failure to eat for more than two consecutive days.
- Significant increase in appetite
- Repeated vomiting
- Diarrhea that persists more than two days.
- Lameness that lasts for more than three or four days.
- Lumps or masses in or under the skin,
- Open sores or multiple scabs in the skin, especially if they seem to be getting larger or worse.
- Hair loss, especially if accompanied by scratching.
- Persistent coughing or gagging
- Excessive panting
- Sudden collapse or weakness
- Inability to chew dry food
- Seizures, convulsions or sudden changes in behavior

Many of the diseases associated with aging can be easily diagnosed and treated, providing comfort in the dog's senior years.

End of Life Considerations for Sled Dogs:

Injuries and illnesses can threaten dogs quality of life. Whether your dog is injured during the prime of life or debilitated due to the diseases of old age you may have to decide whether or not to euthanize your dog.

Animal care experts agree that it is appropriate to humanely kill a dog rather than to prolong suffering. There are no hard and fast rules regarding when it is or is not appropriate to do so. Here are some considerations you can use to help make your own decision:

- Is professional veterinary care available in your community?
- Can you afford to pay for the necessary veterinary care?
- How likely is your dog to recover from the problem?
- Is your dog in pain? If so, can the pain be effectively controlled?
- Is your dog able to eat and digest enough food to remain properly nourished?
- Is your dog mobile enough to move around its housing area?
- Is your dog able to breathe without difficulty?
- Does your dog behave as though it still enjoys living?

Once you have considered the above, establish a euthanasia baseline condition. These are best established before the animal reaches the euthanasia threshold. It is much easier to establish these before human emotion becomes the deciding factor. It can be stated as simply as: When the dog is not longer able to..., then we will euthanize it. It is very easy to change this threshold as a dog approaches it. Experience has shown that as one "quality of life" measurement goes by, another threshold is established and so on. When this happens, it is only avoiding the inevitable.

Whenever possible, animal control shelters or veterinarians should be used to perform euthanasia as necessary. In isolated rural areas where such facilities are not available you must still make sure your dog is killed humanely, with no suffering. Consult a veterinarian or animal control officer for advice.

In some regions local or state/provincial laws or regulations regulate body disposal. Many veterinarians and animal control shelters can cremate the body for you at little or no cost. If the law permits and you wish to bury your dog's body at your home or kennel it is recommended you place the body in a heavy duty plastic bag encased in a secure receptacle such as a wooden or metal box. You should bury the body under at least 3 ft of earth to prevent other animals from digging at the gravesite.

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Additional recommended sources of information on sled dogs and mushing equipment include mushing workshops and seminars, sled dog symposiums and trade fairs, and listservs such as *Sled Dog Central* (<http://www.sleddogcentral.com>).

Appendix

Courtesy Nestlé Purina


Nestlé PURINA

BODY CONDITION SYSTEM

| | | | |
|------------------|--|---|--|
| TOO THIN | 1 Ribs, lumbar vertebrae, pelvic bones and all bony prominences evident from a distance. No discernible body fat. Obvious loss of muscle mass. | 
 1 | |
| | 2 Ribs, lumbar vertebrae and pelvic bones easily visible. No palpable fat. Some evidence of other bony prominence. Minimal loss of muscle mass. | | 
 3 |
| | 3 Ribs easily palpated and may be visible with no palpable fat. Tops of lumbar vertebrae visible. Pelvic bones becoming prominent. Obvious waist and abdominal tuck. | | |
| IDEAL | 4 Ribs easily palpable, with minimal fat covering. Waist easily noted, viewed from above. Abdominal tuck evident. | 
 5 | |
| | 5 Ribs palpable without excess fat covering. Waist observed behind ribs when viewed from above. Abdomen tucked up when viewed from side. | | |
| TOO HEAVY | 6 Ribs palpable with slight excess fat covering. Waist is discernible viewed from above but is not prominent. Abdominal tuck apparent. | 
 7 | |
| | 7 Ribs palpable with difficulty; heavy fat cover. Noticeable fat deposits over lumbar area and base of tail. Waist absent or barely visible. Abdominal tuck may be present. | | |
| | 8 Ribs not palpable under very heavy fat cover, or palpable only with significant pressure. Heavy fat deposits over lumbar area and base of tail. Waist absent. No abdominal tuck. Obvious abdominal distention may be present. | | |
| | 9 Massive fat deposits over thorax, spine and base of tail. Waist and abdominal tuck absent. Fat deposits on neck and limbs. Obvious abdominal distention. | | |

The BODY CONDITION SYSTEM was developed at the Nestlé Purina Pet Care Center and has been validated as documented in the following publications:

Marley D, Barlges JW, Mayers T, et al. Comparison of body fat estimates by dual-energy x-ray absorptiometry and deuterium oxide dilution in client owned dogs. *Compendium* 2001; 23 (9A): 70

Lafontaine DP. Development and Validation of a Body Condition Score System for Dogs. *Canine Practice* July/August 1997; 22:10-15

Keady, et al. Effects of Diet Restriction on Life Span and Age-Related Changes in Dogs. *JAVMA* 2002; 220:1315-1320

Call 1-800-222-VET5 (8387), weekdays, 8:00 a.m. to 4:30 p.m. CT


Nestlé PURINA

Ravenwood Veterinary Clinic 2015 Kennel Management

I. Record Keeping

1. Medical
 - a. Vaccination chart
 - b. Deworming chart
 - c. Female heat cycle chart
 - d. Individual - lameness, major surgery, etc.
2. Pedigrees on each animal
3. Mileage log in training
4. "Weekly Minder" - to keep track of breeding, weaning, when dogs are sick, etc. The above is a list we organized from experience. This insures you have the information, in writing, that your veterinarian needs to help you trouble shoot specific problems.

II. Vaccinations

1. Miscellaneous notes on vaccinations:
 - a. Dogs can be vaccinated after training for the day, but not immediately before running. Body temperatures over 104° F can inactivate the vaccines before they are absorbed.
 - b. Do not vaccinate pregnant females. Little information is available on the affects of modified live vaccines on fetal development.
 - c. Booster vaccines should not be given any closer than 10 days apart. The body's normal immune response can inactivate the second vaccine during this period.
 - d. I prefer you vaccinate in the fall since the dogs have been resting and their immune systems are at their peak. This will ensure highest vaccine titers during the winter months.
 - e. There are a variety of thoughts on the vaccine frontier and each year we receive more information. Most of the upheaval in vaccine considerations has come from speculation that cats who are genetically predisposed to a type of subcutaneous tumor may be stimulated to grow these tumors in chronic vaccine reaction sites. There has not been any evidence of this in dogs; however, this has prompted the universities to review the vaccine protocols. The heart of the confusion is vaccine companies are only required to show that their vaccines will protect a majority of vaccinated dogs for the time interval they state on their vaccines. This means no one really knows how long the vaccines will protect the individual animal and most company's only license for 12 months. Here at RVC we follow the guidelines laid out by the AAHA (American Animal Hospital Association) vaccine committee in program for core vaccine recommendations. These are reviewed by a panel of specialist every other year.
2. Competitive dogs, like competitive human runners, are training to a level that creates some immunosuppression. This means that the dogs are more susceptible to "bugs" and infections. It also means that an individual dog may not have as high a response to the vaccine as expected. (Thus, the reason for vaccinating in the fall before heavy training starts!). Because our sled dogs are exposed to a lot of different animals from all over the country the risk of exposure to these viruses is high. At this point in time the American Animal Hospital Association core vaccine guidelines recommend three year programs for dogs that have had a

solid initial series of vaccines. One exception would be breeding females. Since you want high antibody titers in the milk from the female, they should stay on annual combinations. Older sled dogs that are not traveling as much are good three year candidates and many events and kennels are moving to the three year recommendation for racing dogs as well. You will want to make sure you have maximized your dogs' health in other areas if you are going to the three year program.

3. On a similar line of thinking, we want to be sure our puppies are protected, but we don't want to add stress by over-vaccinating (i.e. you can't overcome parasite, poor feeding, or management problems by giving more vaccines). In light of the newer, low passage-high titer Parvo vaccines we are making the following recommendations:

Puppies (assuming dam is current on her vaccines at the time of breeding)

6 weeks - DA₂PV-CPV (Combo distemper/parvo/adenovirus)

10 weeks - DA₂PV-CPV

14 weeks - DA₂PV-CPV

Have a veterinarian vaccinate against rabies anytime after 12 weeks, as required by state and local laws.

4. Adults

- a. DA₂PV-CPV - annually, preferably in August or September, unless you want to go to the three year program after 2 years of age (puppy series and one annual).
- b. Rabies (administered by a veterinarian) - initially after 12 weeks of age, and repeated one year later. Subsequent rabies vaccinations are good for three years in the State of Alaska.
- c. Bordetella (Kennel Cough) - I now recommend the oral vaccine. To maximize protection, give in November and again two weeks before your major race. Do not run the dog for 24 hours after the oral vaccine. Some people have given the injectable vaccine two weeks after the intranasal for added protection. The injectable vaccine needs to be boosted two weeks after the first injection. If an outbreak of any respiratory infection occurs in your area, it helps to booster your non-symptomatic dogs immediately with the oral. At best, this respiratory vaccine is about 20% effective.
- d. Influenza vaccine – as of 6/2015 there has not been a documented case of canine influenza in the State of Alaska. Considering our high rate of traveling dogs, it is only a matter of time. This vaccine needs a primary and then 2 to 6 weeks later a booster to be protective. At this time we are only giving these vaccines on special request.

III. Worm Control

1. Types of worms:
 - a. "Rounds" (toxocara species, hooks, whips) – fecal-oral transmission, or nursing on infected mothers.
 - b. Tapeworms (Taenia, Echinococcus)- from fleas, raw meat, voles, and fish
2. Damage caused by:
 - a. Migration of larva through the liver and lungs causing scarring
 - b. Competition for calories in small intestines (creates a thin, unkempt appearance)
 - c. Indigestion from presence in small intestines creates stress diarrhea
3. Open sores in intestine allows other viral, bacterial, and parasitic agents

into the body.

4. Humans can be infected with "round" and tapeworm parasites. Small children are especially at risk from diminished hygiene awareness. There have been cases of migrating roundworm larva causing blindness in humans. Transmission is through ingesting eggs (fecal/oral) or picking up larva while walking barefoot. Both can be prevented with good hygiene.

5. Prevention:

- a. Deworm puppies with a thorough schedule.
- b. Deworm entire kennel regularly.
- c. Have stool samples checked at least once a year. We recommend samples representing 20% of your kennel at a time i.e., 0 to 10 dogs - 2 samples, 10 to 20 - 4 samples, etc.
- d. CLEAN YARD AT LEAST ONCE A DAY to prevent larvae and eggs from becoming infective.
- e. Don't feed raw game meat.
- f. Move where there aren't any fleas.
- g. Get ready to deal with ticks. Ticks have been documented over wintering in both the Kenai and Fairbanks areas.

6. Types of dewormers:

- a. Fenbendazole (Panacur) - kills some migrating larvae as well as adult worms, effective against rounds, whips, hooks, and certain tapes (not the flea tapes), **MUST** be given daily for three days in a row. While not FDA approved for giardia, it does appear to show promise in treating giardia infections if given for 5 or more consecutive days.
- b. Piperazine (numerous brand names) - only kills roundworms, one dose, repeat in three weeks.
- c. Pyrantel Pamoate (Nemex, Strongid T) - kills only rounds and hookworms, one dose, repeat in three weeks.
- d. Praziquantel (Droncit) or Epsiprantel (Cestex) - kills only tapeworms. This drug is more toxic than previously mentioned dewormers. Research has not established use in pregnant or breeding animals.
- e. Praziquantel/Febantel/Pyrantel (Drontal-Plus)- a broad-spectrum product that eliminates roundworm species and both species of tapeworms in a single dose tablet.
- f. You should check with your veterinarian on ages of animals and doses of all dewormers since some can be toxic.
- g. Many of the spot-on treatments for fleas/ticks/lice/heartworm will treat some internal parasites as well.

7. Puppies:

Panacur to dam (for best elimination the specialists recommended every day from day 45 to day 63. There is evidence that treating day 45, 46, 47 will help decrease the overall amount of transplacental spread of the feti).

- a. Panacur to dam at day 45, 46, 47 of pregnancy to whelping.
- b. Panacur at two weeks of age (deworm dam each time as long as she is with pups)
- c. Pyrantel P. at five weeks of age

- d. Pyrantel P. at seven weeks of age
- e. Pyrantel P. at nine weeks of age
- f. Pyrantel P. at 11 weeks of age
- g. Pyrantel P. at 13 weeks of age
- h. Pyrantel P. at 15 weeks of age
- i. Panacur three consecutive days when put out of puppy pens
- j. Fecal tests should be run at three, six, and 21 weeks of age to evaluate your deworming success and check for other parasites such as coccidia and giardia.

7. Adults

- a. *Small yards* - one to four dogs with no litters on the ground. Treat once a year with 3 days of Panacur or one dose of Pyrantel P followed by a second dose three weeks later. Run fecal tests every 6 months. We recommend samples representing 20% of your kennel at a time i.e., 0 to 10 dogs - 2 samples, 10 to 20 - 4 samples, etc.
- b. *Medium and Large Yards* - five or more dogs with no more than one litter per year, or any kennel that is racing at the local tracks. Treat all dogs three times a year (September, January, and March). Panacur should be one given at one of those times and Pyrantel P. the other two. Run fecal tests every 6 months. We recommend samples representing 20% of your kennel at a time i.e., 0 to 10 dogs - 2 samples, 10 to 20 - 4 samples, etc.
- c. *Other situations* - five or more dogs with multiple litters each year, and any kennel that is racing at both local and state wide tracks. Treat five times a year (April, August, October, December and February). Again, at least one treatment with Panacur and the others with Pyrantel P. Run fecal tests every 6 months. We recommend samples representing 20% of your kennel at a time i.e., 0 to 10 dogs - 2 samples, 10 to 20 - 4 samples, etc.
- d. *Mega yards* - lots of dogs and lots of puppies. Given the high, essentially uncontrollable incidence of infection in the soil, it is best to deworm all dogs once a month. Deworm at least twice with Panacur and the rest with Pyrantel P.
- e. Consider Drontal-Plus deworming once a year in any kennel feeding a meat diet. Though higher in price per dog, this product effectively kills all species of tapeworms in one dose, including Echinococcus. Iditarod requires all competitors to deworm with Drontal-Plus due to the potential for human infection from this parasite. Kennels north of Talkeetna and anyone feeding raw game meat should consider annual tape deworming for Enchinococcus.

IV. Other Parasites to Watch for:

Coccidia, giardia, lice, mites, ticks, fleas, heartworm.

V. Nutrition

Rules of thumb

- a. Use a proven diet (one that someone else has been using for a while with success) or go straight commercial to prevent nutritional deficiencies or toxicities.
- b. If formulating your own diet, work with a veterinarian or a nutritionist.
- c. Make any changes in diet gradually, change only one ingredient at a time, and give 4 to 6 weeks to see any difference (skin and coat can take up to eight weeks to show the effects of diet changes).
- d. If feeding more than 25% meat (this is straight muscle meat, not necessarily commercial meat mixtures) in a diet, be sure to calculate the calcium: phosphorus

ratio (preferably 2 calcium to 1 phosphorus). Ensure that the levels are within a safe range for the dogs' age by consulting with the veterinary clinic or finding an AAFCO (American Animal Feed Control Official) table for the daily requirements.

- e. Puppies (dogs ten months and younger) should be on straight commercial puppy food. We have found a number of race diets are too "hot" and the calcium: phosphorus ratio is not balanced for proper puppy bone growth.
- f. You will get into trouble by heavily supplementing with any one vitamin or mineral. If you want to supplement, get hold of a current NRC table and calculate how much is in the total diet.
- g. Look for articles written by Dr. Arleigh Reynolds on sled dog nutrition.
- h. Because vitamins/supplements are not regulated by FDA you have to be cautious of your source. One recent study found out of 10 samples of glucosamine tested only 3 actually had what and the amount stated on the packaging. The following is an independent laboratory that you can join and have access to products that have been tested and proven to have what they label. ConsumerLab.com

VI. Miscellaneous

1. If you do have a fatality with a puppy, I highly recommend you aggressively pursue an accurate answer. The cost of a necropsy with viral, parasitic, bacterial, and histological exams may appear expensive at the time; however, it won't be nearly as expensive as losing all of your pups. Owners are often surprised by how correctable the problem can be.
2. Nails - trim nails before the start of cart training to prevent breaks. Leave them sufficiently long to account for hard surface wear. Once the snow falls, check nails the first of each month you are running. Do not trim nails the day before a major race! If your race surface includes snow on pavement, do not leave the nails too long but do not cut them any closer than two weeks to the event.
3. Safety - always carry a knife, extra double snaps, and common sense when running dogs. Be sure you can safely carry a dog in your sled. Remember, you put the dog in the situation and it is up to you to safely get him out.
4. Ask a lot of questions all the time!
5. Don't cuss out your veterinarian if you don't follow their instructions.
6. "The one important thing I have learned over the years is the difference between taking one's work seriously and taking one's self seriously. The first is imperative and the second is disastrous." – Margot Fonteyn (1919-1991).

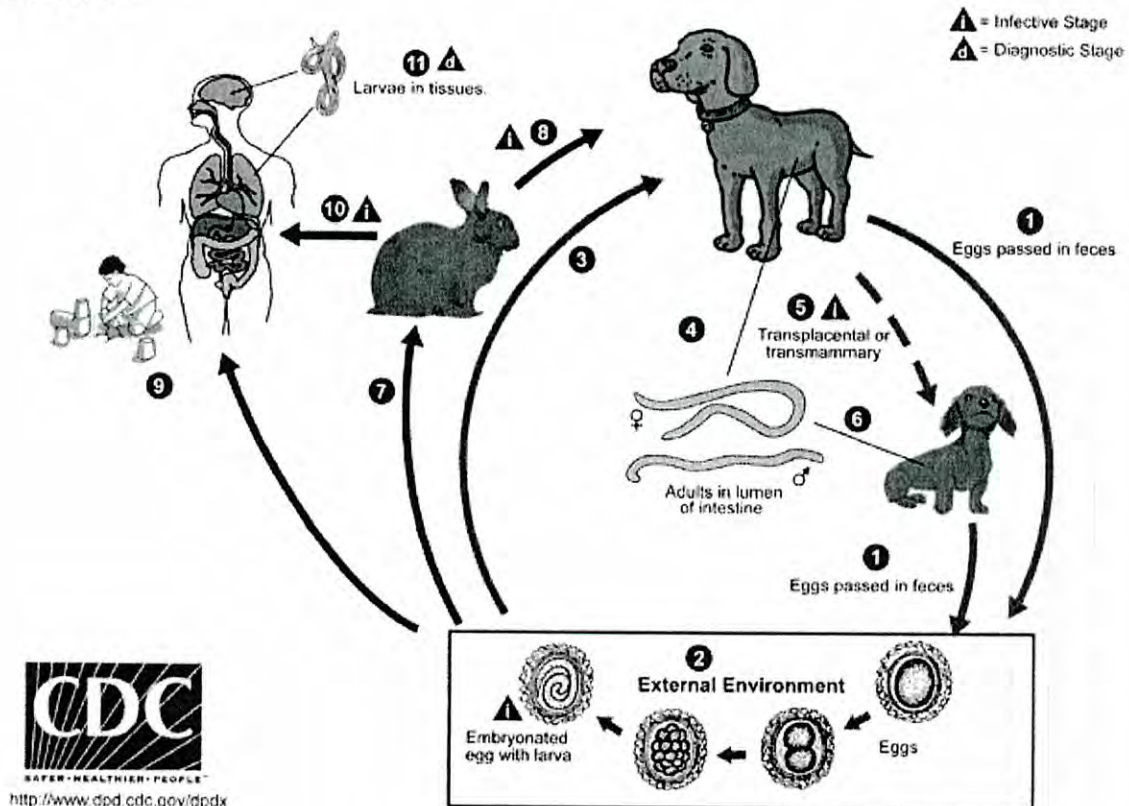
Toxocara canis – Life Cycle and how it transmits in the environment (and to people).

Biology

Causal Agents:

Toxocariasis is caused by larvae of *Toxocara canis* (dog roundworm) and less frequently of *T. cati* (cat roundworm), two nematode parasites of animals.

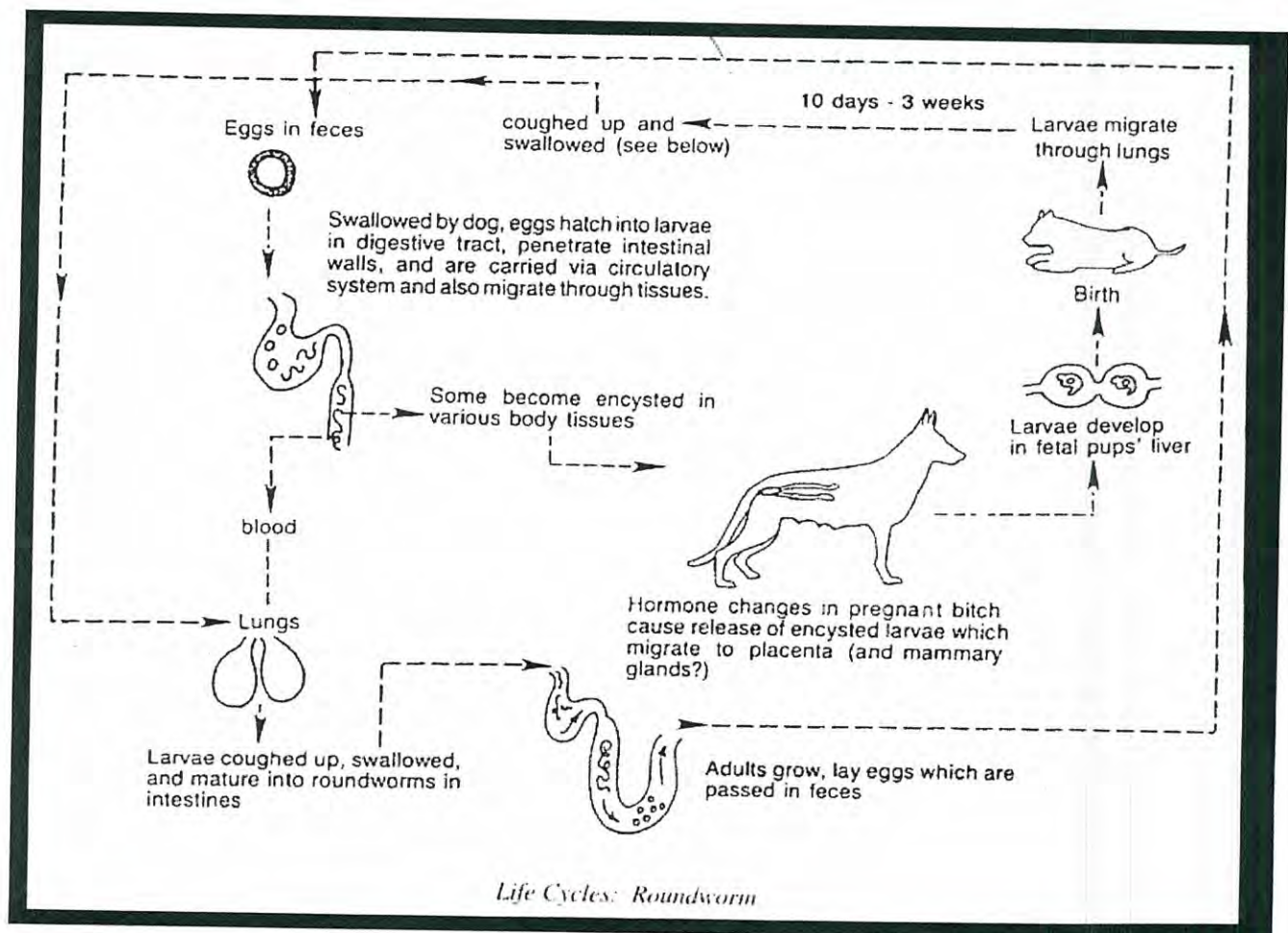
Life Cycle:



<http://www.dpd.cdc.gov/dpdx>

Toxacara canis life cycle in the body of the dog:

With a heavy roundworm parasite load it is possible for a three month old puppy to be suffering from liver, lung, and intestinal damage. Not to mention all the encysted parasites that could end up in the muscle tissue, which is suppose to be moving the dog!



Dog Mushers' Emergency Kit Continued

| Use | Administration | Instructions |
|--|----------------|---|
| Eye injuries | For eyes only | A bland trithalamic Eye Ointment - Apply to injured eye every 4 hours until examined by a Veterinarian. |
| Ear infections | For ears only | Choose a mild ear or otic cleanser. Epi-Otic advanced is the only one approved in ears with ruptured ear drums. |
| Mild stress diarrhea, or after eating sharp objects or garbage | Oral | Metamucil (plain) - 1 tsp. powder or 1 wafer per 30 lbs of dog. (Or pumpkin or beet pulp.) Mix powder with moist food or cottage cheese. Give 4 to 6 hours before a run to help with chronic stress diarrhea. |

- ◆ Minimize all licking of ointments and creams by the dog. Many foot ointments contain zinc oxide, which if ingested, in volume and/or over a period can cause serious medical problems.

Bandage Material and Other Useful Items to Have on Hand

- ◆ Telfa pads - apply over open or draining wounds before bandaging
- ◆ Gauze brown cling - use to hold telfa pad in place
- ◆ Vet wrap - to support or protect affected areas
- ◆ Neoprene wraps with "mineral ice"
- ◆ Dental acrylic with gel caps or "Soft Paws" nail caps
(Sterilize nail tip before applying and always remove immediately after race)
- ◆ Bandage scissors
- ◆ Nail trimmers
- ◆ At least one good leash
- ◆ Muzzle
- ◆ Extra booties

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Ravenwood Veterinary Clinic PC
 16743 Coronado Rd
 Eagle River, AK 99577

2015
Controlling Stress Diarrhea in Working Dogs

1. A complete and thorough parasite control program should be in place to help rule out parasitism. Even so, multiple fecal samples may be needed to detect parasites. A good sample – fresh not frozen- centrifuge method for the float, a giardia snap tests and cytology are the basics to evaluate the situation.
2. A thorough and accurate history of the diet is needed to rule out its involvement in the situation. Not what you meant to do but what actually happened with the diet and changes.

If the history clearly demonstrates that this particular animal only has diarrhea on or immediately after a run and all later bowel movements are normal, this can be initially treated as stress or exercise induced diarrhea. Stress diarrhea is not associated with vomiting or lack of appetite but often worsens as the ambient working temperature increases.

The following is a simple outline we use:

Try to feed these dogs 24 hours before they run to minimize stool in colon when they exercise.

Add one tsp of plain Metamucil per 30# of dog to the broth on the day of exercise. Do not add the Metamucil to each feeding and brothing. Beet pulp may have a similar beneficial affect.

Give a probiotic daily while the patient is running. Culterelle is a tested and proven probiotic. Be cautions when using supplement/vitamin products – these are not FDA regulated.

If the above does not work, a fecal stain for Clostridial overgrowth may be indicated. In addition there are several tests that are specialty tests that have to be sent to an outside lab. These tests often have very specific requirements to be accurate.

Often working dogs will be more predisposed to stress diarrhea after they have recovered from a "gut" bug. The intestinal lining can take up to 21 days (or longer in chronic cases) to fully recover.

Psyllium powder (water soluble fiber) on husks, Prostora (probiotic) are the gold standard created specifically for dogs.

Louse Infestations

Louse infestation (pediculosis) is usually seen during the winter or on young puppies. The rate of infestation is much higher in larger kennels, more mobile populations, or in severely debilitated animals.

Transmission: Lice are very species specific and will only live and feed on their correct host. The transmission must be direct, hair-to-hair. If off the intended host, lice will die within a few hours. Some dogs can be carriers. This means there are just a few lice that do not cause any symptoms on that host, but can infest other dogs that come in contact with this particular dog.

Lousy facts: There are two types, chewing and sucking lice. The chewing variety cause the dog to itch a little more and the sucking type can cause anemia if in large enough numbers. A louse-infested dog can have anything from a normal coat to being bald with open sores and a greasy, smelly coat. The animals are usually "itchy". Lice spend their entire time on the dog, making them much easier to eradicate than other insect parasites. From the time an egg is laid until the louse is an adult is 14 to 21 days.

Diagnosis: The best way to diagnose lice infestations is to look around the head, shoulders, and chest. Puppies that have "lots of scratches from playing with other puppies" should be closely examined. Lice can commonly be seen running across the bridge of the nose where the hair is very thin. They are much slower moving than fleas. If a litter of puppies "comes down" with lice, the female is infested and you have lice in your dog yard. Don't argue, it just is.

Treatment: Because lice cannot live off the host and they are sensitive to most insecticides, lice are easy to eliminate. Currently, there are newer flea products on the market that only require topical applications. Not all are FDA approved for lice so you may need to consult your veterinarian. (They may even repel mosquitoes!) Be sure to read the labels on these products as far as pregnant females and puppies are concerned. Follow the recommendations on the packaging for repeat treatments. To eliminate lice from your kennel you should treat all dogs and clean and disinfect all housing. Did you forget the dog truck?

Ravenwood Veterinary Clinic 2015

Ravenwood Veterinary Clinic 2015

[illegible]

Ravenwood Veterinary Clinic Kennel Management 2015

Heat Cycle Monitoring Chart

[illegible]

Ravenwood Veterinary Clinic

Whelping Information Sheet

When is my female's due date?

Hopefully, the breeding dates were accurately noted. The normal gestation for a dog is between 58 and 65 days. More accurately it is 59 days from the day the ovaries shed the eggs. If she was bred every other day and you have the first date she refused the male, you can call this the 59th day. Since, however, this is not easily identified and the majority of females ovulate at the end of their fertile week (the middle week), we pick 63 days from the middle breeding.

How do I know if my dog is pregnant?

A majority of the time the embryos can be palpated between 21 and 35 days of pregnancy. Before this they are too small and after this the fluid in the uterus connects together and makes palpation unreliable.

Sonograms can be done from day 28 on and give us an accurate diagnosis of pregnancy as well as a fair indication of number. The whole procedure takes 5 to 10 minutes, but may require a small square of hair to be shaved.

A blood test called Relaxin can be run; it is most accurate if the sample is drawn 30 days after the last breeding. This test is sent out to our lab in Seattle and takes one to three days to get results back.

The bones of the fetuses will start taking calcium up after day 43 of pregnancy. We can X-ray the female after 45 days and give you a fair count of the puppies. If there are five or more feti, however, even x-rays will give variable numbers.

How will my dog act while she is pregnant?

The majority of females will not show any change in behavior. Some may have a clear mucus discharge from the vulva. The important point is that the discharge is clear! Some dogs may have some mild vomiting, equivalent to morning sickness in women. And a few may try to worry you with a decrease in appetite, the important note here is to watch their weight and be sure they do not lose weight. If they are vomiting it is a good idea to monitor their temperature for a few day and have them examined by your veterinarian to make sure a more serious condition is not occurring.

What should my dog be fed while she is pregnant?

This is the area where the most mistakes are made. Your dog should not be bred until her weight is perfect. Nor should you see any signs of weight gain before 30 days of pregnancy. The dog's diet should be strictly commercial dog food, with a high digestibility. High protein type foods for lactating females do not need to be started until the last 2 to 3 weeks of pregnancy. I recommend, gradually over the last three weeks of pregnancy, weaning the female onto the puppy food you plan to feed the puppies. I do not recommend feeding meat mixture during pregnancy. There are a growing number of bacteria/parasites being detected in raw meat diets and pregnant females are a prime target. The important thing to remember is that we are putting her on a high quality diet, but we do not want her to get fat!

The dog's body will not require extra energy until a slow increase starts three weeks from whelping. It does not take much energy to make puppies! A good diet before whelping will play a key roll in preventing milk fever in the bitch and ensure healthy puppies. Never supplement a pregnant dog with one particular mineral or vitamin!! Ratio and balances are essential for normal fetal development.

How should I prepare for the puppies?

If she is having the puppies indoors, 2 to 3 weeks before the due date you should construct a whelping box. There are numerous designs. Mainly, be sure she can stretch out in either direction and that the sides are at least as high as her shoulders. A thick layer of old newspaper makes for good and easy to clean bedding. Indoor/outdoor carpeting, cut to match the box floor, is a great surface for nursing pups. Be prepared to change it every other day until the pups can leave the box.

If she is going to whelp outside, be sure her doghouse is long enough for her to lie flat at least along one of the sides. Be sure you have the fence constructed so she cannot get out, since you will have to remove the chain. Put fresh straw in a week before she is expecting.

When will we see her milk let down?

In first-litter females the milk usually does not drop until 24 to 48 hours before whelping. In females that have had previous litters this can occur up to 10 days before the actual delivery.

How can I tell when it is "time"?

Most female's body temperature will drop 48 to 24 hours before whelping and then return to normal. This, however, can last as short as 12 hours. If you want to monitor this, you should plan on taking the temperature every 12 hours from the first 63-day count. Normal body temperature for a dog is from 100.0° F to 102.5° F. It can drop at this stage to as low as 98.0° F.

Stage 1: This stage can start as early as 24 hours before actual delivery. The mother will usually become restless, pace, rearrange the "nest" constantly, and may not be interested in food.

Stage 2: This stage is the state where the actual contractions start. The female starts straining and pushing with her abdominal muscles. Within 20 minutes of actual contractions she should deliver a pup. First thing visible will be the "water sac" (amniotic sac). This will usually break as the puppy is expelled. Puppies are easily born head or feet first. Once in the birth canal the puppy is usually quickly expelled and the mother starts immediately to remove the membranes and stimulate it to breath. The female usually chews the umbilical cord and cleans the puppy.

Stage 3: This is when she passes the placenta and the uterus contracts to clean itself out. Most females actually eat the placentas as they come down the vaginal canal. She may have two puppies before she passes two placentas. Normally, the puppies will come within 15 minutes to an hour between each other. After the last puppy, the female should settle down and nurse quietly. Offer her water or broth in the box often. After 24 hours the appetite should return for the female, but it is good idea to leave lots of fresh water close to the box.

What are danger signs I should watch for?

1. If the female goes beyond 63 days from the last breeding date.
 2. If you notice a vaginal discharge which is "milky", yellow or green in color.
 3. If the female has been having contractions for over an hour with no results.
 4. If you see the "water break" and do not have a puppy within 30 minutes.
 5. If the female is contracting and it has been longer than 2 hours between puppies.
 6. When the whelping is over, if the puppies do not feed and sleep contentedly. Healthy puppies will be relatively quiet and have plump abdomens.
 7. If the mother does not clean the membranes off of the puppy as soon as it is born, you will have to. Clean the puppy off with a dry towel. Do not be afraid to rub briskly to induce breathing.
 8. If the mother does not chew the umbilical cord, DO NOT PULL! Tie off the cord with a little thread, 2 to 3 inches from the body and cut with a pair of scissors. It is best to dip the cord attached to the puppy in iodine.
 9. The mammary glands on the female will change in size as the pups nurse. You should always be able to express some milk from all glands. If you cannot or if the gland is hard, or red, she needs to be seen immediately.
- It is a good idea to have the female and puppies examined 24 to 48 hours after whelping to check their well-being.
 - If you want tails docked or dewclaws removed, then this needs to be done 3 to 5 days after whelping.
 - The female may have a bloody/greenish discharge for up to 6 weeks after having the puppies.
 - Most females can have plain dry commercial puppy food left in their pens for free choice feeding if this is started while they are whelping.
 - It is not recommended to change diets once the puppies are born, until after weaning.

Ravenwood Veterinary Clinic Kennel Management 2015

Litter Management Chart

(Year and Dam for title)

Dam _____ Sire _____ Whelping Date _____

Number in Litter _____ Number of females _____ Number of males _____

| <u>Date due</u> | <u>Date done</u> | <u>Timing</u> | <u>Activity</u> |
|-----------------|----------------------|------------------|--|
| _____ | <input type="text"/> | 45 days pregnant | Panacur dam |
| _____ | <input type="text"/> | 3 to 5 days old | Dewclaw removal on pups |
| _____ | <input type="text"/> | 2 weeks old | Panacur dam and pups |
| _____ | <input type="text"/> | 3 weeks old | Fecal _____ |
| _____ | <input type="text"/> | 5 weeks old | Pyrantel P. pups and dam |
| _____ | <input type="text"/> | 6 weeks old | DA ₂ PV-CPV-CV vaccination |
| _____ | <input type="text"/> | | Fecal _____ |
| _____ | <input type="text"/> | 7 weeks old | Pyrantel P. pups and dam |
| _____ | <input type="text"/> | 9 weeks old | Pyrantel P. pups and dam |
| _____ | <input type="text"/> | 10 weeks old | DA ₂ PV-CPV-CV vaccination |
| _____ | <input type="text"/> | 11 weeks old | Pyrantel P. pups and dam |
| _____ | <input type="text"/> | 13 weeks old | Pyrantel P. pups |
| _____ | <input type="text"/> | 14 weeks old | DA ₂ PV-CPV-CV Rabies vaccs |
| _____ | <input type="text"/> | 15 weeks old | Pyrantel P. pups |
| _____ | <input type="text"/> | 21 weeks old | Fecal _____ |

Date weaning complete _____ Panacur three days

Date put out _____

Any pups lost by six months? _____ Why? _____

Ravenwood Veterinary Clinic
Kennel Management 2015

Puppy Weight Chart

(Year and dam for title)

(Whelping date)

| Description | Birth | 2 weeks | 4 weeks | 6 weeks | 8 weeks | 12 weeks | 6 months |
|-------------|-------|---------|---------|---------|---------|----------|----------|
| 1 | | | | | | | |
| 2 | | | | | | | |
| 3 | | | | | | | |
| 4 | | | | | | | |
| 5 | | | | | | | |
| 6 | | | | | | | |
| 7 | | | | | | | |
| 8 | | | | | | | |
| 9 | | | | | | | |
| 10 | | | | | | | |

“Wheezer” – Disease (Laryngeal Paralysis)



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Laryngeal Paralysis (“Wheezer-Disease”)

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Laryngeal paralysis (LarPar) is a dysfunction of the nerves which control opening and closing of the larynx (voicebox). With this disorder, the upper airway fails to open adequately during inhalation. LarPar occurs in the Alaskan Husky. These dogs are frequently called “Wheezers” because of the typical breathing sound. The causes, signs, and treatment of LarPar are described in this chapter.

GENERAL INFORMATION AND CLINICAL SIGNS

The larynx is located at the beginning of the trachea (wind pipe). It controls airflow, voice production, enables effective coughing, and facilitates increased abdominal pressure during whelping and defecation. Specific nerves control the muscles of the larynx. With LarPar, the larynx cannot be opened or closed appropriately, secondary to nerve dysfunction. This results in inadequate oxygen supply and development of respiratory distress. It may predispose affected animals to aspiration pneumonia.^{1,2}

Clinical signs include exercise intolerance, stridor, cough, hoarse bark, choking/gagging, inspiratory dyspnea (difficulties breathing), cyanosis (blue mucous membranes), and sometimes collapse.

LarPar can be congenital (the animal is born with it), inherited (genetic defect, transmitted from one or both parents), or acquired. The acquired form can be due to trauma to the nerves (bite wounds to the neck), surgical disruption of nerves (cardiac or tracheal surgery), secondary to polyneuropathy (generalized nerve disorder), or idiopathic (no identified cause). Alaskan Huskies can be born with LarPar (“Wheezers”). It is currently suspected that this type of LarPar is an inherited condition, but a study evaluating a genetic defect in this breed could never be completed.⁴ Most affected dogs are described to have a black and/or white hair coat, “freckles”, blue eyes, and hard oral mucosal tags (Figures 1,2A,2B).



Figures 1,2A,2B: Typical appearance of Alaskan Huskies, possibly carrying the gene for LarPar (“Wheezer”): Black and/or white hair coat, “freckles”, blue eyes, and hard oral mucosal tags.

Severely affected “Wheezers” are reported to “run about 100 yards, then collapse gasping and turning blue”.⁴ They may be more prone to overheating.⁴ Apparently, some “Wheezers” compensate with age. Indeed, some Iditarod finishers were “Wheezers” at a young age.⁵ The reason for improvement cannot currently be explained. Conversely, some affected puppies require surgery to survive. Thus, the term “one-year-death-sentence” is occasionally heard.⁶ There is definitely a need for more scientific investigation focusing on LarPar in Alaskan Huskies.*

An inherited form of LarPar is described in Bouvier des Flandres, Siberian Husky-Alaskan malamute crosses, Bull Terriers, and white-coated German Shepherds.^{1,2,3} A developmental LarPar-polyneuropathy complex is described in Dalmatians, Rottweilers, Pyrenean mountain dogs, and Leonbergers.³ An acquired form affects mostly older Labradors or other large breeds.^{1,2}

* The authors are currently (2010) developing a protocol to collect DNA and pedigrees from affected dogs. Mushers with “Wheezers” are encouraged to contact Dr. von Pfeil or their regular veterinarian.

DIAGNOSIS

Diagnostics include complete physical (including auscultation and palpation of larynx and trachea) and neurological examinations. Blood work (+/- thyroid profile; Ach-Receptor-antibody test for myasthenia gravis, a neuromuscular dysfunction sometimes associated with LarPar and megaesophagus), and radiographs of chest and neck are recommended.^{1,2} Evaluation of swallowing using contrast fluoroscopy helps to determine esophageal function and may rule out hiatal herniation secondary to LarPar. Electrodiagnostic assessment of muscles and nerves are other available tests. The definitive diagnosis of LarPar is made under light anesthesia and direct visualization of the larynx. The affected larynx does not open normally during inhalation (Figure 3).



Figure 3: View of the laryngeal inlet in a dog. Normal movement of the laryngeal inlet while breathing is depicted by the white arrows and black dotted line. In dogs with LarPar, there is no such movement. Compare with Figure 5.

TREATMENT

Treatment for LarPar depends on the symptoms as the disease progresses. Thus, mildly affected animals have to be monitored carefully to avoid an emergency situation later. In very mild cases, decreasing body weight, limiting activity level, and preventing exposure to high temperatures may be sufficient. In moderate cases, mild sedatives help (Acepromazine). In severe cases, the animal is brought to a cool, non-stressful environment. Water or alcohol may be poured over the body, icepacks placed in axilla/groin for cooling. Sedation and oxygen delivery provide the most immediate relief. Occasionally, a temporary tracheostomy may be necessary. The most commonly performed surgical treatment is arytenoid lateralization (aka: "Laryngeal Tieback"). Other techniques are associated with increased complications, such as web formation later in life, obstructing airflow and leading to recurrence of symptoms. Arytenoid lateralization is technically demanding, especially when performed in a puppy, and must be performed correctly to assure acceptable outcome. (Figures 4,5).^{1,2}



Figure 4: Model of the larynx. Placement of sutures (dashed lines) during "Laryngeal Tieback", resulting in an opened airway (arrow).



Figure 5: View of the laryngeal inlet after "Laryngeal Tieback". Note the opened airway in comparison with Figure 3.

POSTOPERATIVE CARE

Surgery is usually associated with significant, immediate improvement of breathing and thus quality of life. After surgery, occasional coughing may be seen. This may improve over time if associated with irritation of the tracheal mucosa from intubation/surgery. As the larynx is permanently opened after surgery, food or water can accidentally enter the airway, and aspiration pneumonia may develop. This may be prevented by changing feeding habits and preventing swimming after surgery.

Aspiration pneumonia was evident in 18.5% of dogs with acquired idiopathic LarPar after arytenoid lateralization in a recent study.³ All dogs with pneumonia had more severe esophageal dysfunction compared to dogs that did not develop pneumonia. All patients recovered well with treatment, and no patient died from pneumonia. Interestingly, all dogs in that study developed some degree of neurological dysfunction within one year.³ Feeding from an elevated position, and keeping the animal on the hind legs for some minutes after eating is recommended. This may reduce the risk of aspiration pneumonia.³ It appears that Alaskan Huskies, operated early on in life, have a normal life expectancy and do not show the signs of polyneuropathy, seen in older dogs with acquired idiopathic LarPar. It is not known if affected Alaskan Huskies are more susceptible to aspiration pneumonia, nor has their esophageal function been evaluated.

CONCLUSION

In many dogs, LarPar appears to be a complex dysfunction not only of nerves supplying the larynx, but also of other nerves. Intensive research is performed to further investigate this potentially generalized neuromuscular disorder. The cause for LarPar in Alaskan Huskies is unknown. Although a genetic trait has not been proven yet, "Wheezers", even if clinically normal at a later stage in life, should not be used for breeding. Instead, they should be neutered in an effort to eliminate the disease from the bloodline. Surgical treatment is recommended as it usually provides fast and long standing improved quality of life.

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Injuries to the Axial Skeleton of Sled-Dogs



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Injuries to the Axial Skeleton of Sled Dogs

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Orthopedic injuries of sled dogs can occur during training or racing, and frequently result in lameness. While there are numerous possible causes for lameness in the dog, this chapter will discuss clinical signs, diagnostic findings, treatment, and prognosis of the most commonly observed orthopedic injuries in the sled dog. A thorough knowledge of canine anatomy is essential to understand, diagnose, and treat orthopedic injuries. Interested readers are referred to the recommended literature on canine anatomy in the chapter on orthopedic examination of the sled dog in this book. That chapter also provides a glossary of terms for non-veterinarians.

Prevention of orthopedic injuries may be achieved by appropriate training methods and conditioning of the dogs. Although scarce, there is some literature available on the prevention of orthopedic injuries in the canine athlete. These suggestions are included in the beginning of this chapter. Some orthopedic injuries may be treated without surgery. Massage therapy or other non-invasive modalities may lead to a successful outcome. Several courses to learn these techniques are offered to people without a professional background in veterinary medicine, and to people who have been formally trained as veterinarians or veterinary technicians. Participants are provided with knowledge on canine anatomy, physiology, biomechanics, massage techniques, physical rehabilitation, behavior, handling and first aid.¹⁻² Other orthopedic injuries need to be treated surgically. The goal of this chapter is to help the reader understand how to prevent orthopedic injuries, which injuries may be treated on the trail, how appropriate treatment is performed, and which injuries may require surgical therapy.

PREVENTION OF INJURIES TO THE AXIAL SKELETON OF THE SLED DOG³

Selective breeding may help to prevent injuries from occurring. Although there is no proof yet, various orthopedic diseases or injuries may be based on a genetic predisposition. More and more attention is given to this in veterinary orthopedic research.⁴

Choosing the right equipment (neck lines, tug lines, tow line, harness, etc) may prevent injuries from occurring. Neck injuries may be prevented by the use of bungee cord necklines. These are elastic bands that will stretch out like a thick rubber band, indicating to the dog that it will soon reach the point of maximal pull. A similar device is frequently applied as the connection between tow line and sled. Some teams do not use necklines at all (Figure 1). This provides a larger range of motion of the neck and allows the dog to select its footing on the trail better. Not using neck lines requires the dogs to be well trained and disciplined, or bit wounds may occur.

The correct harness choice is also important. There are three main types of harness styles, the H-back, X-back, and Collared-Neck Harness (Figure 2). Each has its specific advantages and disadvantages (Table 1). These harnesses all consist of straps that go around the neck, meet in the middle of the chest between the front legs, and then extend over the lateral aspect of the chest towards the flank of the dog to finally end caudal to the base of the tail, where they are attached to the tug line. Several variations of these harness types can also be found.

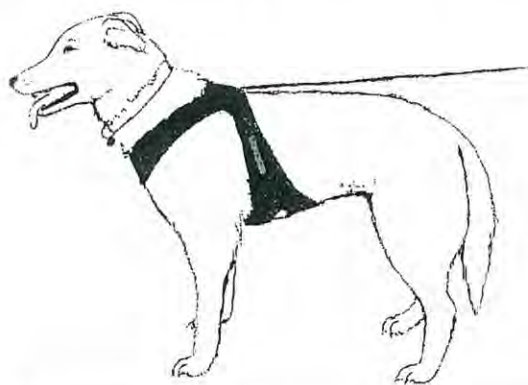


Figure 1: A short harness without neck lines provides more freedom for the dog and *may* decrease the risk for back injuries. Dogs using this type of harness need to be very disciplined.

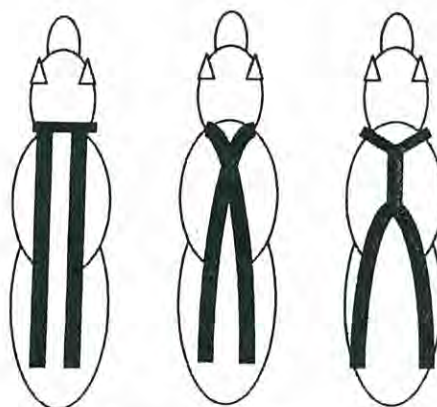


Figure 2: Commonly used harness types. H-Back, X-Back, Collared Neck (left to right).

Efficiency of all harnesses depends on proper fit. The musher must remember that the dog's body is constantly changing with weight, muscle tone, and hair coat so the fit must be monitored constantly even during races. In general, short harness types may decrease the risk of back injuries (Figure 1). These types of harnesses *may* decrease the risk of back injuries, but not other types of injuries. The authors have still seen lumbar pain in dogs with short harnesses – typically more cranial lumbar pain, versus the caudal lumbar pain that is more common with the standard harness types. Short harnesses do best without necklines. In fact, much of the benefit may be from not using necklines, rather than directly from the short harness. There is a newer type of harness being used which has better padding than the Man-Mat skijoring harness that has been used commonly in the past (Figure 1). The newer harness has rings to attach the tug-line on the side of the dog. On the other hand there may be less pulling power, because the range of motion of the hind legs is decreased. Most mushers experiment during training to find the best fit for every single dog. Some dogs may benefit from rotating different harness styles because of structure and/or gait. It is an advantage during a long race to have alternate sizes and styles available at some checkpoints for this reason.

| H-Back | |
|--|---|
| Advantages | Disadvantages |
| Allows rib cage expansion | Neck straps may accelerate muscle fatigue especially if poor fit |
| Tug line angle reduces pressure on lumbosacral area, especially for long distances at a trot | Dog may be able to escape from harness |
| Elastic band strap alleviates pressure on the spine | Neck straps do not conform to body |
| Angles neck straps are cone/collar shaped | Neck strap edge may dig into neck muscles if poor fit |
| A scapula plate provides similar even pressure distribution as the breast plate | Neck straps bear weight over scapulae |
| Less stress and less rub potential over spine | Neck straps impede movement of shoulders |
| | Neck straps accelerate muscle fatigue |
| | Harness rub can cause localized hair loss / skin abrasion |
| X-Back | |
| Advantages | Disadvantages |
| Easy to put on and off | Side straps may hinder chest expansion |
| Difficult for dog to escape | Too short cross strips increase pressure at loin arch instead of tail base |
| Neck straps conform to curvature of body | Neck straps bear weight over scapulas |
| Optimal length of cross straps contribute pressure from loin arch to tail base | Neck straps may impede movement of shoulders |
| | Neck straps may accelerate muscle fatigue especially if poor fit |
| | Harness rub causes localized hair loss / skin abrasion |
| Collared neck (improved X-Back design) | |
| Improved movement of forelimb decreases pressure on scapula – scapula plate | More difficult to fit properly |
| X-straps around body caudal to ribcage increase ribcage expansion | If not fitted properly, can put pressure on trachea, neck or shoulders |
| Not fit tightly around the neck | If fit tight enough to gain advantages of this style, can be difficult to put on and remove |
| | Medium and heavy coated dogs may be uncomfortable during harnessing and unharnessing due to tight fit |
| | Harness rub can cause localized hair loss/ skin abrasion |

Table 1: Advantages and disadvantages of various harness types.

The dog should be able to move and breathe comfortably when in harness. A correctly fitted harness will allow the dog to perform to its full potential, and prevent injuries. Each dog has a slightly different anatomy, and therefore each harness should be matched to the individual body. A perfectly fitted harness is not too tight, not too loose, and wraps snugly

around the dog's body when pulling. The forces when pulling should be evenly distributed among all straps. There should be no bulges, wrinkles, sudden angles in the straps, and points of pressure on the skin or bony prominences when the harness is properly applied. Occasionally it may be necessary to add custom designed buffer material to the straps to prevent pressure sores (Figure 3). The use of sliding O-rings on the tug line may help to adjust for different body lengths and may decrease shock to slower starting dogs (Figure 4).

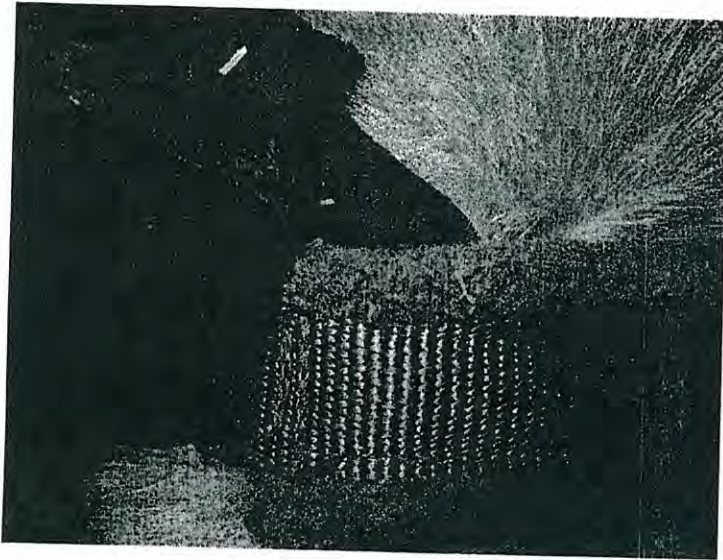
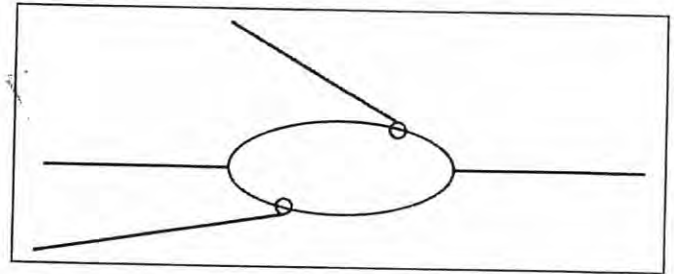


Figure 3 (left): Insulation to prevent skin sores.

Figure 4 (below): Tug line with O-rings.



Most dogs vector away from the gangline when pulling. This results in a force that is not parallel to the tow line but at a slight angle to it. Using parallel bars that are attached to the harness behind the dog at a distance long enough to not interfere with full range of motion of the hind legs, may prevent uneven pull. Especially if used without necklines, the dogs may travel in a straighter path, resulting in more even stress on the limbs.

It may be necessary to change harness types occasionally, depending on the terrain the team is travelling on. This may help to change pressure points and relieve stress on specific areas. Finally, rotating the position of the dogs may help to prevent monotonous stress on one or the other side of the dog's musculoskeletal system, thus preventing overuse injuries. Vectoring of the dogs away from the gangline while pulling creates increased stress on the outside (farthest away from the line) forelimb and inside hindlimb.

Conditioning of sporting dogs, including sled dogs, plays a major part in prevention of injuries.⁵ There is no scientific information describing the best training method or conditioning protocol for sled dogs or any sporting dogs. Therefore it is important to pay close attention to every dog during training. Screen the team for any metabolic or orthopedic diseases regularly during training and racing, and provide adequate rest to the canine athletes between runs. Conditioning starts at a young age. Animals should be evaluated for evidence of developmental orthopedic diseases between 4-9 months of age (hip dysplasia, osteochondrosis dissecans and others). Playful activities are allowed and encouraged until growth plate closure occurs, which is usually around 12 months of age. Until that time, the dog can be made familiar with the equipment. Light weights may be pulled for short intervals. Exposure to excessive stress in younger dogs may lead to fractures of the immature bone or to the growth plates. At approximately one year of age, more intense training can be initiated. The effect of conditioning is the adaptation of different body systems to the increased work, such as the gastrointestinal tract (gastric emptying, absorption of nutrients), hormonal systems (thyroid hormones), body fat metabolism, and the musculoskeletal system. Conditioning influences the musculoskeletal system by improving strength, endurance, and conscious proprioception. Strength is needed mainly in speed racing, while endurance is needed in long-distance races. The ratio of muscle mass to body weight is higher in dogs, trained for speed racing, compared to dogs trained for endurance. Training protocols and nutrition should be adjusted to reach the specific goals. It is interesting that there is no difference in performance between male and female dogs, despite differences in muscle mass and body size. Training affects all musculoskeletal tissues, and makes cartilage and ligaments stiffer, -and bones, muscles, and tendons stronger. If balance and conscious proprioception are well developed, the athlete will be able to adjust better to the challenges of different terrains during the race. Conscious proprioception can be improved by specific exercises, such as walking in specific patterns, on various flooring, over obstacles, or on grounds with different, fast changing slopes. Other exercises include rapid changes of direction, the use of cavaletti rails, exercise balls, ball playing, and walking on front or back legs,

while supported by a physical rehabilitation therapist. Conditioning also results in a decreased duration of circulating lactic acids in the body after training. Therefore the athlete may experience less or no muscle soreness when appropriately trained. Adequate rest between training intervals will decrease the concentration of lactic acid in the tissues. As a result, muscle and overuse injuries may be prevented. In addition, the application of specific treatments after training runs or during a race, such as massage, stretching, or if indicated the use of other modalities (see muscle injuries) may help to prevent injuries in the sled dog.

It is the impression of the authors that injuries are less frequently seen in teams that have covered between 3000-4000 training miles, when compared to teams that covered 1000-2000 miles during preparation for a long distance race, such as the Iditarod. It is possible that such intense training programs help to select for animals that are not very prone to injury. If a dog went through a reasonable training program over the year, and competed in several races of increasing length in the racing season without a significant injury, it is possible that it will successfully complete marathon races such as the Iditarod. However, this hypothesis needs to be proven.

Despite all preventative measures, injuries occur occasionally. It is important to detect these injuries, and to apply appropriate treatment as soon as possible.

FRACTURES

Any fracture makes it necessary to remove the dog from the race. Emergency treatment for fractures on the trail is difficult to apply. If possible, pain medication should be administered. Oral medications for pain relief and/or to reduce inflammation include tramadol, amantadine, and non-steroidal anti-inflammatories (NSAIDs), such as Rimadyl, Metacam, Etogesic etc. It is important to follow the instructions on the prescription label to prevent or decrease possible negative side effects with any drug. Trail veterinarians may have injectable analgesics such as Morphine derivatives available. If there is an open wound associated with the fracture (Figure 5), the damaged area should be treated as indicated in the chapter on wound treatment in this handbook. If this is not possible, the application of a sterile bandage, or at least clean layer of cotton should be used to cover the wound. Fractures of the humerus or femur are difficult to stabilize using a bandage. With these injuries it may be adequate to limit supportive care to the administration of analgesics and careful handling. In contrast, fractures below the elbow or stifle can usually be stabilized by application of a "Robert-Jones-Bandage" (Figure 6). This bandage consists of a sterile layer if there is a wound. Then a thick layer of soft cotton, followed by a layer of stretch gauze, and a splint are applied. A light aluminum or stable plastic rod, a straight piece of wood, or rolled newspaper could be used in addition to, or in lieu of commercially available splints. The final layer consists of vetrap, or carefully applied tape. Animals with fractures should be transported to the closest veterinary facility as soon as possible.



Figure 5: An open fracture has to be treated following the principles of open wound management.



Figure 6: Robert-Jones Bandage.

Radiographs are necessary for a definitive diagnosis (Figure 7A). Depending on various factors, definitive treatment methods include the application of a professionally applied cast or stabilization using orthopedic surgical devices, such as pins and wire, bone plates, pin-plate combinations, interlocking nails, or external fixators (Figure 7B). The prognosis for fracture healing depends on the severity of the injury, the age of the animal, the time between incident and treatment, and most importantly the post-operative care. Strict limitation of exercise for at least 8 weeks after surgery is imperative for initial bone healing to occur. With severe fractures this time may be longer. Radiographs are taken at specific time intervals to monitor bone healing. A specific activity protocol should be designed for each patient. A quick return to full activity may be achieved by using appropriate physical rehabilitation modalities. Bone plates used to stabilize fractures of bones lacking soft tissue coverage, for example the distal radius, may get very cold when the dog is outside for an extended period of time. This cold can lead to a painful sensation and lameness. To prevent or treat lameness that occurs secondary to thermal conduction, surgery to remove the implant is usually recommended. While bone healing is frequently adequate for normal leg use after approximately 8 weeks in most cases, it may take 1 year or longer to achieve adequate bone stability that allows removal of the implant. At that point, the bone should be able to withstand forces while running without the support of a bone plate.



Figure 7A:
Comminuted
femur fracture

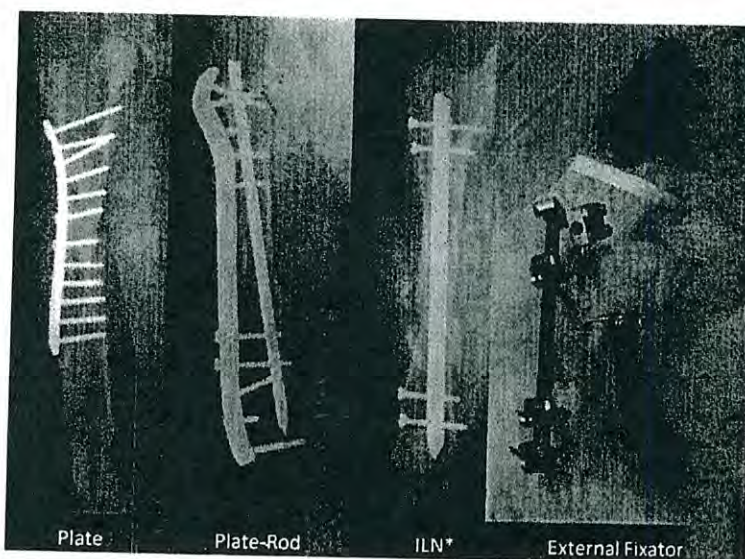


Figure 7B: Fracture repair options. *ILN=Interlocking nail.

MUSCLE INJURIES

Muscle injuries occur frequently in the sled dog. They include exertional rhabdomyolysis and acute muscle injuries such as muscle strains to certain muscle groups. Information on exertional rhabdomyolysis during endurance sled dog racing is provided elsewhere in this book. The following paragraph is written to inform the reader about possible treatment options for acute muscle injuries. The chapter on orthopedic examination provides information on how to detect such injuries. Over the last several years various techniques, used successfully in humans and horses, have been implemented in the treatment of muscle injuries in the canine athlete. However, their application may be limited during a race. The most commonly used techniques to treat muscle injuries on the trail are application of various liniments, thermal treatment, and massage. Changing harness type or fit and changing the side of the gangline the dog runs on can help reduce stress on an injured leg. More advanced therapies include range of motion exercises, exercises to improve balance and proprioception, hydrotherapy, laser therapy, neuromuscular stimulation, ultrasound, or acupuncture. Many of these modalities can also be applied during rehabilitation of other orthopedic injuries.^{1,2,5}

Liniment application

The benefit of liniment or ointment application is probably one of the most discussed topics among mushers and trail veterinarians. It is important to understand that there is no miracle drug that can be used during a race that will cure orthopedic injuries. Appropriate rest and supportive modalities are the best treatments for most soft-tissue injuries. Appropriate conditioning and careful racing strategies, especially in rough terrain and hills, are a better choice to likely

prevent injuries, rather than relying on the effect of a liniment once an injury has occurred. Various liniments are available, and it appears that each musher or veterinarian swears by one or the other. There is no scientific evidence proving one liniment superior to the other. It is imperative to verify that the ingredients of a liniment are allowed during a race prior to application of a liniment. Liniments that can be used without interfering with doping regulations contain various substances, which may or may not be beneficial to the musculoskeletal system of the sled dog. The beneficial effect of a liniment may arise solely from the additional massage that accompanies application of the liniment. Rubbing liniment on the legs, shoulders and flanks may encourage additional blood flow to the muscles, resulting in increased relaxation and elasticity. Several liniments seem to be used frequently among the sled dog community in North America. These include Algyval, Zalox, Accede, Emu oil, and others.

Algyval (Figure 8) contains mainly hyperoxygenated peanut oil and rosemary oil extract as active ingredients. The mechanism of action is not scientifically proven or understood. It has been suggested that the active ingredients inhibit leukotriene and prostaglandin receptors, resulting in an interruption of the inflammatory cascade. However, no studies are available to solidify these hypotheses. Algyval seems to have a better effect when applied warm. A possible negative side effect is that it tends to make the skin “greasy”. These areas may freeze at very low temperatures, possibly leading to skin damage from hypothermia.

Zalox (Figure 9) is also frequently used to relieve joint swelling, inflammation, and muscle soreness. Active ingredients are essential oil of wintergreen, menthol, camphor, eucalyptus oil, and tea tree oil. Other ingredients include deionized water, aloe vera, kosher glycerin, stearic acid, mineral oil, glyceryl monostearate, glycol stearate, isopropyl palmitate, dimethicone, and triethanolamine. The same precautions for preventing skin freezing as with Algyval should be taken with Zalox, especially as it has water in its base. Aloe vera may help to treat minor cuts and skin injuries.

The application of Accede or Emu oil may also have beneficial effects to treat muscle injuries, but reliable information on ingredients and their effect is scarce. Accede is oil based also, and contains several herbal extracts including rosemary. As a safety precaution, any handlers with allergies should check ingredient lists. For example, both Algyval and Accede contain peanut oil or peanut extract, which could trigger a reaction for someone with a peanut allergy. The authors know about at least one dog with confirmed allergic reaction to these products that did not react with an olive oil based liniment – *so peanut allergies appear to happen in dogs too*. That dog had redness, pruritus and alopecia in the areas it was applied.

Other liniments are available and may contain the following ingredients: willow, arnica, wormwood, devils club, and/or buckbean, grape seed, apricot kernel oil, calendula, vitamin E, or tea tree oil. It must be emphasized, that these ingredients may show beneficial effects for some dogs, but scientific data supporting the use of these substances is lacking. The use of liniments or supplements containing willow bark may cause a positive drug test for salicylates. *Always check before the race to make sure the products are allowed.*



Figure 8 (left) and Figure 9 (right): Algyval or Zalox are frequently used during massage of sore muscles. Their use may improve muscle function or prevent injury.

Thermal treatment⁵

The application of cold helps to control pain and inflammation immediately after injury. If decreased range of motion is a result of pain or swelling, apply cold. Snow or crushed ice can be placed in a plastic bag. A thin cotton cloth is placed over the injured area, and the cold pack applied for 15-20 minutes after injury. It can be reapplied after a pause of the same time. Icing the damaged tissue provides an analgesic effect, decreases blood flow, edema and muscle spasm. There is some controversy about the use of cold packs in sled dogs because the injured areas may already be hypothermic. As such, cold therapy may be reserved for surgical patients, for patients that have a very acute injury and can be treated in a warm environment, or during races in temperatures above freezing.

Heat therapy is frequently used to treat injuries in sled dogs such as strains of the shoulder muscles (see below – Figure 10). If decreased range of motion is a result of stiffness, apply heat. On the trail, custom made sleeves or wraps for the specific anatomic location are equipped with chemical handwarmers to provide heat to the injured tissues. These hot or warm packs apply superficial heat. If available in the kennel, therapeutic ultrasound or laser (see below) can be used, and may effectively provide deep heat penetration. Heat causes vasodilation which increases blood flow, resulting in increased metabolism of the treated tissues, relaxation, increased elasticity and facilitation of tissue healing. Similarly, the range of motion of treated joints may increase, resulting in more comfortable ambulation. Heat should be applied for approximately 30 minutes. However, many mushers leave packs on for longer, often the entire 6-8 hour rest period, and some even use them while running. Temperatures should not rise above 113°F (45°C) in order to prevent thermal injury.

Range-of-motion therapy⁵

Similar to stretching, range-of-motion exercises help improve muscle elasticity, decrease the risk of adhesion development between soft tissue and bone, and prevent further injury to joints, ligaments, tendons, and muscles. Range-of-motion exercises may not be as important to perform on the trail, as they are to an injured animal recovering at the kennel. Frequency and intensity of range of motion exercises depends on the specific injury and should be performed following the recommendations of a veterinarian experienced in canine sports medicine.

Massage therapy¹

Massage therapy is a technique that can be done on the trail and may result in significant improvement of the animal's performance or recovery. Several mushers have asked specifically for more information on "how to get a dog through an injury". To gain the necessary knowledge in these disciplines, we recommend that specially tailored courses be attended.^{1,2} It is not possible to perform a complete "text-book" massage on every animal at every checkpoint. However, some of the provided information may be applicable to certain animals, which will hopefully benefit from massage, and finish the race successfully. On the other hand, massage therapy could be implemented in training, possibly helping to prevent acute injuries during a race.

Benefits of massage therapy include stimulation of the lymphatic system and increased localized blood circulation, resulting in increased cellular activity and metabolism, removal of metabolic waste products, and thus decreased localized inflammation. The release of naturally occurring endorphins stimulates the parasympathetic nervous system, resulting in relaxation. A quick return of muscles to a physiologic resting length, early detection of injuries, and assisting in the promotion of healthy scar tissue formation are further benefits of massage therapy. These benefits result in improved immune function, reduced recovery time between workouts or from injuries, decreased post-workout discomfort, improved proprioceptive skills, improved work efficiency, stability and flexibility, decreased risk of stress-related injuries, and prevention of chronic conditions. Although there is no proof in the literature yet that massage can diagnose, treat, reverse, affect, or help to prognose an existing condition in the canine athlete, the benefit of massage therapy in long distance races, such as the Iditarod, seems to be undeniable. Several mushers are convinced that dogs with certain injuries, such as carpal or some shoulder problems would not have been able to finish the race without applying massage therapy at every checkpoint or even intermittently on the trail.

It is important to avoid forcing an animal to receive a massage, and never to massage directly over an open wound or skin infection, or over a suspicious lump. The animal's response must be observed, and the massage modified as needed for comfort and safety.

Every massage session consists of five stages: Opening, palpation, body of the massage, closing and stretching. Opening involves greeting, initial touching to quiet the dog, and to encourage relaxation. Palpation consists of running the hands over the entire body of the animal using mild strokes. It will help to relax the animal and provide the therapist with information on general health of the tissues. Tension, swelling, spasms, or heat may be felt. The body of the massage is the application of selected strokes and techniques, applied based on the findings from palpation. There are three categories of body massage: Maintenance of health and fitness (1-2 treatments per month), optimizing performance potential (3-4/month), or rehabilitation from injury (daily to 3-5/week). Stretching is used after the massage to restore good circulation to all parts of the body, and to help solidifying the work performed. Closing is performed using light strokes to reconnect

all aspects of the body, and to provide a calm and relaxed feeling. Ending the massage too abruptly may lead to an unresolved or confused conscious state in the animal.

There are six strokes that are all part of the body of the massage. They can all be modified in speed and pressure to alter the effect that is desired for each individual patient. Faster strokes are more stimulating, slower strokes more sedating. The therapist will have to strive for excellence by practicing, practicing, and practicing again. Results will improve with time and experience.

Passive touch is an excellent opening stroke. It is sustained contact, using the hand, fingers or fingertips. It can be used to feel for tissue irritation, it can help to create good contact between therapist and animal, and has a calming effect. It can be applied with only one hand, while the other hand is applying another stroke elsewhere. Passive touch is applied with minimal pressure and increases neural awareness and the sensation for warmth.

Effleurage describes passing of the hand over tissues with moderate pressure. It warms an area prior to advanced work and can be used for opening or closing, and transition between areas. It stimulates lymphatic flow and superficial circulation. If massage therapy is performed with focus on one area, effleurage is a good technique to provide the remainder of the body with a few comforting strokes.

Petrissage is also known as kneading. It describes the application of deep pressure to the tissues or lifting of tissues, leading to increased circulation. This helps with removal of waste products. It should only be applied after passive touch and effleurage have been performed. If petrissage is too aggressive, it may lead to the damage of tissues. Always begin with gentle pressure or lifting and a slow rhythm. One-handed petrissage is the lifting and squeezing of tissue into the palm of one hand. The other hand can be used to apply a different stroke, or to restrain the animal. Small muscle groups or the limbs may be massaged using this stroke. Two-handed petrissage describes the placement of a muscle belly between the palms of the therapist's hands. The muscle is lifted and squeezed carefully. The amount of applied force has to be monitored carefully. Open-C / Closed-C petrissage describes a two-handed petrissage in which the thumbs and forefingers are held in the shape of a C. Tissues are lifted and pressed together with one toward the other hand, resulting in the creation of an effective fluid pump within the muscle belly. This technique is useful to relieve a mild cramp. Several more variations of petrissage are available.

Compression describes the application of pressure with a flat or balled hand, the arm, elbow or fingertips of the therapist to the underlying tissues. It is a useful stroke for muscles that are difficult to lift. Pressure intensity should be increased carefully from light to deep. Depending on the body part used to apply pressure, the effect will differ: A flat hand provides pressure over a large area, whereas the elbow creates a more focused pressure point. Application of compression using the fingertips over a joint may help to stimulate fluid activity and relax surrounding tissues.

Friction strokes increase warmth and localized circulation, relax restricted muscles and lengthen muscle fibers. This stroke needs to be applied with caution as not to damage healthy tissue. It should only be used after opening strokes such as passive touch or effleurage. Direct pressure can be applied over areas of tenderness, adhesions, or as part of professional acupressure. Repetitions with increasing pressure can be applied. However, this stroke should be limited to five applications for not more than 30 seconds. If no improvement is observed, the area should be left, and repeated treatment applied at another time. Overly aggressive treatment may result in tissue damage. Several sub-categories of friction are described: Rubbing is used to build up heat in the superficial tissues. Chucking is used on the limbs and consists of holding tissues with the whole hand, and carefully pulling the tissues proximally and distally. Twisting is performed by the same hand placement, but the tissues are carefully twisted. These techniques also increase localized circulation. Circular friction is used to warm and mobilize underlying soft tissues, and supplies a relaxing effect on the nervous system. Linear friction helps to move waste products out of muscle bundles. The stroke follows the direction of the muscle fibers. Cross-fiber friction describes pressure application across the direction of muscle fibers. It is used to loosen adhesions or hardened muscle areas. Direct pressure for a few seconds drains blood from the area. A short release of pressure is then followed by movement of the fingers across the muscle fibers for a few seconds. These steps are repeated several times with the goal being to resolve the adhesion or "knot". Cross-fascicular friction is the spanning of muscle fiber bundles using the fingertips. This technique leads to a more diffuse distribution of pressure and is frequently used to treat adhesions, which cannot be released using other techniques.

Tapotement is the application of short percussive strokes. Indications for tapotement are stimulation of weak or atrophied tissue, numbing of tissues, engaging the attention of the animal or loosening and warming tissues up prior to work. It can be applied by drumming the fingers on certain areas (tapping) or by applying short strokes with the flat hand onto the abdominal or thoracic muscles (slapping). To warm tissues up prior to exercise, the edge of the hand can be used to carefully hit the underlying tissues (hacking). To possibly relieve congestion in the chest or to decrease joint swelling, the hand can be formed to a small cup and light blows are applied to the area of interest (cupping). Finally, the fingers can be moved over the tissues in a "chopstick" – like fashion. This stimulates superficial circulation and lymphatic drainage, helping with removal of waste products. Contraindications for tapotement are muscle spasms, cramps or hypertonicity. The application of this technique to such conditions may worsen the severity of the problem.

Finally, the limb, skin, or certain body parts can be loosened by shaking it. "Chucking" describes moving a fascial plane up and down. "Vibrating" is the application of short pulses from the palm or fist of the therapist to joints or deeper tissues. The pulse waves penetrate the tissues and lead to a sensation of warmth. They also stimulate joint fluid production, which may be beneficial prior to work. "Brushing" is a short stroke in which the whole the hand is used to touch the skin rapidly, with the effect of restoring sensation and increasing circulation. These techniques are good for closing the massage. After each massage, the optimal effect can be reached by stretching the tissues for five to ten minutes and a period of easy walking.

Stretching results in increased muscle length, and thus flexibility. Increased flexibility results in a decreased amount of injuries of both, muscles and joints. Therefore, good stretching is imperative for injury prevention. The following guidelines should be followed when stretching: *Never stretch a cold muscle*. Some light work or massage will increase blood flow and decrease muscle fiber rupture from sudden application of stretching to cold tissues. *Always stay within a comfortable range of motion*. Passive repetitive stretches consist of fast movement of muscle through its complete range of motion. This stretch does not involve excessive tension, which is held for an extended period of time, as is done with static stretching. Passive repetitive stretching creates a fluid pump within the muscle, leading to removal of waste products and supplying oxygen and nutrients to the muscles. Animals seem to be more receptive to this stretching technique than to static stretching. From neck to hind legs, several stretches are available for the major muscle groups, and can be applied safely on a regular basis to improve flexibility. If possible, stretching should be performed 3-5 times per week. Neck stretches consist of careful lateral flexion, extension, and dorsal/ventral flexion. Front leg stretches consist of flexion/extension, abduction/adduction, internal/external rotation of the shoulder, flexion/extension of the elbow, wrist, and digits. Stretches for the spine and abdomen consist of careful dorsal/ventral flexion and extension as well as lateral flexion. Hind leg stretches consist of flexion, extension, abduction, adduction, internal, external rotation of the hip and extension/flexion of stifle, tarsus, and digits.

Laser Therapy (LT)⁵

LT uses electromagnetic radiation that enhances energy production (Adenosine-Tri-Phosphate, ATP) and metabolism within cells that may ultimately result in an increase of events that accelerate tissue healing (increased amount of growth factors, cytokines, and cell replication). It also seems to provide analgesia, to preserve cartilage properties, and to help resolve peripheral nerve injuries. The U.S. Food and Drug Administration (FDA) has approved 635nm low-level lasers for the management of chronic minor pain, such as osteoarthritis and muscle spasm. The use of laser type (helium-neon or gallium-aluminum-arsenide with 632.8nm to 780-904 nm, respectively), and applied energy ($4-11 \text{ J/cm}^2$) vary, depending on the injury. LT can be used daily during week 1 after injury, and every other day during the second week. The use of LT remains controversial, but is noninvasive and has not been associated with side effects, when used properly. Higher doses of laser energy may inhibit tissue healing. Therefore, this treatment should only be applied by specifically trained personnel.

Neuromuscular electrical stimulation (NMES) and Transcutaneous electrical nerve stimulation (TENS)⁵

NMES is used to stimulate skeletal muscle contractions through excitation of nerves. Differently pulsed patterns are used to treat various orthopedic and occasionally spinal cord injuries. The application of NMES results in increased blood circulation, muscle strength, joint mobility, and decreased tissue contracture, edema, and pain.⁵ TENS is the use of electrical stimulation for pain control. The effectiveness of these treatments has been proven for dogs. Specific protocols for strengthening or pain control need to be followed to avoid injury. These treatments are only to be applied by specifically trained personnel.

Therapeutic ultrasound (TUS)⁵

The application of energy through an ultrasonic beam leads to compression and separation of the exposed tissues. As a result, deep tissue heating occurs, which has been shown to result in tissue healing. TUS may be of benefit during the rehabilitation of muscle injuries, joint contractures, or during the healing process of bone, cartilage, or tendons, when applied properly. The intensity, frequency, and duration of ultrasonic waves can be adjusted, depending on the treated injury. Frequencies from 0.25-MHz to 3-MHz with an intensity of $0.5-2.0 \text{ W/cm}^2$ are frequently used. The recommended transducer speed movement is 4cm/s. Treatment is usually applied for 10 days daily, then on an as needed basis. It is recommended not to exceed two 10-day courses without a 3-week rest. Additional therapies such as massage and stretching aid to achieve the optimal effect of TUS. This treatment is only to be applied by specifically trained personnel.

Extracorporeal shock wave treatment (ESWT)⁵

ESWT is the application of acoustic waves initiated outside the body. The clinical effects include decreased swelling and inflammation, realignment of tendon fibers, analgesia, increased bone and soft tissue healing through increased

neovascularization and vascularity. Appropriate use is imperative to prevent negative side effects. ESWT has been used successfully in horses for treatment of tendinopathies, back pain, suspensory ligament desmitis, osteoarthritis, and stress fractures. In dogs, ESWT has been applied successfully to treat shoulder tendinopathies with tissue calcification, and osteoarthritis of hips and elbows. Both, subjective and objective evaluation showed improvement of the underlying condition after ESWT was applied.⁵ The effect was compared to the effect that is typically expected after the use of NSAIDs. Although promising, further studies need to be conducted to support this treatment modality for muscle injuries in the canine athlete. This treatment is only to be applied by specifically trained personnel.

Acupuncture

Acupuncture may improve healing of muscle injuries or other orthopedic conditions in some selected patients.⁶ The musculoskeletal system is like a pulley and lever mechanism - a system of agonists and antagonists. Orthopedic injury may result in a conflict between the opposing agonist and antagonist muscle groups, which creates undue stress on the joints, muscles, tendons and ligaments. Obscure muscular pain found in association with either acute traumatic or chronic/overuse injuries can result in weakness at the injured site, and to the onset of lameness after a certain degree of activity. In traditional Chinese medicine, acupuncture points are utilized to address these complaints. Acupuncture points, important for the treatment of muscle injuries, are trigger points and motor points. A trigger point is defined as a "hyperirritable spot in the skeletal muscle that triggers a characteristic referred pain or tenderness phenomenon." Some of the indicators of a trigger point include pain referral, muscle twitch response and tight palpable bands. These tight and frequently painful palpable bands may be treated effectively with the acupuncture needle. A motor point is defined as "the most electrically excitable area of the muscle, and represents the greatest concentration of nerve endings", medically speaking the neuromuscular junction. The insertion of an acupuncture needle into a motor point may elicit a change in length of the muscle, resulting in muscle relaxation. Acupuncture to the motor points seems to correct abnormal muscle function and reflexive spasm. The motor point serves as a way of balancing the agonist and antagonist muscle groups. Acupuncture performed by an experienced therapist may help to decrease discomfort, and possibly treat muscle injuries in the canine athlete. However, there is no scientific data available to support this treatment method, and skepticism in its effectiveness may be warranted.

Hydrotherapy

Although this treatment method can only be applied if an appropriate beach, river, pool or rehabilitation center for canine athletes are available, its benefits justify a short description in this chapter. It is becoming more common for mushers to use a pool to exercise their dogs during the off season to maintain conditioning. Water temperature can influence the cardiovascular system. It is generally recommended to keep the water temperature during hydrotherapy for dogs in the range of 78.8° to 82.4°F (26° to 28° C). Studies to determine the optimal water temperature for sled dogs need to be conducted. The effect of gravity is reduced in direct relation to the level of submersion. Body weight decreases by 9%, 15%, or 62% if the water level rises to tarsus, stifle, or greater trochanter, respectively. This effect is described as buoyancy. It describes the force lifting the body upward while submerged. With increased buoyancy, there is less impact on the musculoskeletal system when walking. This prevents injury to the joints or injured tissue. The amount of hydrostatic pressure, which increases with submersion, may improve lymphatic or venous drainage from injured tissues or joints. The viscosity of water, the depth of immersion, and turbulent drag resists limb and body movement. This may result in muscle strengthening and improved cardiovascular function. It has been shown that with movement just beneath the surface of the water, approximately eight times the amount of energy are required to double forward motion. Some specifically designed hydrotherapy treadmills offer the use of jets, which can be used to increase the aforementioned effects. Thus, muscle workout can be increased in a very controlled manner, and re-injury prevented by appropriate application. In summary, hydrotherapy assists weightbearing, increases range of motion, balance, proprioception, strength, endurance, and cardiovascular fitness, and decreases swelling of tissues or joints.

Exercises to improve balance and proprioception⁵

Several exercises are available to improve balance and conscious proprioception (see above, conditioning). These exercises not only help to prevent injuries, but can also be applied during the recovery period from an injury. Most of these techniques cannot be performed during a race. Their application is limited to use at the kennel. Explaining the specifics of these techniques is beyond the extend of this chapter. However, the interested reader is encouraged to increase his knowledge through readily available literature, or by participating in specially designed courses or classes (see above).

SHOULDER INJURIES

The diagnosis of shoulder injuries is made by an appropriate orthopedic examination (see previous chapter).

Differential diagnoses for shoulder injuries include: biceps tendon injury, triceps muscle, pectoralis muscle, infraspinatus, supraspinatus, or other cuff muscle injury, medial or lateral glenohumeral ligament damage, trauma to the scapula, proximal humerus, or entire joint including luxation, degenerative joint disease, osteochondrosis dissecans, panosteitis, joint infection, or arthritis.

Injuries to the biceps are among the most commonly seen orthopedic shoulder injuries in the racing sled dog. Injuries to the triceps muscle or the pectoral muscles can also be observed frequently. In general triceps and pectoral injuries can more often be massaged and treated on the trail and dogs can finish the race, when compared to biceps muscle injuries. Shoulder injuries may be more frequently observed in speed racing sled dogs, but are also commonly seen during long distance races. The authors have noticed these injuries to occur frequently in dogs that covered long distances running on ice, or after high-stepping through deep snow. Injured dogs show a gait with the head going down when placing the healthy foot on the ground, and moving the head up when placing the injured leg on the ground ("head-bob", also see the chapter on orthopedic examination of the sled dog). Dogs with biceps injuries are painful upon extension of the shoulder while simultaneously applying pressure on the tendon within the bicipital groove. Triceps injuries are diagnosed by eliciting a painful response when extending the leg, and when deeply palpating the different heads of the triceps muscle. In severe cases, triceps avulsion injuries from the olecranon can be seen. These dogs show pronounced swelling just proximal to the olecranon, and are painful upon palpation of this area. Treatment for biceps or triceps injuries include massages, application of warm packs (Figure 10), and possibly more sophisticated treatments as described in the section of "general treatment for muscle injuries".



Figure 10: Warm pack to prevent or treat shoulder injuries.

It may be helpful to remove the dog from the team and walk it for 10-20 minutes prior to leaving the checkpoint. This will help the muscle to warm up, and possibly prevent more damage. Some mushers perform massages even when on the trail at various intervals. Once a dog had a muscle injury the treatment should be automatically performed at every following checkpoint. Dogs with biceps or triceps injuries may sometimes "warm out" of the injury, and may finish the race. However, most dogs with injuries to these structures will have to be dropped. Some mushers will drop dogs with injured shoulders as soon as they are aware of this cause for lameness to minimize possible long-term damage. The key for a good recovery is appropriate treatment for muscle injuries, rest, and appropriate rehabilitation. Dedicated protocols for physical rehabilitation to treat shoulder injuries are available, and the application of a specifically designed pair of shoulder hobbles may improve outcome.⁷ Repetitive injuries of the biceps tendon, the infra- or supraspinatus tendon, or the glenohumeral ligaments may result in chronic damage. In these cases it may be necessary to apply advanced treatment methods, such as arthroscopy, and / or surgical repair of the damaged tissues, for example placement of prosthetic ligaments in cases of shoulder instability. The recovery time for shoulder injuries can be as long as 3-6 months. Shoulder injuries may have a significantly better chance for rapid recovery if diagnosed within 12 hours of occurrence, if non-steroidal anti-inflammatory is administered within 24 hours, if the dog receives a thorough, deep massage (see above) of all shoulder muscles, and if the dog is confined to a small space for 48 hours.⁸ If surgery may be necessary the use of arthroscopy may allow a quick diagnosis and possible simultaneous treatment during one session.

CARPAL INJURIES

The diagnosis of carpal injuries is made by an appropriate orthopedic examination (see previous chapter).

Differential diagnoses for carpal injuries include: Severe exercise induced joint inflammation, complete or partial luxation, cartilage damage to the joint surface, fissures or fractures of the carpal bones, severe ligamentous injuries, such as hyperextension damage, or degenerative joint disease.

Carpal injuries are common during the first longer training runs in the season or during the first 1-500 miles during long distance races. These injuries are commonly observed when the trail is hard, or leads through rough, hilly terrain. Animals with severe discomfort will show the typical “head-bob”, typical for front leg lameness. Most wrist injuries arise secondary to heavy exercise and present as marked swelling of the wrist from significant inflammation of the joint. Repetitive high impact trauma leads to increased blood flow and production of inflammatory joint fluid. The joint capsules of the carpus are tight, and an increase in joint fluid leads quickly to increased pressure on pain receptors within the joint capsule, possibly leading to clinical signs of lameness. The joint should be thoroughly massaged at the checkpoint, and range of motion exercises should be performed. The carpus is flexed and extended several times. Furthermore, the application of joint health supporting liniments or ointments (see above) in combination with some cellophane wrap and a custom made wrist wrap, or “sweat” wrap (neoprene and Velcro™ straps) seem to frequently improve the condition (Figure 11). Wraps may also be applied as a preventative measure at every checkpoint to interrupt the mechanism for joint swelling before it begins. The wraps are left in place for the duration of rest at the specific checkpoint. It is imperative not to place the wrap too tight and to cover the leg completely all the way to the toes in order to prevent venous obstruction. If the dog’s foot swells up after application of the bandage, it was too tight! A dog must not run with the wrist wrap on! The application of wrist wraps in combination with an ointment may help reduce the risk for injury. However, appropriate conditioning and slowing down on hills may be more helpful in preventing wrist injuries from occurring. Furthermore, some mushers believe that there is a relation between the degree of hydration status of the animal and the occurrence of carpal injuries. There is no scientific evidence for this statement, but it is always recommended to maintain the best possible hydration status in racing sled dogs to decrease the risk of any exercise related disease or injury. Most sled dogs with carpal swelling improve throughout the course of the race. On the other hand, repeated or prolonged trauma, cartilage wear and development of arthritis may result in consistently decreased range of motion and early onset of lameness after running comparably short distances. The prognosis for chronic carpal injuries in the sled dog is guarded. With chronic lameness, arthrodesis (joint fusion) may be the only way to provide the animal with pain free ambulation. Fractures, luxations, or ligamentous injuries frequently need to be treated by surgical stabilization, such as arthrodesis (Figure 12). Dogs with fused carpi can often return to an athletic life style. The specific form of treatment and the prognosis depend on the underlying condition and will have to be discussed with an appropriately trained veterinarian or veterinary surgeon.



Figure 11: Wrist wraps often help to overcome swollen carpi.



Figure 12: Carpal arthrodesis may be necessary to treat severe injuries of the carpus. Dogs with fused carpi can often return to an athletic life style.

ELBOW INJURIES

The diagnosis of elbow injuries is made by an appropriate orthopedic examination (see previous chapter). Differentials for lameness arising from an elbow problem include: Fracture, luxation, infection, arthritis, ununited coronoid process, ununited anconeal process, incomplete ossification of the humeral condyles, ununited medial epicondyle, osteochondrosis dissecans, or elbow incongruency. Most of these diseases require a thorough diagnostic work up (radiographs, computed tomography, arthroscopy). Treatment consists mostly of supportive care and/or surgery. Sled dogs with elbow problems that arise during a race are unlikely to finish the event. Depending on the underlying disease, the prognosis after surgical treatment varies, depending on the underlying disease.

HIP INJURIES

The diagnosis of hip injuries is made by an appropriate orthopedic examination (see previous chapter). The most common cause of hip injuries in the sled dog is muscle strain. Psoas muscle injuries can be seen and may be confused with injuries to the hip joint. The psoas muscles consist of a group of deep stomach muscles attaching at the pelvis, femur and ventral lumbar spine. It is not possible to palpate most of these muscles, because of their deep location. This muscle group is contracted to bring the hind legs underneath the body when running, resulting in forward propulsion. Contraction of these muscles arches the lumbar spine and changes the angle of the long axis of the pelvis to a steeper slope. This movement may be compared to winding up a spring for an explosive release. If this muscle group is injured, the dog may present with a pronounced upward bend of the lower back. The dog shows decreased ability of hip extension. Severe injury of these muscles may result in uncomfortable gait for several months. Appropriate conditioning and stretching at least every other day may prevent this injury from occurring. Muscle strains to the hip may be treated using conventional methods. Massage, warm packs, and careful stretching may help to improve the condition. Similar treatment can be applied to other muscle injuries of the hip joint. More information on muscle injury is provided in the section of "general treatment for muscle injuries". Administration of non-steroidal anti-inflammatories, muscle relaxants such as methocarbamol or application of TENS (see above) may also help.

Differentials for surgical problems of the hip include: Fracture, luxation, and degenerative changes such as arthritis or hip dysplasia. Treatment and prognosis depend on the underlying condition. A quick return to full activity may be achieved by using appropriate physical rehabilitation modalities.

STIFLE INJURIES

The diagnosis of stifle injuries is made by an appropriate orthopedic examination (see previous chapter). Most stifle injuries are secondary to trauma or advanced degenerative changes, and conservative treatment is unlikely to return the animal to racing conditions. Cranial cruciate ligament rupture (CCLR) is the most common orthopedic injury of the dog. History, signalment, gait, and orthopedic examination usually provide good information on the specific stifle injury. To confirm the diagnosis, radiographs are frequently taken. Further diagnostic imaging may include arthroscopy (Figure 13) or magnetic resonance imaging.

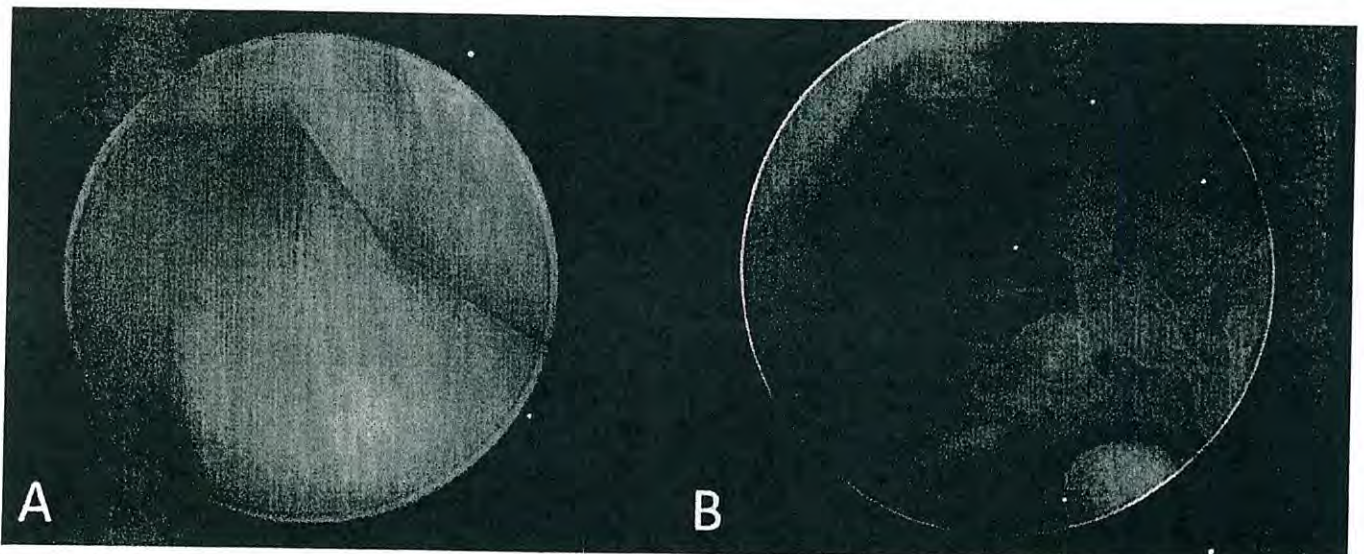


Figure 13: Arthroscopy of the stifle. Intact cranial cruciate ligament (A), severe joint inflammation secondary to ruptured cranial cruciate ligament (B). (Pictures: Brian Beale, DVM, MS, DACVS, Gulf Coast Clinic, Houston).

The surgical procedure of choice for CCLR in the first author's hands is a Tibial Plateau Leveling Osteotomy (TPLO). In a nutshell, with this procedure the direction of forces within the stifle joint is altered, resulting in a stable knee during weight bearing. There is a high likelihood for operated animals to return to racing activity if the procedure is performed accurately, and once healing is completed (Figure 14). To achieve best results, it is highly recommended to follow a strict physical rehabilitation protocol during the recovery period.

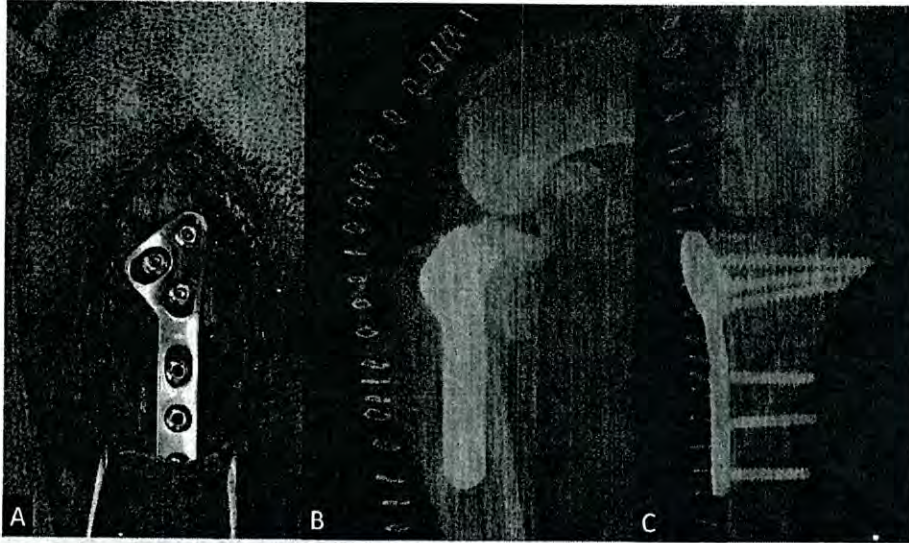


Figure 14: TPLO provides usually excellent return to function after a cranial cruciate ligament rupture. During surgery (A), post-operative radiographs lateral view (B) and anterior-posterior view (C).

Patella luxations (PL) are less likely to be a cause for lameness in a racing sled dog. However, it is a common orthopedic condition in the dog, and may be the cause for lameness in a young dog that just started training. Treatment is surgical and the prognosis is good, if the procedure is performed appropriately. To determine the best treatment for PL, several factors have to be taken into consideration. A grading system is described based on the amount of PL. Grade I: Luxation occurs only intermittently. The patella can be luxated manually, but replaces itself immediately. Grade II: After manual luxation, the stifle is flexed and extended, and the patella remains luxated. Grade III: The patella is luxated at all times, but can be pushed into normal anatomical position. Without manipulation of the stifle joint, the patella returns to the outside of the knee joint. Grade IV: The patella is luxated permanently and cannot be reduced manually. PL can occur medially or laterally. Most small breeds have medial luxations, most large breeds lateral luxations. Lateral luxations seem to be more common in dogs with straight legs and high riding patellas (Akita, Malamute). The treatment for PL depends on the degree of lameness and grade of PL, the chronicity, and the findings during surgery. In general, during surgery the goals are to achieve good seating of the patella within the femoro-patellar joint and anatomical alignment of the patellar tendon with the direction of quadriceps muscle pull. The surgical steps necessary to achieve these goals differ on a case to case basis.

Avulsion injuries to the tibial tubercle have been described for young racing animals. The growth plates in these patients may not be closed when heavy training is initiated. Chronic strain from the patellar tendon to the cranial tibial tuberosity may lead to avulsion injuries of that structure. Treatment depends on the severity of the injury.⁹ Further possible injuries of the stifle include collateral ligament damage, caudal cruciate ligament damage, fractures of the distal femur, the patella, the proximal tibia or fibula, or avulsion injuries of the cranio-proximal tibial tuberosity in young dogs. Most of these injuries need to be addressed by surgical stabilization and placement of screws, prosthetic ligaments, or bone plates. The prognosis depends on the degree of injury. A quick return to full activity may be achieved by using appropriate physical rehabilitation modalities.

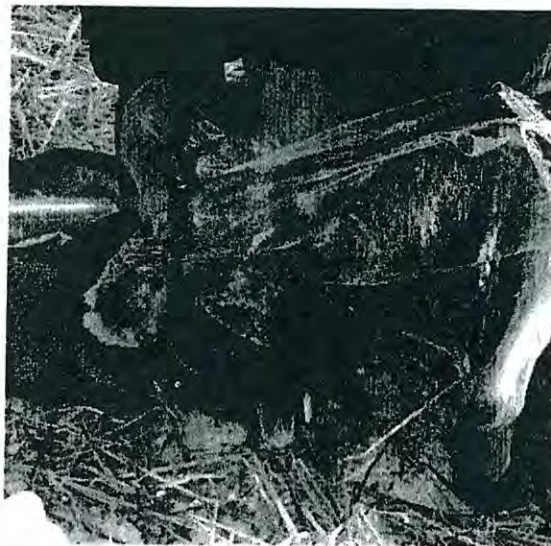
ACHILLES TENDON INJURIES

The diagnosis of Achilles tendon injuries is made by an appropriate orthopedic examination (see previous chapter). Achilles tendon injuries can be long-lasting problems. A dog with an Achilles tendon injury should be dropped. These dogs do not “warm out of the injury”. In contrast, continued exercise will worsen the condition. While weight bearing is possible at 6-8 weeks after injury, complete healing of a tendon may take as long as one year. Tendons never regain 100% of original strength after injury.⁵ If surgical repair is necessary, additional stabilization is achieved by application of a cast, or external fixator. Duration of post – operative care is usually 6-12 weeks. Dogs with chronic Achilles tendon injuries have a guarded prognosis for complete recovery. In contrast, acutely injured animals seem to have a better outcome.¹⁰ In general, the earlier treatment is performed, the better the prognosis. A quick return to full activity may be achieved by using appropriate physical rehabilitation modalities. Ultrasound treatment (pulsed 3.3 MHz) of Achilles tendon injuries has been shown to improve results through stimulation of collagen repair. Gastrocnemius muscle tears have a better outcome than tendon injuries. Many will heal with 6-12 weeks of strict confinement and appropriate rehabilitation when healed. Some dogs benefit during the initial weeks of healing from a supportive splint.

TARSAL INJURIES

The diagnosis of tarsal injuries is made by an appropriate orthopedic examination (see previous chapter). Differentials for tarsal injuries include: Ligamentous injuries, fracture, luxation, immune-mediated arthritis, osteochondrosis dissecans, or infection. Similar to carpal injuries, tarsal swelling may be treated successfully with tarsal wraps (Figure 15). For more severe injuries or diseases there is no treatment that can be performed successfully on the trail. With severe instability it may be helpful to apply a splint in an appropriate manner. Depending on the specific injury, surgical repair may allow the animal to return to racing activity. A quick return to full activity may be achieved by using appropriate physical rehabilitation modalities post surgery. With some tarsal injuries however, the prognosis for competitive racing activity is guarded.

Figure 15: Tarsal wraps may improve swelling. Cellophane band is applied to cover the liniment, and vetrap is used to wrap the tarsus.



NEUROLOGIC INJURIES

Injuries to the neck can result from acute trauma such as a fight or severe pull on a neck line. Soft tissue injuries such as a muscle strain may respond to massage therapy, rest, and the administration of a muscle relaxant, such as methocarbamol, and / or a non-steroidal anti-inflammatory. Primary spinal cord injury may be treated using steroids. It is contraindicated to administer NSAIDs and steroids together or within a short period of time. This can lead to significant damage to the gastro intestinal tract, the kidneys, liver or other organ systems. If there is severe, non-ambulatory lameness due to a compression of the spinal canal, for example from a protruded disc that has been diagnosed and localized appropriately via myelogram (Figure 16), computer tomography (CT), or magnetic resonance imaging (MRI), specific surgical procedures are indicated.



Figure 16: A myelogram can help to localize a lesion of the spinal cord. The dog in this figure suffered from intervertebral disc disease (white arrow head).

The location of the lesion often dictates the procedure. A ventral slot may be recommended for treatment of compression of the spinal cord in the neck area. A hemilaminectomy is recommended for treatment of spinal cord compression in the lumbar area. Depending on the underlying cause, the surgical approach and treatment method may differ. Fractures or luxations of the vertebral bodies may be treated using internal fixation methods. It is highly recommended to have a highly trained and experienced small animal surgeon perform these procedures in order to increase chances for an acceptable outcome. Recovery is improved when surgery is followed by a specifically designed protocol under the guidance of a compassionate veterinarian trained in veterinary physical rehabilitation. The prognosis depends on the severity of trauma, the time since injury, and the preoperative condition and neurological status of the animal. Various exercises, especially the use of hydrotherapy during the postoperative period, are likely to help the animal to improve steadily during the recovery period.

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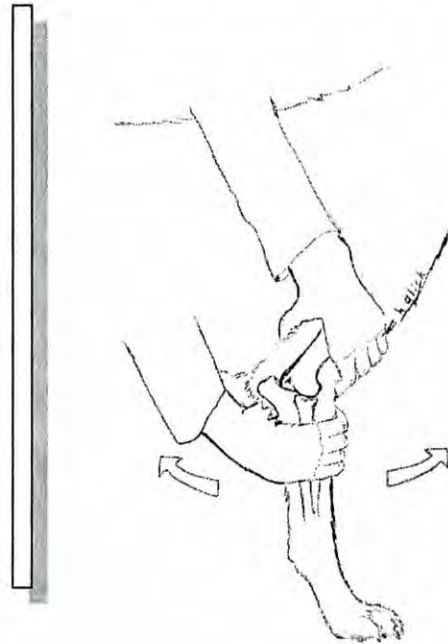
Orthopedic Examination Basics of Sled-Dogs

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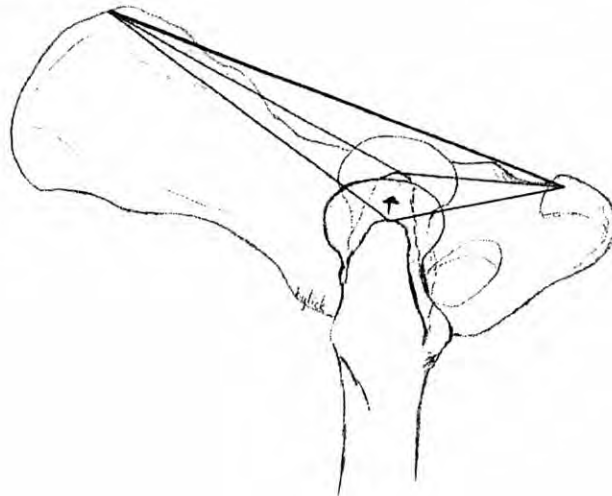
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Orthopedic Examination Basics for the Sled-Dog

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Term glossary for non-veterinarians:

Cranial – Towards the head
Caudal – Towards the tail
Proximal – Towards the body
Distal – Away from the body
Medial – Towards midline
Lateral – Towards the outside
Dorsal – Towards the back, or front aspect of the lower front or hind limb
Palmar – Caudal aspect of the lower front limb
Plantar – Caudal aspect of the lower hind limb
Radiographs – X-rays
Scapula – Shoulder blade
Humerus – Upper arm
Radius/Ulna – Lower arm
Carpus – Wrist
Metacarpus – Bones beneath the wrist
Pelvis – Hip
Femur – Thigh bone
Patella – Knee cap
Stifle – Knee
Tibia – Shin bone
Tibial tuberosity – Crest of the tibia in the front of the bone
Fibula – Small bone next to the tibia
Tarsus – Hock
Metatarsus - Bones below the hock
Flexion – Bending
Extension – Straightening
Crepitus – Grinding
Muscle atrophy - Wasting or decreased amount of muscle
Interdigital webbing - Skin between the toes
Vertebral column - back bone

Lameness may be the result of injury to bone, joints, tendon, muscle, ligaments or neurovascular structures. Therefore lameness may be variable, depending on localization and severity of injury. The orthopedic examination of sled dogs will mostly reveal injuries to the musculoskeletal system. This chapter focuses on techniques to diagnose an orthopedic injury or underlying orthopedic condition. For comparative reasons, a table showing normal joint range of motion values in the dog is provided (Table 1). In rare cases disorders of the nervous system may be the cause of lameness. For that reason, basic steps of a neurological examination are explained at the end of this chapter.

It is imperative to have a basic knowledge of the anatomy of the canine musculoskeletal system when performing an orthopedic examination. This will also help to understand the anatomic terms used in the paragraphs below. Providing the reader with this information is beyond the scope of this chapter. However, there is a tremendous amount of literature available regarding the anatomy of the dog. Canine skeletons can be purchased through various sources.¹ Books on canine anatomy are readily available.^{2,3} If you have the chance to observe a necropsy of a dog, ask the veterinarian to show you the most important muscles, tendons, bones and joints. The more anatomic knowledge you have, the easier it will be to localize and possibly treat the area of discomfort.

The goal of this chapter is to improve your skills when performing an orthopedic examination and to help diagnose orthopedic injuries in sled dogs. Never forget that orthopedic injuries are usually not life-threatening. In contrast, dehydration, poor nutritional status or diseases of the heart, lungs or gastro-intestinal tract are more dangerous to the racing animal than a sore muscle. *Never forget to evaluate these body systems thoroughly prior to the orthopedic examination.*

BASICS

The orthopedic examination includes several steps: 1.) Observation of the gait and obtaining information about the time of onset of lameness. 2.) Performance of a quick overall flexion/extension test. 3.) If necessary, thorough evaluation of the whole leg and all muscles. 4.) Performance of a basic neurologic examination, if indicated.

The gait is best observed when the dog team arrives at a checkpoint. A trail veterinarian should always make a major effort to see the team coming in. It may help to stand at a slight angle to the team when it arrives, and let it pass by. This permits evaluation of the dogs from cranially, laterally, and caudally. Dogs with frontleg lameness usually show a so called "head bob". This is the description of the head going down when the dog places his sound leg on the ground. In contrast, the head goes up when the dog places the injured leg on the ground. In addition, the stride is shortened on the affected side. Just imagine how an injured person walks if hurt on one foot. He or she won't place much weight on the injured leg, but will put more weight on the sound one. *The head falls on the sound leg.*

Dogs with hind leg lameness are a little more difficult to evaluate. The hips of the animal have to be observed. The sore leg prevents full weight bearing, and the hip drops on that side, or the knee may be bent more when compared to the other side. Obviously with very severe injuries, the dog will only toe-touch the ground, or holds up the injured leg.

The history of lameness should be reviewed. The musher should be asked if there are any other dogs he or she may be concerned about. Sometimes dogs are not pulling well or show signs of mild lameness during the run, but perk up when reaching the checkpoint. Some dogs may only show a gait change, not an obvious lameness. For example, loping instead of trotting. If there is a problem, most mushers gladly share their concerns about a dog that may have an orthopedic injury. Occasionally there may be different opinions on the significance of lameness. It needs to be remembered that the musher has spent a very long time with his dogs, and should know their gait very well. On the other hand, mushers should not forget that trail veterinarians are professionals who have learned to examine dogs over years of study, and have gained a great deal of experience in their professional lives. Some have even specialized in orthopedic injuries. Over time, both mushers and veterinarians learn to associate specific gait abnormalities with certain orthopedic injuries, and are able to determine the significance of each gait abnormality.

While the musher takes the booties off, snacks the dogs, and prepares the straw beds, the "Vet-Book" (if available at the race) can be checked for remarks from the last checkpoints. Also evaluate the overall appearance of the dogs, and observe appetite, urination, and the way they lie down. Some dogs may favor one leg, some may tremble or seem otherwise uncomfortable. Prior to starting the examination, it is never wrong to ask the musher if all dogs can easily be examined or if a dog needs to be held while being examined. Better safe than sorry.

At most checkpoints it isn't possible to take a dog out of harness to perform a thorough evaluation as is taught during veterinary school. However, in some cases it may be best to remove the lame dog from the team, and do a complete, "text-book" orthopedic examination. Depending on the situation, the examination can be performed by one person within a few minutes, or may require the help of an assistant, and the complete process may last longer.

As with every time an animal is approached, it is helpful to talk to the dog making it aware of the presence of the veterinarian. Most dogs will be very tired when arriving at a checkpoint and lie down. It may be necessary to prop them up so you can perform your exam. For safety reasons, it is best to straddle the dog. Always try to keep one arm between the dog's head and the examiner's face. Use the left (right) arm to hold the right (left) front leg of the animal, and always keep an eye on the head of the dog. This may decrease the chance of getting bitten. In the vast majority of cases however, sled dogs are used to being handled and the veterinarian does not have to be concerned.

A dog that is not in harness should be restrained appropriately by an assistant. To hold a dog in an appropriate manner on the ground, the assistant should kneel with the knees perpendicular to the back of the animal. Appropriate placement of the hands is important. For example, if the animal is in left lateral recumbency, the assistant's left hand is placed on the left front leg as close to the body as possible. The left elbow of the assistant is placed onto the neck of the animal. The right hand reaches through the hind legs from cranially, and the left hind leg is grasped firmly as proximal as possible to the body of the dog. This will prevent the dog from getting up, and the veterinarian can thoroughly evaluate the animal.

PROTOCOL AND QUICK EXAMINATION

It is imperative to develop a protocol that is followed during every orthopedic examination. This will prevent forgetting a step and decreases the chance of missing a possibly important injury. In general, the right leg should always be compared with the left. This helps to pick up injuries. Being firm during the examination is recommended, as most sled dogs have a high threshold for pain. Some dogs may show only very subtle signs of pain such as facial wincing, withdrawal of the limb or flinching during the exam. However, too much force should not be applied either. Most mushers do not appreciate hearing their dog crying out when being examined. On the other hand, despite gentle handling, some dogs may be severely injured and react very quickly with vocalizing. To prevent misunderstandings between musher and veterinarian it is important to discuss the cause for discomfort on a professional level.

Obviously, if there is evidence of a fracture or another obvious injury, it may not be necessary to perform a complete orthopedic examination. A dog with an orthopedic injury should show one of the following: swelling (soft or firm), pain on flexion, pain on extension, decreased range of motion, crepitus, muscle atrophy, muscle cramping or spasm.⁴ Pain is elicited easier with acute injuries rather than with chronic ones. Dogs that get up after a rest may show lameness that disappears after warming out of the problem. Some dogs may not show pain when they first arrive, but stiffen as the muscles cool down. Therefore, if the musher has seen a problem with a dog, and on initial examination no injury is found, it is recommended to re-examine the dog in a few hours.

As an overall "quick-check" it may be permissible to flex the carpus, elbow, and shoulder at the same time by grasping the dog's metacarpus, and pull it up towards the lateral chest wall. A healthy dog should be able to flex all these joints without any resistance. A similar test can be performed for the hind leg. The metatarsus is grasped and the leg flexed. An injured animal may resist these tests and the examiner should be alerted. The downside with this test is that several areas are evaluated simultaneously. If a pain response is elicited it will be difficult to assign it to a specific part of the leg. It is then important to isolate the area of interest. When performing a more thorough examination, one hand should always be placed above, and one below the joint or area of bone that is examined.

ORTHOPEDIC EXAMINATION

To perform a detailed orthopedic examination, the following steps should be followed:

- 1.) The toes, pads, and skin of the front legs are examined as follows: The foot is picked up and snow and ice is removed. The toes are spread and the interdigital webbing is evaluated. The pads are inspected as they may be injured (Figure 1). The paws are examined for abrasions, discolorations, puncture wounds or swelling. The toe nails are evaluated. Every digit is flexed and extended, and medial and lateral stress are applied. As an experienced trail veterinarian put it once: "The feet of sled dogs are like the tires of race cars. Only minor changes in quality can make huge differences." This should be kept in mind. *Be very thorough during the examination.*

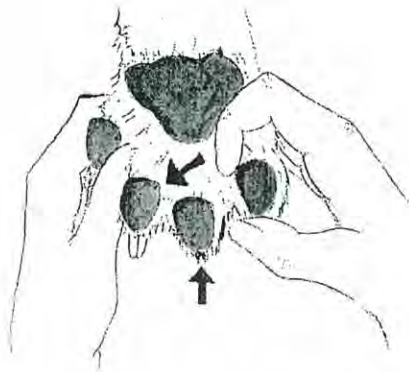


Figure 1: Webbing and pads of the foot are evaluated. Look for lesions on the pads and toe nails (red arrows)

- 2.) Carpal injuries are among the most common problems in the sled dog. The stance and gait of the dog are observed first. It needs to be determined if there is more swelling on one side, or a different angle of the wrist when standing, when compared to the other side. Careful palpation of the dorsal surface of the carpus with the finger tips will help to detect minor swelling (Figure 2). To flex and extend the carpus, the leg is held with one hand proximal the carpus, and the other hand is placed distal to it (Figure 3). Movement of the elbow and shoulder joints should be avoided when performing the evaluation of the wrist. The normal carpus can be flexed to a point where the pad touches the caudal area of the lower leg (Figure 4). The normal extension of the carpus is 10-12 degrees. With the carpus in extension, lateral and medial stability are tested. Instability may be indicative of collateral ligament damage. To do this test, the leg is held above and below the carpus. A certain degree of laxity laterally and medially is normal. It is important to compare the right and left carpus to determine whether the instability that may be felt is normal or pathologic. To detect damage to the accessory carpal bone, pressure is applied to this structure while carefully extending the carpus (Figure 5). A pain response should be followed by taking radiographs. The flexor tendons caudal to the carpus should also be palpated for swelling.



Figure 2: Evaluation for a swollen carpus.



Figure 3: Extension of the carpus.



Figure 4: Flexion of the carpus.



Figure 5: Evaluation for injuries of the accessory carpal bone.

3.) Elbow problems are uncommon in sled dogs. However, this part of the orthopedic examination should not be overlooked. Elbow swelling can sometimes be observed by just looking at the joint and comparing it to the other side, while the dog is standing. This may be difficult in animals with a thick hair coat. Palpation for swelling is done by placing the index finger between the lateral epicondyle of the humerus and the olecranon, and placing the thumb between the medial epicondyle and the olecranon. Both elbows are palpated at the same time. Swelling leads to a bulge between the two bony landmarks when the animal puts weight on the affected leg (Figure 6). The elbow is placed through a range of motion (Figure 7). To evaluate discomfort on the level of the medial coronoid process of the ulna, the carpus can be flexed and rotated internally while holding the elbow slightly flexed. Application of pressure on the medial coronoid process with the thumb of the other hand may elicit a pain response (Figure 8). Joint effusion in association with a pain response to this test is frequently indicative of an injury to the medial coronoid process of the ulna. Collateral ligament injuries are examined next. The carpus and the elbow are flexed and then the paw is rotated internally and externally. With intact medial and lateral collateral ligaments the paw can be rotated 40-50 degrees and 80-90 degrees, respectively. Again, the right and left leg should be compared.



Figure 6: Elbow swelling is evaluated by moving the finger from epicondyle to olecranon and back (blue arrow).



Figure 7: Evaluation for range of motion of the elbow.

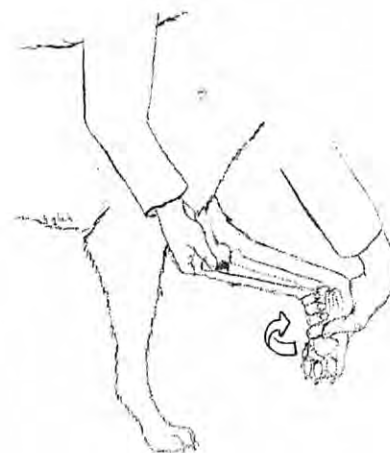


Figure 8: Evaluation for injuries of the medial coronoid process.

4.) Shoulder injuries are among the most common injuries of sled dogs. To examine the canine shoulder, one hand is placed proximal to the elbow on the caudal aspect of the humerus. The other hand is placed over the cranial aspect of the scapula. The shoulder is extended by pushing the humerus forward, and the shoulder joint is stabilized by holding the cranial part of the scapula. Animals that react upon extension of the shoulder may have a joint injury or a strain to the triceps muscle, or other muscles in the caudal area of the shoulder joint. Some biceps injuries will also result in a pain response on extension. The position of the hands is then changed. The shoulder is flexed by pulling the humerus caudally and placing the index- and middle finger of the other hand over the shoulder joint. The bicipital groove is palpated, and some pressure is applied over the biceps tendon. Pain upon flexion of the shoulder can be associated with biceps tendon or biceps muscle injury. The response for these injuries can be exaggerated by deep and thorough palpation of the biceps tendon. With a ruptured biceps tendon, the elbow is straight, compared to an angled elbow joint with intact biceps tendon (Figure 9). Occasionally a pain response when performing this test can result from injuries of the supraspinatus muscle. Injuries to the glenohumeral ligaments or of the joint cartilage may be difficult to diagnose via an orthopedic examination. Some subjective tests to evaluate these structures are described. However, more sophisticated diagnostics such as arthroscopy, computed tomography (CT), or magnetic resonance imaging (MRI) done at a specialty center will likely provide more objective data to diagnose such problems:

Figure 9: Biceps tendon rupture results in a straight elbow when flexing the shoulder joint (red), compared to normal (green).



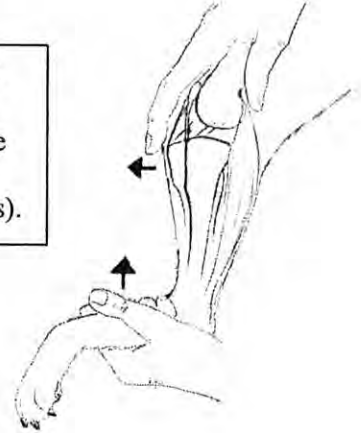
- 5.) The toes on the hind legs are evaluated as described for the front legs.
- 6.) Tarsal (hock) injuries are not commonly observed in sled dogs. The evaluation of this joint follows for the most part the description given for the carpus (see above). Swelling of the tarsal joint is best felt by palpating the area just proximal to and on each side of the calcaneus (Figure 10). The collateral ligaments can be evaluated with the hock held in extension. The joint is opened laterally and medially.
- 7.) The Achilles tendon consists of five tendons (tendon of the gastrocnemius muscle, common tendon of the biceps femoris, semitendinosus, and gracilis muscles, and the tendon of the superficial digital flexor muscle). A partial Achilles tendon rupture (only the superficial digital flexor tendon is intact) usually results in a typical stance of the animal, with the tarsus hyperflexed (dropped) in comparison to the non-injured leg, and the toes in a flexed "crablike" position. Complete Achilles tendon rupture results in a plantigrade stance, or extreme hyperflexion, of the hind leg (the dog seems to walk on the hock instead of on the toes). Achilles tendon injury or strain will lead to swelling of the tendon. To detect swelling, palpate the gastrocnemius muscles and Achilles tendons on each side simultaneously (Figure 10). The tendon should also be palpated thoroughly during flexion and extension of the hock. Shortly after injury, it may be possible to detect a gap between the calcaneus and the avulsed end of the tendon several centimeters above the calcaneus. Over time this gap will be filled with fibrous scar tissue, possibly giving the examiner the false impression of examining an intact tendon. Radiographs (x-rays) or ultrasound of the tendon may help to support the diagnosis.

The “Indirect Drawer Test” (Figure 13) is performed by placing the palm of one hand over the patella, while extending the index finger of that hand and placing it on the most proximal aspect of the tibial tuberosity. The palm of the other hand is placed on the plantar (caudal) aspect of the metatarsus (bones below the hock). The stifle joint is extended or just slightly flexed, while the hock is flexed to approximately 90 degrees. With a CCLR, the index finger of the hand that is placed around the stifle will move forward. This test frequently allows detection of CCLR in heavily muscled dogs that may not show instability with the “Direct Drawer Test”. Sometimes a mild degree of tibial rotation is observed when performing the tests for “Drawer”. This should not be mistaken for evidence of CCLR. Dogs with CCLR, especially chronic cases, are at risk for the development of meniscal injuries. The meniscus is a structure that is located between the femur and the tibia. It acts as a cushion between these two bones. Meniscal injuries may occasionally be detected by an audible “click” when the dog is walking or be felt as a “pop” during examination.

Figure 12: “Direct drawer test.” The tibia moves cranially with respect to the femur.



Figure 13: “Indirekt drawer test.” While flexing the tarsus the tibia is pushed cranially (red arrows).



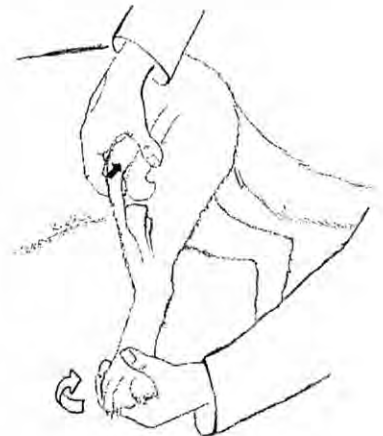
Joint swelling without instability of the stifle joint in young dogs may be associated with other etiologies that may need to be diagnosed using more sophisticated tests such as radiography or arthroscopy.

Instability of the collateral ligaments is evaluated by extending the joint and trying to open it laterally or medially. For this test one hand holds the femur above the stifle joint and the other hand holds the tibia below the stifle joint. Then the examiner tries to open the joint medially or laterally. Compare the two legs and be sure to have the hind leg extended. Differences in the degree of joint laxity associated with discomfort are likely the result of collateral ligament damage. Patella luxations are uncommon in active racing sled dogs. However, they can be the cause of lameness in young dogs that just started training. The tests for patellar luxation should be performed with the stifle in different degrees of flexion and extension. Medial patellar luxation is diagnosed by extending the stifle and partially flexing the tarsus while rotating the later internally (Figure 14). The thumb of one hand is placed on the lateral aspect of the patella, and the patella is pushed medially out of the femoro-patellar joint. The opposite is done for lateral patellar luxations (Figure 15). Dogs with patellar luxations can obtain a secondary CCLR (see above). Therefore it is imperative to always examine dogs with patellar luxations for evidence of CCLR.



Figure 14 (left): Test for medial patella luxation.

Figure 15 (right) : Test for lateral patella luxation.

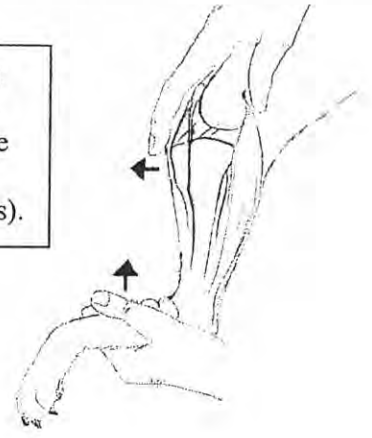


The “Indirect Drawer Test” (Figure 13) is performed by placing the palm of one hand over the patella, while extending the index finger of that hand and placing it on the most proximal aspect of the tibial tuberosity. The palm of the other hand is placed on the plantar (caudal) aspect of the metatarsus (bones below the hock). The stifle joint is extended or just slightly flexed, while the hock is flexed to approximately 90 degrees. With a CCLR, the index finger of the hand that is placed around the stifle will move forward. This test frequently allows detection of CCLR in heavily muscled dogs that may not show instability with the “Direct Drawer Test”. Sometimes a mild degree of tibial rotation is observed when performing the tests for “Drawer”. This should not be mistaken for evidence of CCLR. Dogs with CCLR, especially chronic cases, are at risk for the development of meniscal injuries. The meniscus is a structure that is located between the femur and the tibia. It acts as a cushion between these two bones. Meniscal injuries may occasionally be detected by an audible “click” when the dog is walking or be felt as a “pop” during examination.

Figure 12: “Direct drawer test.” The tibia moves cranially with respect to the femur.



Figure 13: “Indirekt drawer test.” While flexing the tarsus the tibia is pushed cranially (red arrows).



Joint swelling without instability of the stifle joint in young dogs may be associated with other etiologies that may need to be diagnosed using more sophisticated tests such as radiography or arthroscopy.

Instability of the collateral ligaments is evaluated by extending the joint and trying to open it laterally or medially. For this test one hand holds the femur above the stifle joint and the other hand holds the tibia below the stifle joint. Then the examiner tries to open the joint medially or laterally. Compare the two legs and be sure to have the hind leg extended. Differences in the degree of joint laxity associated with discomfort are likely the result of collateral ligament damage. Patella luxations are uncommon in active racing sled dogs. However, they can be the cause of lameness in young dogs that just started training. The tests for patellar luxation should be performed with the stifle in different degrees of flexion and extension. Medial patellar luxation is diagnosed by extending the stifle and partially flexing the tarsus while rotating the later internally (Figure 14). The thumb of one hand is placed on the lateral aspect of the patella, and the patella is pushed medially out of the femoro-patellar joint. The opposite is done for lateral patellar luxations (Figure 15). Dogs with patellar luxations can obtain a secondary CCLR (see above). Therefore it is imperative to always examine dogs with patellar luxations for evidence of CCLR.



Figure 14 (left): Test for medial patella luxation.

Figure 15 (right) : Test for lateral patella luxation.



9.) Pelvic (hip) injuries of sled dogs are mostly limited to muscle strains. Dogs with hip dysplasia or degenerative changes are unlikely to enter a race on a competitive level. Other injuries such as hip luxations or fractures may occur after accidents on the trail and could be the cause of lameness. Evaluation of the hip joint should begin with simultaneous palpation of both hind legs. Muscle swelling or atrophy can be evaluated by letting both hands glide simultaneously from proximal to distal with one hand on each hind leg. An evaluation for hip luxation or other causes for hip bone asymmetry is done next. The index fingers are placed on the tuber coxae, the thumbs on the tuber ischii, and the middle fingers on the greater trochanters. Asymmetry of the bones of the pelvis may be an indication of dislocation, fracture, or chronic coxofemoral arthritis. The three fingers of each hand should form a triangle. With the most common form of hip luxation (craniodorsal), the middle finger on the affected side will be positioned more proximally, when compared to the contralateral side (the triangle is “flattened”) (Figure 16). Animals with hip luxation frequently are toe-touching lame, and the affected leg is carried in a non-weight-bearing position, with internal rotation of the foot.

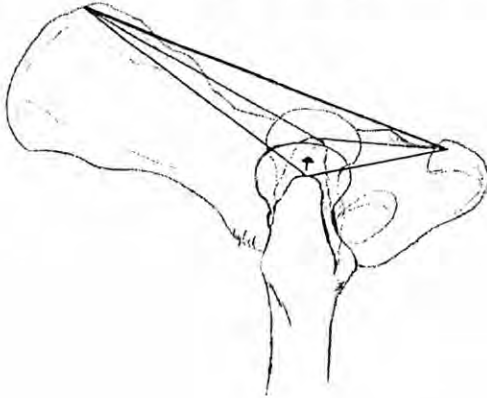


Figure 16: With hip luxation the triangle formed by the examiner's fingers is flattened (red) when compared to normal (black).

Extension of the hip is performed by placing the hand on the femur, just proximal to the cranial aspect of the stifle. The other hand is placed flat on the caudal lumbar spine or hip. The femur is then carefully pulled caudally (Figure 17). A healthy dog should be able to extend the hip joint to a point at which the hind leg is approximately on one level with the hip. Animals with hip injuries may resist this amount of extension. Flexion of the hip is performed by placement of the palm of the examiner's hand just caudal and proximal to the stifle joint, and pulling the femur dorsally and cranially (Figure 17). A consistent method to elicit pain from the hip in animals with a hip joint problem is to abduct the femur. One hand is placed on the medial aspect of the stifle, and the other on the lower back of the animal. Then the leg is carefully pulled laterally. The iliopsoas muscle courses from the transverse processes of 5th to 7th lumbar vertebral bodies to the third trochanter on the femur. To test for iliopsoas muscle injury, the examiner places his arm between the hind legs of the dog, and puts the palm of the hand on the medial aspect of the stifle. The cranial aspect of the hock will automatically rest on the dorsal aspect of the examiner's elbow, while the hind leg is extended. With the fingers of the contralateral hand, the examiner places pressure on the origin of the iliopsoas muscle (Figure 18). A healthy animal will allow this test to be performed without reacting uncomfortably. Fractures of the pelvic bones may be detected during a rectal examination.

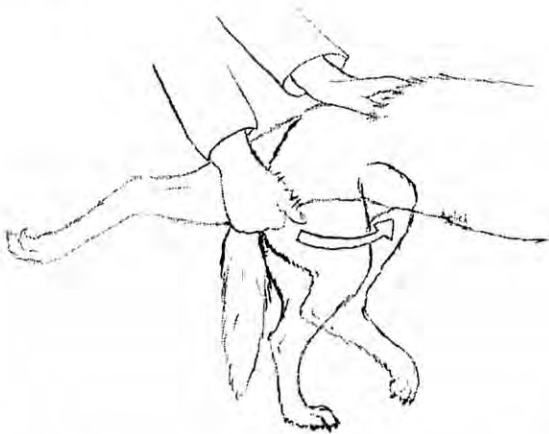


Figure 17: Hip flexion and extension.

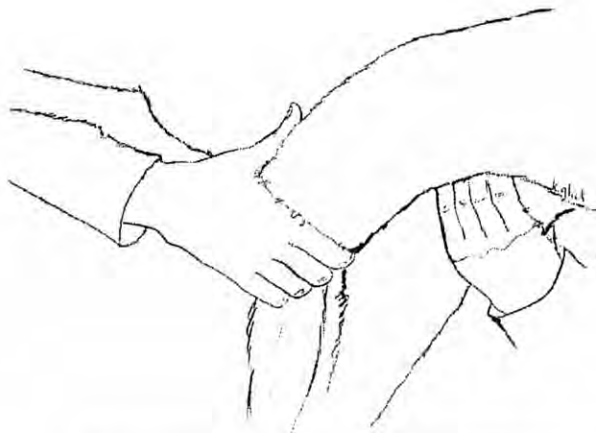


Figure 18: Test for iliopsoas muscle

10.) Injury to the sacro-iliac joint or the lumbo-sacral joint can lead to severe hindleg lameness. These areas are exposed to tremendous stress while racing, as all the force arising from the hind legs is translated into cranial movement of the body through these structures. Examination is performed by extending both hind legs simultaneously and applying pressure to the junction between the lumbar spine and the sacrum. Healthy animals will not react uncomfortably. Dogs with lumbo-sacral disease frequently react by sitting down suddenly when pressure is applied to the caudal back area. Severely affected animals may also experience pain upon lifting the tail, or show decreased anal sphincter tone.

11.) If the above listed examinations do not reveal a "sore spot", the cause of lameness may be secondary to a muscle injury, or secondary to pain from primary bone disease. While the former is common in racing sled dogs, the latter is not to be expected on the trail, but could be the cause for lameness in very young, severely injured, or in older, retired sled dogs. To localize the affected muscle, it is necessary to know its anatomical origin, course, and insertion. Closely associated with this knowledge is the nomenclature of bones and bony prominences of the canine skeleton. The most commonly injured muscles of the front leg are the biceps, triceps and extensor muscles. In the rear leg, common muscle injuries are the extensors, flexors, and rotators of the hip joint, as well as the gastrocnemius muscle. Deep, careful palpation of each of these muscles is necessary to localize the injured region. Chronic injuries (3-4 weeks) may lead to muscle atrophy on the affected leg. This may be detected when comparing the right and left leg. Chronic front leg lameness frequently results in atrophy of the shoulder muscles, which may be detected by a more pronounced acromion on the affected side. Primary bone diseases, such as panosteitis (inflammation of the bone) in young dogs, osteomyelitis (infection of the bone), for example after an open injury or surgery, or bone cancer (mainly in older animals) are frequently associated with a pain response upon deep palpation of the affected area of the long bones. Palpation of the long bones should be reserved for the last part of the orthopedic examination as bone pain can be extreme at times. The best locations to palpate the long bones without interfering with overlying musculature are the distal radius, proximal ulna, distal and proximal humerus, medial aspect of the femur, medial and lateral aspect of the stifle, and medial aspect of the tibia. Increasing pressure is applied to these areas and the animal is carefully observed for a reaction. In most cases with a positive response, radiographs should be taken to obtain a definitive diagnosis.

12.) Injuries to the spinal cord or peripheral nerves can also be the cause for lameness in sled dogs. The description of a complete neurologic examination is beyond the scope of this chapter, but information on this examination is widely available in the literature.⁶ A few basic steps include acute neurologic dysfunction, which may arise secondary to neck injuries which are seen mainly in teams using neck lines or after a tangle with dogs in the gangline; dogs may also get a leg caught in a loop of the gangline resulting in nerve damage or severe orthopedic damage. The composition of the harness may also play a role in the etiology of both neurological and orthopedic injuries of sled dogs. Animals with neck pain won't pull well, and will guard their neck when being examined. Besides ataxia (a wobbly gait), conscious proprioceptive deficits (the dog cannot feel where his feet are) are the earliest symptoms associated with neurological dysfunction. To evaluate conscious proprioception, the paw is knuckled over on its dorsal aspect, one foot at a time. Normal animals replace the foot immediately. If the flow of information from the paw to the brain is impaired, the paw will not be replaced to a normal stance immediately.

Some dogs may also injure the muscles surrounding the vertebral column or the vertebral bodies themselves. Depending on the degree of injury, affected dogs may show only pain upon palpation of the back, may not be able to coordinate all limbs, or may have difficulties moving one or several legs normally. To determine whether there is back pain, the examiner should palpate and apply pressure to the spinous processes of the vertebral bodies. The epaxial muscles on either side of the spinous processes should be palpated next. As there are no spinous processes on the cervical vertebral bodies, palpation of the neck is limited to deep palpation of the epaxial muscles, and the vertebral bodies. The vertebral bodies may be easier to locate in thin racing dogs, compared to regular pet dogs. Further examination of the neck includes careful flexion (nose down to the chest), extension (nose up into the air), and bending to the right and left (nose should touch the lateral chest wall on each side in a normal dog). It is critical to be very careful with these tests, as they may result in worsening of the condition when too much force is applied. Tumors of the nerves that supply the front leg may occur in older dogs and result in severe muscle atrophy of the front legs. Frequently a large, painful mass in the axillary area can be found upon deep palpation. Once more it is recommended to compare the right and left side of the standing dog to confirm the finding. Dogs with lumbo-sacral disorders may also show neurological abnormalities. The examination of this area is described above. Radiology, myelography, CT, or MRI may be necessary for a definitive diagnosis.

ADDITIONAL DIAGNOSTICS

It requires much practice to become an experienced examiner for orthopedic injuries. Using the techniques described above, it may not be possible to identify some injuries clearly. With unresolved, obscure lameness it may be worthwhile considering a consultation with an experienced veterinarian or veterinary surgeon who is interested in orthopedic injuries of the sled dog. Diagnostic imaging techniques, such as radiography, ultrasound, arthroscopy, CT, or MRI, as well as laboratory tests, such as blood work and/or joint fluid examination, may be helpful in definitively diagnosing the cause for lameness in the sled dog.

Table 1: Normal Joint Range of Motion in the Dog (Average of 10 mixed-breed dogs).⁷

| Joint | Range of Motion (°) | Joint | Range of Motion (°) |
|-------------------|---------------------|-------------------|---------------------|
| Shoulder | | Hip | |
| Flexion | 60-70 | Flexion | 70-80 |
| Extension | 65-75 | Extension | 80-90 |
| Hyperextension | 0 | Adduction | 30-40 |
| Adduction | 40-50 | Abduction | 70-80 |
| Abduction | 40-50 | Internal rotation | 50-60 |
| Internal rotation | 40-50 | External rotation | 80-90 |
| External rotation | 40-50 | | |
| Elbow | | Stifle | |
| Flexion | 70-75 | Flexion | 65-75 |
| Extension | 70-75 | Extension | 65-75 |
| Hyperextension | 0 | Hyperextension | 0 |
| Supination | 80-90 | | |
| Pronation | 40-50 | | |
| Carpus | | Tarsus | |
| Flexion | 155-160 | Flexion | 65-75 |
| Extension | 20-30 | Extension | 90-110 |
| Ulnar deviation | 15-20 | Inversion | 40-50 |
| Radial deviation | 15-20 | Eversion | 40-50 |

Acknowledgements: The authors thank Nelson Priddy II, DVM, DACVS and Jim Rearden for editorial work.

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Basics of Wound Care in the Sled-Dog



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Basics of Wound Care in the Sled-Dog

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Skin defects from trauma or mass removal are very common in small-animal practice. Appropriate wound care and techniques to close skin defects are necessary for a good functional and cosmetic outcome. The purpose of this chapter is to provide or refresh basic knowledge in wound treatment, to give an update on recent techniques of wound management and to introduce practical and easily performed techniques for wound closure. Most wounds discussed may be seen in the kennel or on the trail. Specifics on injuries of the pads are covered elsewhere in this book. An outlook on more involved cases is also provided. Finally, complications in wound management and unconventional wound treatment options are briefly discussed.

WOUND HEALING

Wound healing progresses via four phases: Inflammation, proliferation, maturation and remodeling. During the inflammatory phase, the initial clot is formed and neutrophils migrate into the wound. Fibroblasts and epithelial cells migrate into the wound during the proliferative phase. During maturation and remodeling the wound takes on its final appearance. A normal skin wound heals within two to four weeks, depending on size and severity of the wound. However, healing can be delayed by local or systemic factors. Local factors include technique and experience of the surgeon, diminished blood supply, increased tension, presence of infection, mechanical stress, shape of the wound, suture material, foreign material, or radiation injury. Systemic factors include hypoproteinemia, hypovolemia, edema, malnutrition, hypovitaminosis, systemic diseases, age, immunodeficiency, or cytotoxic drugs. Appropriate patient support is indicated to achieve best healing results.

THE USE OF ANTIBIOTICS IN WOUNDS

The definitive choice for an antibiotic should be based on culture and sensitivity results. However as this is frequently not practical in a sled-dog environment, broad-based coverage may be achieved by administration of 1st generation cephalosporins (Cephalexin) or amoxicillin with clavulanic acid (Clavamox). Antibiotics should be given for at least 14 days. To prevent nosocomial or highly resistant infections, radical debridement and lavage is preferred over the inappropriate use of antibiotics. As soon as healthy granulation tissue appears in the wound, the risk of infection decreases, and antibiotics may be discontinued.

TREATMENT OF FROSTBITE

Frostbite is a common problem in the sled dog (Figure 1). The recommended recipe below is based on an original recipe by Dr. Van Pelt of Fairbanks, AK. Ingredients: 4# Zinc oxide ointment, 1# Thuja zinc ointment, 3# Triple antibiotic ointment (Bacitracin/Neomycin/Polymyxin B), 1# 0.1% Triamcinolone ointment, 875cc mineral oil. Preparation: Warm jars in hot water (coffee pot), pour into tub and mix with drillsand paint stirrers (keep this tub within outer larger tub with hot water to keep ointment warm and liquid), pour into cake decorating bag, and squeeze into 4 oz. REI food tubes filling slightly less than three-quarters. Directions: Apply to affected areas 2-3 times daily. Other treatments such as povidone iodine based ointments have also been shown to help treating frostbite.



Figure 1: Frostbite on the scrotum.

TREATMENT OF BITE WOUNDS OR SMALLER LACERATIONS

Bite wounds or small skin lacerations that are observed during training runs or a race should be treated initially as indicated in the first section of open wound treatment (debridement, lavage). Specific information on wound closure is provided in the section on this topic below. Frequently, it is sufficient to clean the wound from debris, hair, and blood, suture the wound under sedation or local anesthesia using absorbable monofilament suture material, and initiate antibiotic treatment. The use of skin staplers is discussed controversial among trail veterinarians. While it is a rapid method to close a skin defect, the metal in the skin may become very cold, leading to localized frostbite. It is obvious that this does not support good wound healing, and may lead to pain from the cold metal.

OPEN WOUND TREATMENT

If a wound cannot be closed, open wound treatment is necessary. It is important to understand that appropriate wound care can only be provided inside. *At temperatures below the freezing point, wet bandage material may freeze and may lead to severe problems.*

Appropriate wound healing requires a clean wound bed. Therefore, if there is any doubt whether a wound is contaminated or not, open wound treatment is indicated prior to closure (Figure 2). Open wound management allows for continued wound drainage, regular wound inspection, lavage and debridement. Sharp debridement under heavy sedation or anesthesia is performed to clean the wound of foreign material and necrotic tissue. Wound lavage is imperative during open wound management and can be performed by either using a syringe system, consisting of an 18g needle and a 60ml syringe, or by using a clean bottle with 18g needle holes in the top (Figure 3). Alternatively, commercial wound lavage systems are available^a. Ringer's diluted to a 0.05% chlorhexidine solution, is an excellent choice for lavage of open wounds (1# of 2% stock solution and 40# of sterile Lactated Ringer's Solution – don't use Sodium Chloride – chlorhexidine will precipitate in NaCl). Other solutions have been shown to be cytotoxic or to have other negative effects on wound healing.¹ Hydrogen peroxide is contraindicated as wound lavage solution because it kills not only bacteria, but also all reparative cells in the wound, which in turn will delay wound healing.

Traditionally, open wound treatment has been performed using wet-to-dry bandages (Figure 4). These bandages work well and are indicated especially in the early debridement phase, when there is a lot of contamination.



Figure 2: Example of a severe degloving injury on the carpus of a dog. Open wound management is indicated if the leg is to be saved.

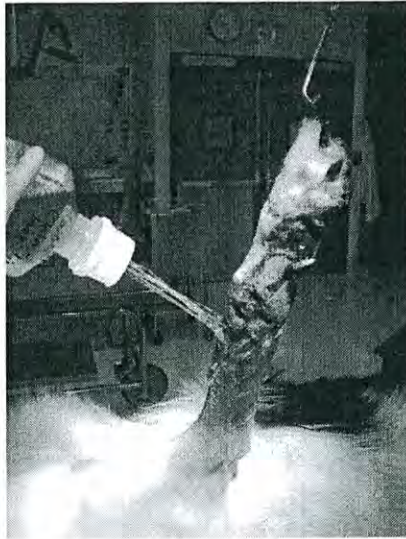


Figure 3: Ringer's diluted to a 0.05% chlorhexidine solution, is an excellent choice for lavage of open wounds.

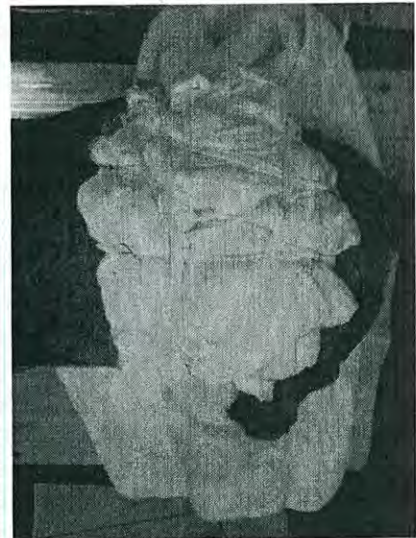


Figure 4: A wet to dry bandage is applied to a severely contaminated wound.

With moderate amounts of exudates, alginate dressings^b may be applied. Recent studies in the human field have shown that a warm and moist environment with a pH around 6 decreases multiplication of bacteria and enhances the formation of granulation tissue. Alginate and other dressings provide this environment^b. The absorptive capacity of an alginate dressing is three times that of an equivalent sized gauze sponge. They can be left on the wound for two to three days. Once the exudation decreases, polyurethane foams^b can be used. They also have larger absorptive capacity than usual sponges and are an effective barrier against bacteria. Finally, once granulation tissue forms, the use of hydrogel dressings^b is

recommended (Figure 5). They are impervious to bacteria, allow rehydration, debride via autolysis and absorb necrotic tissue. They may be left on the wound for two to three days. In comparison to wet-to-dry bandages, there is minimal pain associated with the removal of hydrogel dressings. Unconventional dressings such as honey, sugar, maggots or leeches are shortly discussed at the end of this paper. An intermediate layer, like cast padding and an outer layer to protect the bandage are necessary to complete the bandage. Do NOT use plastic bags or diapers around the wounds. *The wound needs to breathe*. Once healthy granulation tissue is evident, the wound can be closed.

Occasionally healthy granulation tissue won't form, or the wounds are too big for standard closure. In these situations, wound healing may be facilitated by using vacuum-assisted closure, ultrasound treatment of wounds, different skin flaps, and advanced techniques such as free skin grafts or distant flaps using microvascular anastomosis. To provide the reader with specifics about these fascinating wound treatment methods is beyond the scope of this chapter. Adequate training, experience and appropriate equipment to perform these wound closure techniques are necessary and should be done in a veterinary hospital by surgeons, familiar with these procedures.²⁻⁴

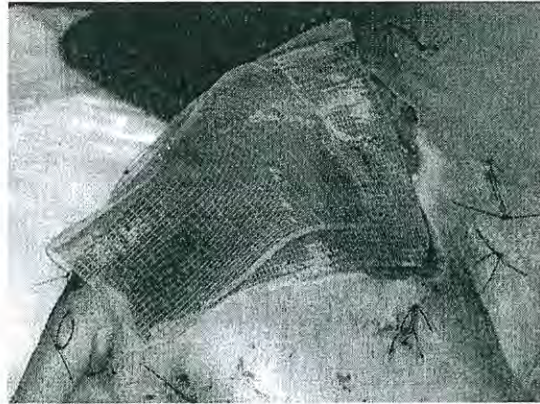


Figure 5: Advanced wound dressings, such as hydrogel, enhance the formation of granulation tissue.

WOUND CLOSURE^{2,3}

The timing of wound closure depends on the amount of contamination and extent of the injury. The four types of wound closure options include: Primary closure, delayed primary closure, secondary closure, and healing by second intention. To avoid dehiscence, it is essential to close all wounds without tension. If necessary, grafting or appropriate flap techniques should be considered. Clean and very fresh wounds may be closed primarily. A commonly used rule of thumb is the “6-hour-rule”. A wound that is older than 6 hours should not be closed immediately. Open wound treatment should be performed first. By definition, delayed primary closure is performed before granulation tissue forms, usually after about five days. Delayed primary closure is performed in clean-contaminated wounds, with questionable tissue viability. Open wound treatment and wound lavage for three to five days is commonly necessary. In contrast, contaminated wounds are usually closed after the appearance of granulation tissue (unless a vital structure is exposed, such as esophagus, trachea, nerves, etc). Initial debridement and open wound treatment are necessary, allowing the elimination of exudates and contamination. Secondary closure can be performed at five to seven days. With large skin defects, severe contamination or the appearance of necrotic tissue, a wound may heal by second intention. Granulation, contraction and re-epithelialization help to cover the wound over time. However, the cosmetic result with this type of wound closure may be inferior, and contracture may cause some functional issues. Small wounds and trunk wounds can more often be treated successfully with second intention healing. As with any surgical procedure, it is important to follow Halstead's principles of surgery when closing a wound. Halstead's principles of surgery include: gentle tissue handling, accurate hemostasis, preservation of blood supply, aseptic technique, wound-closure without tension, careful approximation of tissue, and avoidance of dead space.

The placement of “walking sutures” may help to close the wound without any tension at the wound edge. A heavy gauge monofilament suture material is used to place stitches into the granulation tissue and into the skin, pulling the skin towards the wound edge (Figure 6). If you are uncertain about the risk of infection, or if “dead space” is a concern, place a drain. Closed suction drains^c are preferable over penrose drains in large wounds (Figures 7 and 8). Sled-dogs with drains should be kept in a room to avoid freezing of the exposed tissues or the draining equipment.



Figure 6: Walking sutures help to pull wound edges together and allow closure without excessive tension.

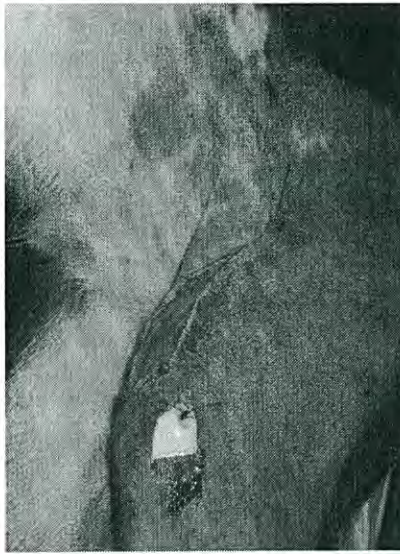


Figure 7: If there is doubt about infection or dead space, a drain should be placed. A penrose drain is shown. Note: This dog chewed on the incision and the wound opened. Re-suturing was necessary. Therefore placement of an E-collar is recommended.

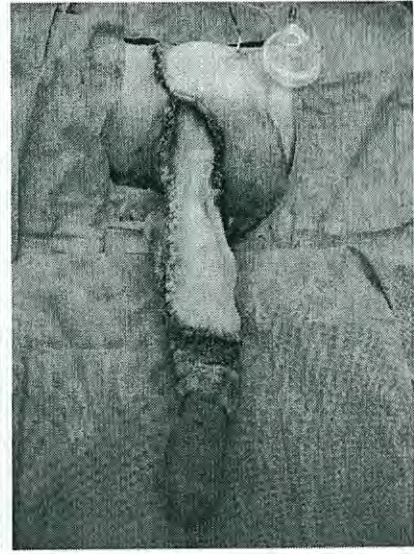


Figure 8: A closed suction drain provides drainage with minimal risk of wound contamination.

COMPLICATIONS WITH WOUND HEALING OR SKIN RECONSTRUCTION¹⁻⁵

Complications with skin reconstructive surgery include hemorrhage or hematomas, which are usually treated with the application of warm compresses. Seromas can form as well. In contrast to infected areas, seromas are not hot or inflamed and a complete blood count will not show signs of infection. Aspiration of seromas increases the risk of infection. Therefore, seromas should not be aspirated. The application of compression bandages and warm compresses will help with the resorption of the seroma. Dehiscence occurs usually three to five days after surgery. If there is too much tension, if there was poor tissue handling, or if there is too little blood supply to the area, dehiscence is likely. The cause for dehiscence should be determined and measures taken to prevent further negative effects on healing. For example, releasing incisions might need to be performed or open wound treatment initiated. Infections of surgical wounds occur in 2.5% of clean surgeries⁵, and may be prevented by the use of strictly aseptic techniques and intraoperative antibiotics. A common protocol is the administration of Cephazolin (22mg/kg IV) at induction followed by intravenous administration every two hours during surgery. Signs of infection usually occur two to three days after surgery. Common cultured bacteria from surgical wounds include: *Staphylococcus species*, *E.coli*, *Bacteroides*, *Clostridium*, and *Pasteurella*. When facing an infection, culture and sensitivity tests should be initiated and surgical drains should be placed. A broad-spectrum antibiotic should be administered until culture and sensitivity tests reveal the most potent and appropriate antimicrobial agent to be used. Finally, problems like delayed- or incomplete wound healing and wound contracture may need to be addressed. If an atypical wound is present, or if the wound is not responding as expected, the following causes have to be considered: Mycobacterial infection, bacterial or fungal disease, underlying host factors (Cushing's disease, diabetes, renal disease etc.), underlying osteomyelitis, or polycythemia. Depending on the cause, medical therapy, tension-relieving techniques, skin flaps or free skin grafts may need to be performed in order to address these problems.

UNCONVENTIONAL WOUND TREATMENT METHODS

Unconventional wound treatment methods include the use of honey, sugar, maggots and leeches. These measures are not adequate for wound treatment on the trail, but may need to be applied occasionally under the guidance of a veterinarian to treat specific wound of dogs in the kennel. Medical grade honey has been shown to have bactericidal effect, and appears in some cases superior to many modern methods of wound treatment. Wounds treated with honey show decreased inflammation and accelerated sloughing of necrotic tissue. Honey provides a local cellular energy source, forms a

protective layer of protein over the wound, and forms a healthy granulation bed. Large case studies in veterinary medicine are lacking, and despite amazing results in the human field the use of honey has been slow to come into acceptance in our profession.⁶

Granulated sugar has also been proven to be beneficial in the treatment of open, contaminated wounds. Sugar reduces the need for debridement because macrophages are drawn into the wound, which cleans the wound and accelerates sloughing of devitalized tissue. Its use enhances healing and eliminates the need for antibiotics because of its rapid antibacterial action. As with honey, its use in animals is not yet widespread.⁷

Maggot debridement therapy is the medical use of specially selected and tested, disinfected fly larvae ("maggots") for cleaning non-healing wounds. In 2004, the Food-and-Drug-Administration (FDA) approved the marketing of medical maggots. Medicinal maggots have been found to have four principle actions: 1) They debride (clean) wounds by dissolving the necrotic, infected tissue. 2) They disinfect the wound by killing bacteria. 3) They stimulate wound healing. 4) They break down and inhibit the formation of biofilm. Maggot therapy was used in the early 20th century until it was replaced by the use of antibiotics. With the increasing amount of antibiotic resistance, this form of therapy for non-healing, infected wounds, is returning. It is uncommonly used in animals, but studies in humans show that this form of wound treatment may help in cases, in which conventional wound treatment methods are unsuccessful.⁸ Leeches can be used to relieve severe venous congestion, following skin-flap procedures. While some reports support their use, other authors raise concerns about the increased risk of infection, or the efficacy of leeches.⁸ It is important to understand that these wound treatment techniques have to be monitored closely by a veterinarian and should not be applied without specific knowledge of wound treatment.

Acknowledgement: The author thanks Nelson Priddy II, DVM, DACVS and Jim Rearden for editorial work.

Footnotes:

a) Stryker Interpulse® System, Kalamozoo, MI 49001, <http://www.stryker.com>. b) The Kendall Company, Mansfield, MA 02048, <http://www.kendallhq.com>. C) Closed "Blake Drain kit": www.ethicon.com.

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References of Cultural Importance of Sled Dogs

SPIRIT OF THE WIND: An instructional unit in Alaska State History

IDITAROD Educational WEB SITE

Healthy Heros ASHOF feature about George Attla

Sled Dog Health and Mushing

Link Between Human-Animal Bond and Mental Health

UAF URSA 388 student paper about sled dogs & FAYSDP

Cultural Importance of Dog mushing

SPIRIT OF THE WIND: An instructional unit in Alaska State History

Written by Dr. Ronald Smith, University of Alaska-Fairbanks Professor of Zoology, Emeritus

Introduction:

SPIRIT OF THE WIND: An instructional unit in Alaska State History is a stand-alone unit incorporating elements and readings about George Attla, perhaps the most famous Alaska musher ever. Aspects of his and Athabascan life and culture are explored through viewing *Spirit of the Wind*, George's story up to 1979, plus related readings.

This instructional unit is not designed as a complete semester-long course in Alaska state history. Rather, it addresses selected aspects of the body of knowledge expected of students completing such a course. It can be used to augment an existing course or as an introduction to Alaska state history.

The elements of Alaska history recognized by the Alaska Board of Education as needed in such a course are listed below. This unit focuses, in part, on aspects of Alaska geography, Alaska's cultures, and modern Alaska.

High School Graduation Requirement:

After a statewide advocacy effort, led by the Alaska Humanities Forum, the State Board of Education amended 4AAC 06.075. This regulation now requires that as of January 1, 2009, the three units of social studies required for graduation must include one-half unit of credit in Alaska history or demonstration that the student meets the Alaska history performance standards. This does not apply to a student (1) who transfers into an Alaska public school after the student's second year of high school or (2) has already successfully completed a high school state history course from another state.

Some elements of Alaska state history:

- Geography: <http://www.akhistorycourse.org/articles/article.php?artID=313>
- Alaska's Cultures: <http://www.akhistorycourse.org/articles/article.php?artID=315>
- Russia's Colony: <http://www.akhistorycourse.org/articles/article.php?artID=316>
- America's Territory: <http://www.akhistorycourse.org/articles/article.php?artID=316>
- Governing Alaska- <http://www.akhistorycourse.org/articles/article.php?artID=318>
- Modern Alaska: <http://www.akhistorycourse.org/articles/article.php?artID=318>

Materials for the unit:

Spirit of the Wind: the life of George Attla and his sled dog racing

Study Guide with additional references to:

George Attla
tuberculosis
dog racing
trapping
subsistence fishing
boarding schools
the Molly Hootsch decision
the relocation of Alaska bush villages

Reading assignments and focus questions for the student to address

An "open book" written assessment

Alaska Native Heritage Center

ATHABASCANS

01-01-2000

Who We Are



The Athabascans traditionally lived in Interior Alaska, an expansive region that begins south of the Brooks Mountain Range and continues down to the Kenai Peninsula. There are eleven linguistic groups of Athabascans in Alaska. Athabaskan people have traditionally lived along five major river ways: the Yukon, the Tanana, the Susitna, the Kuskokwim, and the Copper river drainages. Athabascans were highly nomadic, traveling in small groups to fish, hunt and trap.

Today, Athabascans live throughout Alaska and the Lower 48, returning to their home territories to harvest traditional resources. The Athabaskan people call themselves 'Dena,' or 'the people.' In traditional and contemporary practices Athabascans are taught respect for all living things. The most important part of Athabaskan subsistence living is sharing. All hunters are part of a kin-based network in which they are expected to follow traditional customs for sharing in the community.

House Types and Settlements

The Athabascans traditionally lived in small groups of 20 to 40 people that moved systematically through the resource territories. Annual summer fish camps for the entire family and winter villages served as base camps. Depending on the season and regional resources, several traditional house types were used.

Tools and Technology

Traditional tools and technology reflect the resources of the regions. Traditional tools were made of stone, antlers, wood, and bone. Such tools were used to build houses, boats, snowshoes, clothing, and cooking utensils. Birch trees were used wherever they were found.

Social Organization

The Athabascans have matrilineal system in which children belong to the mother's clan, rather than to the father's clan, with the exception of the Holikachuk and the Deg Hit'an. Clan elders made decisions concerning marriage, leadership, and trading customs. Often the core of the traditional group was a woman and her brother, and their two families. In such a combination the brother and his sister's husband often became hunting partners for life. Sometimes these hunting partnerships started when a couple married.

Traditional Athabascan husbands were expected to live with the wife's family during the first year, when the new husband would work for the family and go hunting with his brothers-in-law. A central feature of traditional Athabascan life was (and still is for some) a system whereby the mother's brother takes social responsibility for training and socializing his sister's children so that the children grow up knowing their clan history and customs.

Clothing

Traditional clothing reflects the resources. For the most part, clothing was made of caribou and moose hide. Moose and caribou hide moccasins and boots were important parts of the wardrobe. Styles of moccasins vary depending on conditions. Both men and women are adept at sewing, although women traditionally did most of skin sewing.

Transportation

Canoes were made of birch bark, moose hide, and cottonwood. All Athabascans used sleds --with and without dogs to pull them – snowshoes and dogs as pack animals.

Trade

Trade was a principle activity of Athabascan men, who formed trading partnerships with men in other communities and cultures as part of an international system of diplomacy and exchange. Traditionally, partners from other tribes were also, at times, enemies, and travelling through enemy territory was dangerous.

Regalia

Traditional regalia varies from region to region. Regalia may include men's beaded jackets, dentalium shell necklaces (traditionally worn by chiefs), men and women's beaded tunics and women's beaded dancing boots.

Assignment 1: Watch the movie Spirit of the Wind. It portrays the life of George Attla up to the year 1979. Be prepared to answer the following questions about the movie and his life:

1. How did George Sr., George's father, run his trap line? That is, what was his means of transportation?
2. What was the source of heat for the family cabin?
3. How did the Attla family buy the goods they couldn't catch, grow, or make themselves?
4. How did the family reach summer fish camp?
5. At what age did George contract tuberculosis?
6. Where did he go for treatment and cure?
7. How long was he away from his home?
8. Where did he attend school while he was being treated for tuberculosis?
9. Where is the Fur Rendezvous World Championship race held?

Assignment 2: Go online and read this 2009 article on the making of the movie:

http://attlamakingofachampion.com/wp-content/uploads/2011/12/2009.10.00-First-Alaskans-Magazine_RJ.jpg

Assignment 3: Read the newspaper article about George's first Fur Rondy win:

http://attlamakingofachampion.com/wp-content/uploads/2011/12/1958.02.24-ADT-Attla-Dog-mushing-Champion-COMP_RJ.jpg

1. Write down the route he took to get from Huslia to the Fur Rondy race site.
2. How many methods of transportation did he use?
3. On a map of Alaska, figure out how many miles he traveled to get to the race.

Assignment 4: If you are interested in sled dogs, read the article on the origins of Alaskan sled dogs: http://www.sleddogcentral.com/features/little_wolf/alaskans.htm

Assignment 5: An important piece of Alaska history was made by both non-native and Alaska native mushers: the 1918 serum run from Nenana to Nome. Read the article about the serum run:

Assignment 6: Working with sled dogs may have unintended or unexpected benefits. Read the interview of George in Mushing Magazine in which he talks about how working with dogs influenced the way he raised his children: <http://attlamakingofachampion.com/wp-content/uploads/2011/12/2012.03-Mushing-Magazine-Part-II-Attla.pdf>

1. Summarize or describe how he saw the connection.
2. Might there be a similar benefit from raising and training hunting dogs? Pet dogs? Explain in a paragraph or two your reasoning.

Assignment 7: Late in life, George used his energies to establish a dog mushing program for middle and high school students in Huslia. This is, perhaps, another unexpected benefit

of working with sled dogs. Read his obituary:

<http://www.adn.com/print/article/20150215/famed-alaska-musher-george-attla-dead-81>

1. Write a paragraph, based on the obituary and perhaps the interview you read, about how the Frank Attla Youth and Sled Dog Care Mushing Program could benefit Alaskan youth. Use your imagination and your understanding about personal accomplishment as it relates to a person's self-image. Please discuss this assignment with your parents. They will have definite ideas to share with you.
2. Develop a short list of activities in which you are or could be involved that might build your confidence and sense of self-worth in a way similar to the mushing program just mentioned in the obituary.

Assignment 8: Recall in the movie that when George finally returned to his family and community he had been removed from his culture for almost ten years. He had a different set of experiences than his brother. He knew less about the fish wheel and its operation than his brother. In many respects, he was caught between two cultures due, in part, to his boarding school experience. Read the article on cultural assimilation of Native Americans as background for the next assignment:

wikipedia.org/wiki/Cultural_assimilation_of_Native_Americans#Non-reservation_boarding_schools

Assignment 9: In the 1960s and, perhaps earlier, Alaska villagers, mostly native, began to push for local village high schools. They were seeking to preserve family and cultural unity. Read *Tobeluk v Lind*, the court case that led to a resolution of this issue:

wikipedia.org/wiki/Tobeluk_v_Lind

See also, the article on the Molly Hootch decision, the legal resolution of the law suit:
http://www.alaskool.org/native_ed/law/mhootch_erq.html

1. How many high school-age students does it take to require the state of Alaska to supply a village high school?
2. Who was Molly Hootch?
3. Make a list of five reasons why you might not like to be sent hundreds or thousands of miles away from home to a boarding school.
4. Suggest one potential advantage that might come to you at a boarding school. You might want to write paragraphs to answer these last two challenges.

Iditarod Sled Dog Education Portal

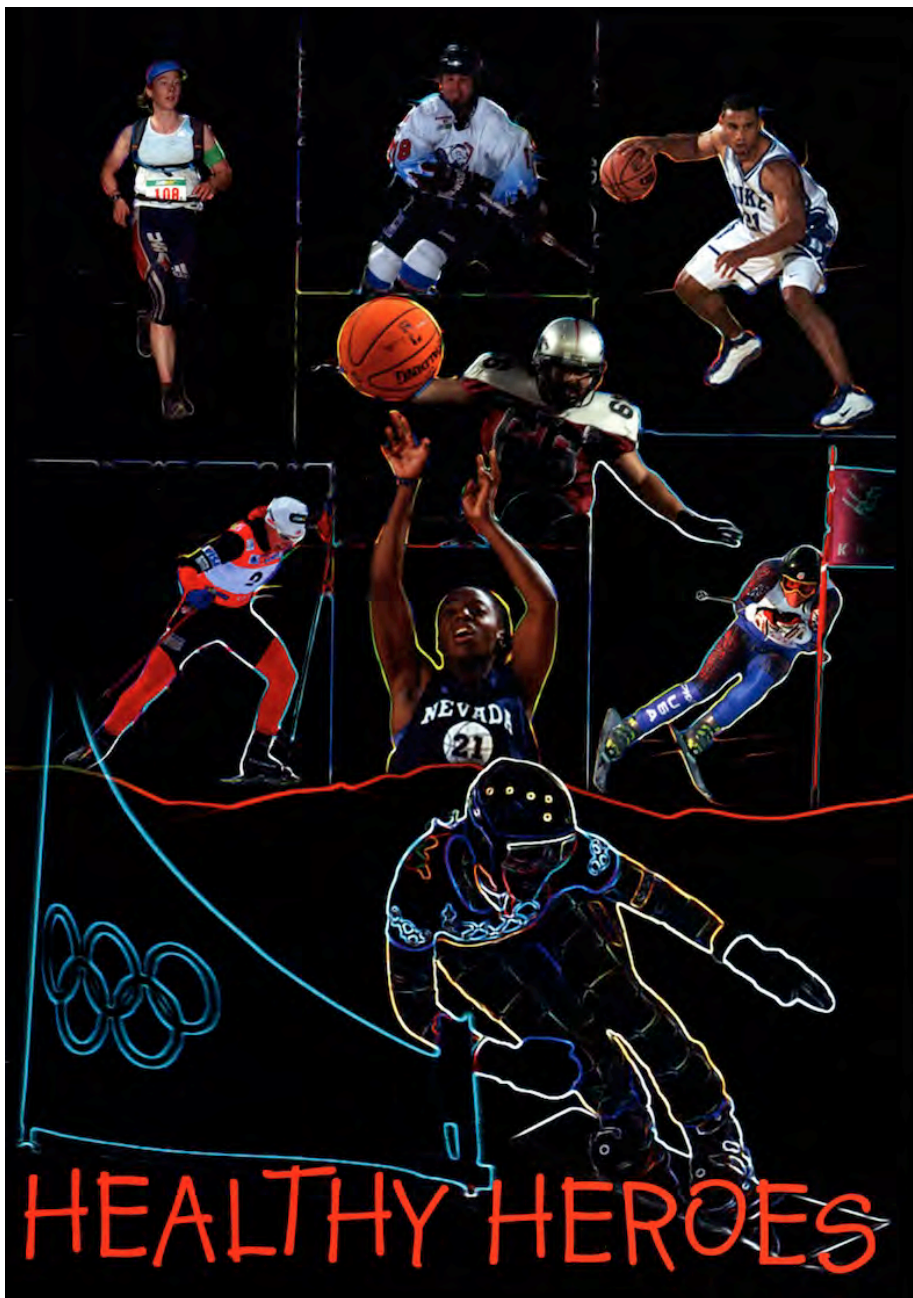
About the Education Department and Contact Information

<http://iditarod.com/teacher/mission-statement-and-goals/>

Portal for Teachers

<http://iditarod.com/teachers/>

Left column has specific content area index of educational ideas and materials



ConocoPhillips Healthy Futures is a community initiative to provide Alaska's youth with greater access to positive role models in sports through the Healthy Heroes project. Founding partners in the project include the Municipality of Anchorage Parks & Recreation, the Anchorage School District, and the Alaska Association for Health, Physical Education, Recreation & Dance ConocoPhillips Healthy Futures initiative and the Alaska Sports Hall of Fame.

This project would not be possible without the efforts of the outstanding Alaskan athletes who contributed their inspiring messages.

The written submissions have been edited here for space considerations. Visit the Healthy Heroes home at www.alaskasportshall.org to read the Healthy Hero contributions in their entirety and find out about opportunities to chat online with your favorite Healthy Hero!

Photos courtesy of Sharbel Dussault, Bob Hill, Erik Hill, Devon Kershaw, Clark James Michler, Eric Teela, Anchorage Daily News, Dallas Morning News, Duke Athletic Department, and the Leadville Times



Anchorage School District
Educating All Students for Success in Life



ConocoPhillips
Healthy
FUTURES
An initiative of the Alaska Association for Health,
Physical Education, Recreation and Dance
www.healthyfuturesak.org

ANCHORAGE



PARKS & RECREATION

EXERCISE

I lived in a remote part of Interior Alaska with my family until I was around eight years old when I contracted tuberculosis in my knee and was sent away to a hospital. At that time the only known way to keep use of my leg was to be put in traction, so I was forced to stay in bed and be totally constricted for over a month. My body became very weak from the inactivity. While stuck in a hospital bed, I was pestered by many kids. When I was finally allowed to get up, I was mad and went after the biggest kid. I was shocked that I was so easily beat up. From that time on, I vowed never to let myself get out of shape. I started working to strengthen my upper body by doing pull-ups, push-ups, and sit-ups. After this first attempt to straighten my leg, I went back to the bush and swung a 10 lb birch hammer every morning. In helping my family live off the land, I did many necessary chores like packing wood. By the next year my leg was bent again. I went back in the hospital for more traction and this time I made friends with a janitor, who kept in tremendous shape by using a spring bar and weights. Once again confined to bed I used his bar and weights daily, while continuing to do pulls-ups, sit-ups, and push-ups. At 15 years old after an operation to permanently fuse my knee, I got out of the hospital but stayed in a boarding school for two years where I learned how to box and play basketball to stay fit. I then went back home and kept in shape all winter by setting beaver traps which entailed cutting and chopping ice and snow shoeing, plus I ran a trap-line and hauled wood using dogs. By spring I felt in such good shape that I thought I could whip the world. In summer, I hauled fish nets, cut 400-500 salmon a day, rowed boats, and used a sledge hammer to pound stakes. — **George Atila**

What do you do on an average day to exercise?

It doesn't matter how long you exercise or what activity you choose for your exercise. It matters that you get out and do it every day! — **Aelin Peterson**

He ran everywhere. If you served under his command in the military, the only time that you were permitted to walk was if you were with a higher ranking officer who was walking. — **Col. Norman Vaughn's wife Carolyn on Norman**

Make sure the activities or sports you choose are fun and enjoyable. Make daily activity a part of your healthy lifestyle. — **Dick Mize**



EXERCISE

I try to go to the gym five days a week. I lift weights and either ride the bike or run for 40 minutes. — **Dean Larson**

I started forming good habits early on, so now it's no problem for me to work out three times a week from 5:30 to 7:00 a.m. — **Eric Heil**

I carry 20 gallons of water everyday to feed my dogs three times a day plus I pick up after them. In winter I cut and haul all my firewood, shovel snow, travel by snow machine and trap beaver under the ice. In summer I set nets to catch salmon. Basically I move all day, everyday. — **George Atila**

When you have a spare hour and can either choose to watch TV or get in a workout, make the decision that will benefit you the most. — **Karen Patten**

My average day of exercise usually includes two workouts. I start my first workout around 8:00 a.m. and train one to two hours. My second workout is at 4:00 p.m. and ranges from 30 min - 1 hr. — **Kikkan Randall**

You don't have to run marathons. Just walking — something almost all of us can do — is great exercise and a great way to start down the path of being active everyday. — **Kris Thorsness**

Alaskans have an advantage in that the wilderness is usually just out our back door, inviting us out. It is easy to stay healthy when hiking, fishing, berry picking and snow shoveling are part of everyday life. — **Libby Riddles**

I try to make up for my bad eating habits by being very active. Basketball is my favorite sport. I try to play basketball at least two to five times per week. — **Ma'o Tosi**

Everybody uses a car, but I used to run, ski, or bike seven miles to work. I wasn't the best dressed person in the office, but I kept a change of clothes I could get in and out of. — **Stan Justice**

For exercise I play basketball, lift weights and do some long distance jogging. — **Trajan Langdon**

I try to ski a lot and go for walks. Nordic skiing is also fun in the winter. — **Tommy Moe**

I stretch, stretch, stretch, strengthening my body to do anything I want it to, which is important for all of the different type of Alaska Native Games events. I build on my core stretches, adding weights to help gravity push me even further. After all of that work, my hope is that I will continue kicking and jumping for years to come. — **Phillip Blanchett**

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Joseph Bifelt

Dr. Taylor

URSA 388

4 May 2015

Introduction

I want to teach Alaska Native youth, not only about United States history or world history but also about their own history. In school, I learned about the western ways of civilization but I did not learn about who my own people are. Are we not important enough to be mentioned? Do we not matter? What this omission does to a young Alaskan Native child is cause confusion and negative effects to their pride. When I would visit my grandpa (George Attla Jr.), I would gain a sense of pride in who I was, where I come from, and in addition, I would feel inspired each time I visited him. There was no mountain too big to climb after hanging with the champion sprint dog musher, George Attla Jr. The sky was the limit for me after hearing stories told by my grandfather, which were about my ancestors and the struggles they had to overcome. After hearing these stories and going through these experiences, I felt like I could be somebody in this world and felt that I could also help inspire other Alaska Native youth. That is why I chose to spend a race season dog mushing under the guidance of George and taking distance education classes through the University of Alaska Fairbanks. Dog mushing has been a learning tool and a way to connect residents of all ages in a rural Alaskan community.

In the Alaska Native culture, dog mushing was originally used for transportation, work, hunting, and trapping. Sled dogs can lead a person out of a storm and also sense problems such as thin ice or perhaps the edge of an upcoming cliff. Evidently, the Alaska Native people trusted their sled dogs with their own lives to guide them through the extreme Alaskan climates in order

to go out and search for food to eat or wood to burn for heat. Since the Alaska Native people trusted their sled dogs with their lives and also had to feed them, not to mention their own families, a sled dog had to be of exceptional quality or else it would not survive. The mushers could not afford to care for a dog that did not work hard or care for itself. For example, if a dog team had crossed overflowing water on the ice, then each dog was expected to immediately clean slush and ice from its paws so that the cold temperatures would not freeze their feet.

With the arrival of the snowmobile, came a decrease in the need for sled dogs. A snowmobile may not be able to guide a person through a storm but it could surely be parked and left in one spot for months at a time (providing that a bear does not tear it up!). The efficient snowmobiles can also generate a higher rate of speed and, depending on the type of engine, just as much if not more horsepower to pull heavy loads. Therefore, the majority of Alaska Natives got rid of their dog teams and bought snowmobiles. The few that still carry on the dog mushing tradition mainly use their sled dogs for racing or recreational purposes.

With new technology and new medicines came an easier life for Alaska Natives, which is both good and bad. When the Indigenous people of Alaska first came in contact with the Russians and the Western world, they were introduced to new illnesses that their bodies were not adapted to and many people died. The new medicines that the Caucasian people brought saved many Indians and Eskimos from diseases such as tuberculosis, influenza, polio, measles, and many more. New modern day technology allows for more innovative ways of living, which has helped the Alaska Native people communicate, learn, and access tremendous amounts of information a lot faster and with far less labor than ever before. The downside of the easier, more modern way of life for the Alaska Native people is that since they do not have to work as hard to accomplish daily tasks, they may become lazy! Training, caring for, and racing sled dogs can

definitely provide work for a person who wants to keep themselves busy both mentally and physically.

All my life, I've felt stuck between two worlds; the Indian world and the Western world. I never wanted to travel too far in either one of them because in order to accomplish big things as a young Native American, I feel that I have to balance the two worlds and get the best of both of them. I can live the Indian way of life by living off the land, but I can also have a stable job as a teacher in rural Alaska, coach basketball (the game that I love), and inspire young rural Alaskans to go out and achieve greatness in which ever field of study they wish. All of this inspiration and fire to go out as a minority and be the best in the world has been passed on to me from George Attla Jr.

Activities

This past fall and winter, I have been teaching Alaska Native youth how to care for, train, and race sled dogs. I began working with the dogs from early-October 2014 until late-March 2015. The dog mushing duties included feeding once a day, watering at least twice a day, cleaning, maintaining straw, maintaining splits/cuts on dog feet, vaccinating puppies, fishing, building dog houses, fixing broken sleds, running the dogs, dropping dogs from their boxes in the dog truck (when on the road in Fairbanks, Willow, Anchorage, North Pole, Tok, and Tanacross, Alaska.), and making sure trails were well groomed. Each day, caring for the dogs was necessary from six o'clock in the morning until around twelve o'clock midnight. The amount of care for the dogs while 'on the road' in the dog truck amplified because every few hours of each day they needed to be "dropped," as I call it when the dogs are let out of their boxes to relieve themselves or to stretch to loosen up their muscles and joints. Every task that I

was required to do was taught to Huslia children through the Frank Attla Youth and Sled Dog Care-Mushing Program.

Approximately sixty children under the age of eighteen, visited the Huslia kennels weekly. It was no shock to have around forty children participate in and watch weekend youth races hosted by the community. Of the children participating in the program, fifteen were male and fifteen were female. Up to four times per week, pre-school/elementary, middle school, and high school groups took turns visiting the kennels and learning through hands on activities and storytelling. They also had the opportunity to help on weekends at any of the three local kennels. For the ninety minute long classes in the dog yard, I would handle dogs and make sure that the kids were staying on task. The local mushers, teachers, participating parents, and myself all act as positive role models for the children. Local youth now know to perform many tasks associated with the care, training, and racing of sprint sled dogs.

The children who have been a part of the youth and sled dog program have also learned how to handle and care for themselves. They have learned how to eat well and care for their body, dress warm, stay in good physical shape, maintain a good attitude for themselves and their dogs, control their emotions, think ahead, manage time, be responsible, and make a commitment in order to care for and train dogs. [borrowed from “20 things that working with sled dogs can teach youth” by George Attla on July 4, 2012]

Caring for dogs requires much work, and it is a job that you have to love in order to do because dogs can sense your feelings and mood. A winning dog team is a happy dog team, and in order to have a happy dog team, a person has to maintain a good attitude each time they step in the dog yard.

Expected Results

I hope to give a charge to Alaska Native youth's cultural activity and awareness by participating with them in working with sled dogs (The main source of transportation for Alaska Natives before the snowmobiles were introduced.). So far, there has been roughly sixty students participating in the youth and sled dog program who, I believe, will feel pride in who they are and where they come from just as I have when working with dogs and the Alaska Native elders. There is a feeling that emerges when running dogs that makes a person feel free and natural in a way that presents a sense of what life was like before the mid-twentieth century. Life was tough in the era before rifles, snowmobiles, running water, and electricity so it seems that stories told of those hard times by the elders can inspire youth nowadays to take advantage of the opportunities set forth in front of them and also take pride in who they are and the tough people they come from.

Dog mushing requires a lot of hard work, dedication, sacrifice, critical thinking, teamwork, communication skills (dealing with the media), and plain old toughness in order to keep your composure in high pressure situations and to be happy to care for over thirty animals for months at a time. I feel that Alaska Native youth today can benefit from working and being around sled dogs because there are all different types of dogs that need to be managed and treated in a dog team. The children and I gained critical thinking and preparation skills, while keeping our emotions under control when managing shy, hyper, lazy, sensitive, and hard-working dogs into one smooth-operating team. Dogs in general are very accepting towards humans of all shapes, ages, color, and sizes, because they do not judge people and in this manner they can create a bond of love and friendship with the children in the youth sled dog program. I expect an increase in cultural awareness in Alaska Native youth and also an increase in

communal activity resulting from my participation in the Frank Attla Youth Sled Dog Care-Mushing Program.

Expected Outcomes

I anticipate that the community will benefit with increased communal activity because during the dog races, there are people of all ages participating and making the race become a success. During dog races, there are the youth racing, the adults grooming the trails, adults officiating the race, and there are the Elders cheering for their grandchildren. With an increase in communal activity comes an increase in community bonding and a commitment to teamwork.

Results and Outcomes

The love of dog mushing has spread throughout the community of Huslia, Alaska and beyond. The Frank Attla Youth and Sled Dog Care-Mushing Program, this URSA 388 undergraduate research course, and the Iditarod sled dog race using Huslia as a halfway checkpoint this year have all contributed to the comeback of sled dog mushing in an Alaskan village, which is giving youth and adults an activity to work on and look forward to in their lives.

Families are now becoming interested in sled dogs for not only themselves but for their children and grandchildren. They not only want to participate in activities that their ancestors have done before them but they also see it as a way to help their children/grandchildren grow into adulthood. There is one grandfather who is buying a pregnant bitch from an Iditarod musher so he can raise the pups with his grandchildren in hopes of decreasing the amount of time they spend playing video games.

It is apparent that villages all throughout the interior of Alaska are also becoming interested in starting a dog mushing program for their community. Dog mushing brings great

pride to rural Alaskan communities and with pride comes confidence, which will hopefully lead to success in other areas of life.

Through the youth and sled dog program, Alaskan Elders also experienced a boost of joy and excitement in their lives, because their future representatives are not only carrying on the dog mushing tradition but the Elders also helped train today's new breed of sprint sled dogs! I gained so many great friends throughout the mushing community and also spent time and got to know my grandparents, Alfred Sr. and Helen Attla. Dog mushing has created a bridge between the youth and the Elders in rural Alaska and has created hopes for a more loving and brighter future.

Reflection

Spending my fall and spring with George Attla, his partner Kathy Turco, my family, the dogs, the community of Huslia, the filmmakers, and everyone who made the race season possible was one of the single best experiences of my life. I am thankful that I was blessed with the opportunity to spend a race season under the training of George while also representing my hometown.

On behalf of the Frank Attla Youth and Sled Dog Care-Mushing Program, I would like to thank Annamaet Dog Food, Ruby Marine, Wright's Air Service, the Alaska Dog Mushers Association, Huslia Dog Mushers Association, and over fifty private donors for their support. I would also like to thank Patty and Glenn Imus (Fairbanks, AK), Chuck and Tracey Schaeffer (Willow, AK), Roni Noonan-Agre (Tok, AK), and Leonard Sanford (Tanacross, AK) for giving myself, my handlers, coaches, and dogs a place to stay during the race season. I am thankful for Floyd Vent Sr. who let me use three of his top dogs in the 2015 GCI North American. Another shoutout goes to mushers who have helped groom and provide training trails such as Marvin

Kokrine, Bill Kornmueller, Wesley Henry, Floyd Vent Sr., Leonard Sanford, and Mary Woods. To my world-class handlers, filmmaker friends, Catharine, Melissa, and Andrew, and my coaches, the Late George Attla Jr., and also Alfred Sr. and Helen Attla, my dad, Fred Lee Bifelt, and fans from the mushing world, I thank you for the great times that you shared with me and all the lessons that you taught me.

Grandpa George passed away in mid-February but with all the stories, knowledge, and inspiration that he has spread throughout the world, I can only feel blessed to have known him and spent time with him. He has truly touched my life in a positive way, and I look at his life as a model of success. Spending time with the ‘champ’ made me feel confident in my abilities for the simple reason that I witnessed the late George go about his business at one hundred and ten percent in all aspects of his life. For George to be passionate and committed to a sport, to be an Alaska Native world champion from Huslia, and also to be my close relative, inspired me to chase my dreams and believe in myself. His mind seemed to always be working and wrapping around problems in search of solutions. He had a powerful imagination to create different scenarios that could potentially hinder the path to success - and then to resolve them. I always wanted to learn how his mind worked and how he always figured out a way to win. What I think was a huge contributor to his success was his open-mindedness and curiosity to learn and try new tactics. It was extremely difficult for me to be in his presence and tell myself that I could not achieve a goal... I mean, here was the greatest dog man who had ever lived, a guy who overcame so many obstacles through the years that it was downright ridiculous to tell him I could not succeed. He was a true champion and a bona fide legend.

“Heroes get remembered, but legends never die.” - Babe Ruth

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